

Periodic Turmites

First Revisits, a Literal Langton Ant and Its Diophantine Certificate

Research Reports 38, 40, 42, 44, 47 and 48

Four Parts

Mathematical research reports

3–4 October 2026

Abstract

[write] This report collects six research reports on binary and cyclic turmites on doubly periodic backgrounds with finitely many defects. **Part I** (Report 38) decides the first revisit of a turmite in polynomial bit time and decides exact first occurrence of finite pattern queries on its certified domain, which ends at the first revisit; hence no undecidable language reduces to globally one-visit runs with such an observation. **Part II** (Report 40) gives the other side: one fixed doubly periodic background for the binary Langton ant (the RL turmite), a fixed start and a finite input loader under which the ant simulates the fixed fifteen-state machine U15 of Neary and Woods, visits every cell at most twice over its whole infinite run, and passes a fixed residue-and-heading port exactly when U15 halts. With Part I this gives a sharp one-versus-two-visit boundary in a precisely stated model. **Part III** (Reports 42 and 44) turns the ant into arithmetic: a paid all-length initializer from raw positive integers, and one fixed polynomial F_2 with 465 positive witnesses, 2,307,457 operations and exact degree 2,304,000 whose positive zeros recognize the U15 sentinel-pair halting language. **Part IV** (Reports 47 and 48) builds the same polynomial from the literals 1 and 3 with ever smaller sources: 554,386,261,710,212,588 operations (Report 44), 31,388,831 (Report 47), 14,658,934 (Report 48), and, in the Hilbert’s-tenth programme’s later packets, 14,620,779 and then 14,620,711. The certificate recognizes a fixed recursively enumerable complete language; it is not an ordinary-input loader for the programme’s input convention, and it is not a universal polynomial in the programme’s sense (preface, item (a)). It does not improve the programme’s 84-operation universal bound.

[write] **Status.** Six AI-assisted research reports, unrefereed, with conventional proofs and finite exact replay checks; several theorems are explicitly conditional on inherited results (a published universal machine, a bounded-history Diophantine relation of the programme’s research tree, Mathlib’s Pell theorems). Nothing in them is formalized in Lean or Rocq. Written into ProveIt’s research-report collection (*hilbert-tenth-problem*): Part I from batch-83 manuscript 10, Parts II–IV from batch-88 manuscripts 03, 04, 07, 09 and 10; see the editorial preface and the provenance of Parts II–IV.

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Editorial preface to the four Parts

[write] This report began as the single-source report *First Revisit Detection and Exact Pattern Queries for Finite Defect Periodic Turmites* (Research Report 38), written in batch 83 (commit c2fe5f03b). That text is now **Part I**, unchanged except for dated notes, with its own editorial preface, numbering and appendices. Five later reports of the same pipeline answer its open question 3 and continue the answer into arithmetic; they were placed by commit ed7c76266 and are Parts II–IV. This preface, the notes marked [write], the Part introductions and the provenance section at the end were written then; everything else is the manuscripts’ own text.

Part	Report	Content	Labels; numbering
I	38	first revisit in P; exact first occurrence of finite observations; no reduction through globally one-visit runs	<code>ptr::</code> ; manuscript’s own
II	40	a literal periodic Langton ant for U15: fixed background, finite loader, at most two visits per cell, exclusive accepting port; the sharp one/two-visit boundary; a paid endpoint selector	<code>ptr:1a::</code> ; 40. <i>k</i>
III	42, 44	paid all-length arithmetic initialization (42); the complete positive polynomial F_2 and its one-input form F_1 (44, the Part’s main theorem)	<code>ptr:ai::</code> , <code>ptr:pc::</code> ; 42. <i>k</i> , 44. <i>k</i>
IV	47, 48	the same polynomial from literals 1 and 3: structured coefficients (47) and fused background polynomials (48)	<code>ptr:sc::</code> , <code>ptr:fu::</code> ; 47. <i>k</i> , 48. <i>k</i>

One theorem, four source sizes. Reports 44, 47 and 48 give the *same* polynomial F_2 (Report 48 writes $\mathcal{P}_2$): the same 465 positive witnesses (467 for the one-input F_1), the same single equation and exact degree 2,304,000, and, by the proofs of Proposition 47.9.1 and Theorem 48.1.1, the same value at every integer assignment. The recognition theorem is printed once, as Theorem 44.1.1; Reports 47 and 48 contribute smaller literal sources for it, and the remark after that theorem lists all of them. None of the later counts is an erratum of an earlier one; each is valid for its grammar.

What the certificate is, and what it is not.

- (a) *The language.* $F_2(x, y, z) = 0$ has a positive solution z exactly when the fixed machine U15, started in state A on the blank-zero tape with the two words sentinel-decoded from x and y , reaches its undefined pair J1. This *fixed* sentinel-pair halting language is recursively enumerable and complete, because Neary and Woods’ published encoding reduces every program to a U15 tape; that encoding is a dependency of all five reports, not implemented by any of them. The programme’s review puts it this way: relative to the inherited proofs, the certificate is “an ordinary-integer Diophantine recognizer of a fixed recursively enumerable complete language”, and it “is not, by itself, an emitted per-program map from the research tree’s ordinary input variable into the published U15 tape encoding”; “the external compilation dependency must not silently become a free arithmetic input loader” [W1]. So the certificate is **not** an ordinary-input loader for the programme’s input convention, and wherever this report speaks of universality it means this qualified sense: never a universal Diophantine polynomial in the programme’s sense without it.
- (b) *The bound.* Its operation counts are far above, and do not improve, the programme’s 84-operation universal representation (commit 20aafb9a5); the cost conventions (prescribed fixed

coefficients versus literal 1 and 3) differ and must not be conflated.

- (c) *The dependencies.* The recognition theorem is relative to the inherited 174-operation bounded-history theorem of the programme’s research tree, Mathlib’s constructive Pell theorems, Report 42’s recoder and initializer, Report 40’s endpoint and literal interface, and the published U15 simulation (Theorem 44.1.1 and its Section 44.3).
- (d) *The ant.* The accepting event is a pre-departure residue-and-heading event; the ant never physically halts and keeps running after it. The background is infinite and periodic: there is no finite blank-background universality.

Already reviewed. The Hilbert’s-tenth programme reviewed all five archives at their arrival commit `c5612efa1`. Its notes, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`: `review_periodic_ant40_48_intake.md` (commit `cfbe37de7`) [W1], `review_ant47_48_intake.md` and `review_periodic_ant_sharing48.md` (commit `5d7e884a5`) [W2, W3], and the successor packets `periodic_ant_endpoint_reuse48.md`, `ant48_rotated_occurrence_sharing.md` (both `5d7e884a5`) and `ant48_rotated_occurrence_cleanup.md` with its review (commit `57a6afbeb`) [W4]. It found no defect in the read proofs, algebra, ledger arithmetic or source binding. Its scope: a bounded proof and source-grammar read of 48 of the 590 archive members, with fresh independent checks (the 1055-row recoder’s 228 residuals, 888 coefficients and three outputs; the 174-row history DAG’s closure, count and degree; 2,296,800 correction-exponent instances and 6,912,000 profile entries of Reports 47–48); no archived program ran, no multi-million-gate stream was regenerated, and the physical templates, the full history theorem and the published simulation were not re-certified. Its findings, each also printed where it applies:

- (i) the scope of the language, item (a) above;
- (ii) the published U15 paper’s last sentence of Section 3.5 names the symbol c (here 0) in the halting transition, while its Table 16 defines that pair and leaves (u_{10}, b) , here J1, undefined; all five reports follow the table (Section 40.5 and Section 44.1);
- (iii) the earlier programme components for a Langton-ant toggle tape (105/158 and 105/155 operations) remain much narrower: they omit planar turns, the background, the loader and acceptance;
- (iv) no change to the universal 84-operation bound;
- (v) Maldonado, Gajardo, Hellouin de Menibus and Moreira give periodic-background universality with at most two visits, not universality from a blank finite-support board;
- (vi) two sharing opportunities in Report 48’s source, since realized by the programme: the endpoint reuses the initializer’s paid W^{576000} (−9 multiplications) and the 24 rotated occurrence Horner chains share six blocks per kind (−38,146), giving 14,620,779 / 14,620,789; a further zero-folding and common-subexpression cleanup (−68) gives 14,620,711 / 14,620,721. These are inherited-plus-delta counts: no new complete stream was emitted or hashed.

The programme also reviewed the placement commit `ed7c76266` (`review_periodic_ant_placement_ed7c76266.md`, commit `70bbc9ce2`) [W5]: every one of the 381 placed files is byte-identical to a member of the reviewed archives, and nothing new was introduced; it noted that the report’s article and README had not yet been written (this write supplies them, with the provenance of

Report 40’s derived cell maps), and that the count 14,620,779 quoted in the placement message predates the cleanup that gives 14,620,711.

What changes in Part I. Part II answers Part I’s question 3 (Section 12) almost word for word, and its Corollary 40.1.2 is the sharp threshold that Report 38 declined to claim. Dated notes in Part I record this beside the statements it makes stale: the question itself, Report 38’s non-claim of a sharp threshold (Sections 1 and 12), the preface paragraph on which side the report answers the programme’s interface obligations, the remark that no other report treats Langton’s ant, the quotation of the programme’s index, and Appendix C. Section 40.14 of Report 40 re-proves, in summary, Part I’s main theorems; a note there says which.

Notation across the Parts. The six reports reuse many letters, and no symbol was renamed. The readings most likely to be confused:

Symbol	Meanings, by Report
y , headings	38: y points north , $e(0) = (0, 1)$; 40: y points south , $d_0 = (0, -1)$ (Section 40.14 reflects one into the other); 42, 44: physical (a, b) east/south, then the rotation $X = b - 75$, $Y = 288650 - a$ makes the start face north
u, v	38: width and height of the explicit tile; 40 (Appendix 40.D), 42, 44: the rotated periods $u = 481238074400$ (vertical in the physical picture) and $v = 576000$; 47: $u = 2V$; 48: $u = 2L$
S V, L	38: $4uv$ projected states; 40, 47, 48: horizontal period 576000 (42’s and 44’s code call it V) 40, 47: $V = 400(R + 2)$, half the vertical period; 48 calls it L ; 40’s V_* and 44’s V_e are endpoint factors; L is also 38’s cycle length, 40’s cable length and left-marker block, and 42/44’s left word length
R	38: rule letter, coordinate bound; 40, 47, 48: $R = 601547591$ instruction rows (40 also: right-marker block); 42, 44: right word length
K	38: number of defects; 40: 958 active columns, and 3^{x_0} in Appendix 40.D; 42, 44: a repunit-quotient witness ($K_e = 3^{x_0}$ in 44; Pell k is different); 47, 48: 957 pair indices
W	40: 960 column slots, and the history radix 3^w in Appendix 40.D; 42, 44: history port $W = 3^w$, an independent degree-one variable in all degree counts; 48: the same W , the spatial variable
G F	40: the radius-one CA map; 42, 44: radix W^v ; 47: $\sum_{r < R} Z^r$ (48: G_Z); 48 also $g = W^S$ 38: defect time, signed indicator; 40: $400(R + 1)$ and a CA map; 42: B^n ; 44: the final polynomials F_2, F_1 (48: $\mathcal{P}_2$); 47, 48: footer profile $F[x]$, and 48’s $\mathcal{F}$ the dense first-column polynomial
H, h	38: history, interval end; 40: headed code $H(q, a)$, halt bit and heading h ; Appendix 40.D: FinalHead ; 42: padding power; 44: halting predicate $H(x, y)$ (48: $\mathcal{H}$) and a geometry witness; 47, 48: header profile $H[x]$
C, T, Q , Z	38: collision time, complexity bound Q ; 40: computation block C , CA rounds T , and in Appendix 40.D $C = 3^u - 1$, $T = W^{y_0}$, $Q = W^h$, $Z = \text{FinalSignPlus}$; 42: radix input C , recoder output T , port Q , $Z = (\Lambda + 1)^U$; 47, 48: correction column $C[x]$, occurrence constants $T_{a,s}$, signed profiles $Q_{a,d}$, $Z = 3^{400}$
x, y	44, 47, 48: the two raw positive inputs of F_2 ; elsewhere coordinates

Macros were reconciled without changing any rendering; the provenance section lists the renamed ones.

Formal status. No statement of any Part is formalized in Lean or Rocq. Report 42’s power macro (Lemma 42.4.1) specializes the Mathlib theorems `Pell.matiyasevic` and `Pell.eq_pow_of_pell`; the repository’s own Lean development uses the first of them (`Computability/HilbertTenthProblem/Lean/Diophantine/Paper1976/Cor26.lean`), but the specialization itself is not formalized. The 174-operation bounded history that Parts II and III inherit is a research-tree note, not a formalized theorem.

Third-party data. Report 40’s seven primitive cell maps are numerical data that its authors extracted from the MetaPost figure sources of Maldonado et al. (Part I, reference [1]; Section 40.3); they are shipped with that provenance and credit, and the report README states that they are not covered by the repository’s MIT-0 licence.

Part I

First revisits and exact pattern queries (Report 38)

Abstract (Report 38). For a cyclic left/right turmite rule, an explicitly listed periodic colour tile, and finitely many binary encoded defects, the first repeated lattice position is decidable in polynomial bit time. The algorithm returns the exact first time and an earlier witness, or certifies that every position is fresh forever. Its affine lane output has polynomial size even when the first revisit occurs exponentially late. We give the complete event driven proof, a numerical first revisit bound, and bounds on temporary Diophantine arithmetic. We also implement exact occurrence and least time for finite unions of heading restrictions, coordinate congruences, finite head site sets, and finite head relative colour stencils. An elementary signed progression calculus replaces the original general Presburger decision subroutine. The observation algorithm terminates without expanding a long flight or a whole residue period, but no polynomial complexity bound is claimed for its Boolean expansion. All observations use the board before departure and stop at the first repeated arrival when a revisit exists. Finite local observations on a certified fresh run are eventually periodic; this is not a periodicity assertion for the whole board. The accompanying frozen source packet and enclosing replay tools separate proofs, finite experiments, historical evidence, and release integrity.

[write] **Status.** An AI-assisted research report, unrefereed, with conventional proofs and finite exact replay checks. Nothing in it is formalized in Lean or Rocq. “In P” refers to the explicit input model of Section 2 (explicitly listed rule and tile, binary coordinates). Written into ProveIt’s research-report collection (*hilbert-tenth-problem*) from batch-83 manuscript 10; see the editorial preface and Appendix C.

Editorial preface

[write] This preface, the notes marked [write] in the text, Appendix C and the bibliography entries marked [write] were written when the manuscript was placed in ProveIt’s research-report collection. Everything else is the manuscript as delivered, with its numbering of sections, statements and equations. Its 59 labels gained the prefix `ptr`; no label was dropped or renamed otherwise.

The source. One manuscript: Research Report 38 of the Hilbert’s-tenth programme’s report pipeline, batch-83 manuscript 10 of ProveIt’s incoming reports (archive `Polynomial_First_Revisit_and_Exact_Pattern_Queries_for_Turmites_Package.zip`, arrival commit 3051d1446, placed by commit a51a439cd). It pins no single repository commit for its mathematics; its frozen source audit read two research-tree notes with search links pinned to commit 5883b08b7, which is taken as its pin (Appendix C). The delivered layout `evidence/source-packet/`, `verification/` and the release tools is shipped flattened: Appendix C and the report README map every delivered name to its shipped name. The frozen packet [3] is shipped except for its two pure checksum manifests, including the `manifest-sha256.json` whose SHA-256 Section 11 prints; the archive of the arrival commit keeps them.

Already reviewed. After arrival, the Hilbert’s-tenth research programme reviewed the archive in commit 308f14072, note `Computability/HilbertTenthProblem/Papers/research-wip/nativ`

`e-stream-queue/review_turmite_first_revisit38_intake.md` [4]. It read the proofs and the active source text; it did not import or run the archived Python, replay the logs, audit the release verifier or certify the delivered PDF. Its findings, each also recorded at the place it concerns:

1. *Verdict.* The first-revisit, polynomial bit complexity and exact observation arguments pass on their stated domains; no mathematical defect was found.
2. *Scope.* The unrestricted turmite substrate is universal, but the certified one-visit portion treated here is decidable and supplies no new universal arithmetic bound (Corollary 12.1).
3. *Complexity.* The proof bounds every intermediate integer (determinants, Bézout coefficients, interval endpoints, reconstructed indices, including discarded candidates), not only the winning event; the coarse $O(N_{\text{in}}^7)$ bound is justified for an explicit rule and tile. Binary coordinates may make τ exponential; succinct tiles and post-revisit dynamics are not covered (Section 6).
4. *Strict first-arrival observation boundary.* The colour identity (16) holds through the first repeated arrival and not after that site’s second departure; zero signed count implies emptiness only because of the Boolean invariant (Lemma 8.2); distributive expansion can be exponential, and no polynomial claim is made for observation compilation; eventual periodicity (Section 9) is of a head-relative stencil, not of the board.
5. *Attribution.* Turmite universality is due to Maldonado, Gajardo, Hellouin de Menibus and Moreira [1]; the review checked the primary paper (Section 12).
6. *A component lead,* Boolean-DAG occurrence certificates, which is unimplemented and is neither a first-hit-minimality certificate nor an ordinary-input universal loader (Section 12).

What the report answers, and from which side. Two notes of the programme’s research tree name the obligation of a turmite-based Diophantine interface that this report bears on: `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md` (line 394: one must “choose the exact macro-pattern whose occurrence is equivalent to simulated halting”) [5] and `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_TOGGLE_ROUTER_UNIVERSALITY.md` (line 128: identifying an actual simulated halt event is an additional compiler obligation) [6]. The report answers them only from the negative side: by Corollary 12.1, no such acceptance event of the finite observation language of Section 7 can be realized on globally one-visit runs, so a universal interface must use revisits. It supplies no interface; the remark at the end of Section 12 gives details.

[write] *Note added with Parts II–IV (batch 88, 3 October 2026).* The positive side now exists in this report. Part II (Report 40) constructs the interface from the upper side: a fixed periodic background, a finite loader and a residue-and-heading port that is passed exactly when the simulated machine halts, on runs that visit every cell at most twice (Theorem 40.1.1). Its Theorem 40.12.2 identifies the event equivalent to simulated halting, its Appendix 40.D pays an arithmetic selector for that event, and Part III pays the rest of the interface for the fixed U15 sentinel-pair language (with the scope stated in item (a) of the preface to the four Parts). The negative statement above is unchanged.

Formal status. No statement of this report is formalized. The standard theorem invoked in Section 7.2, decidability of Presburger arithmetic by Cooper’s quantifier elimination, is formalized in the repository in `Logic/PresburgerArithmetic`: the Lean declarations `PresburgerArithmetic.FoRMula.presburgerArithmetic_decidable` and `PresburgerArithmetic.FoRMula.holds_iff_quantifierEliminate`, and an independent Rocq proof of the one-variable Cooper step

(`cooper_finite_criterion`, `cooper_step_decidable`). The report only invokes that theorem; its executable route (Section 8) avoids it.

Notation. The manuscript reuses letters across sections; no symbol was renamed. The readings most likely to be confused:

Symbol	Meanings, by section
L, R	rule letters (turn left/right, Section 2); L is also the projected cycle length in (4) and (3) and the lower end of the count interval $[L, H]$ of (21); R is also the coordinate bound of (10)
A, B	lanes in Sections 4–5 (“prospective lane A , old lane B ”); $B = R + S + 1$ and $A_0 = Q + 1$ are complexity bounds in Section 6; B_F is the eventual threshold of (24)
Q	complexity bound $S(2B + 1)$; Q_F eventual period; Q_{obs} congruence period factor (Section 9)
q	colour read at departure (Section 2); a position residue (proof of Lemma 3.2); a coordinate index in (11) and Section 9; a modulus in Section 7 and (20); a colour index in Initial_q
a, b, c	lane base a_A , tile b , colour c_t ; also rank-one coefficients $an + bk = c$ of (6), the truncation time b , the congruence $an \equiv b \pmod{q}$, and signed coefficients c_A
s, r	earlier time s of (2) and restart time r ; also the Bézout directions $s = b/g$, $r = -a/g$ and a residue r
C, E, F, H	collision time C , excursion E , defect arrival time F , history H (Section 5); also the inequality $C + Ez \geq 0$, the signed indicator F of (22), and the interval end H
M, N	the intermediate bound M of (14) and the example parameter $M = 10^{100}$; lane endpoints N_A , the input length N_{in} , the threshold N_0 and the example parameter $N = 10^{100} + 7$

1 Conclusions and scope

The central decision problem asks whether a moving head ever returns to a lattice position it previously occupied. Repetition of the heading is not required. In the model studied here, all departures before that event read an unmodified cell. This observation converts the first revisit problem into a static walk problem. Periodicity then compresses each defect free excursion, and a finite number of defect encounters suffices to settle the entire infinite question.

The report establishes three distinct conclusions.

1. **First revisit is in P.** The input contains an explicit rule and tile, a finite defect map, and a starting position and heading. No one visit promise is assumed. The decision, exact first time, repeated position, earlier witness, and compressed valid path are computable in polynomial bit time.
2. **Exact first pattern occurrence is implemented.** On the certified path domain, finite unions of the stated clauses have decidable occurrence and an exact least time. This includes genuinely changing board colour tests, not merely head position tests. The implementation uses integer arithmetic, generalized CRT, and finite signed sums of progression indicators.
3. **Finite observations stabilize by phase.** On a globally fresh run, every fixed finite head relative colour stencil and the heading are eventually periodic. Coordinate congruences supply a further computable period factor. This concerns the finite observation, not the entire evolving board.

The polynomial theorem applies only to the first item. General first hit compilation can create exponentially many signed terms; neither that procedure nor general Presburger elimination is claimed to run in polynomial time. The algorithm does not analyze arbitrary post-revisit dynamics and does not make an ordinary ant physically halt.

Scientific provenance. This article consolidates the frozen original boundary proof, the complete observation implementation proof, and the separate bit complexity review dated 3 October 2026. The original boundary program left the general first hit subroutine unimplemented; the active extension supplies it. The historical original is preserved unchanged. A limited source search is evidence about what was inspected, not a priority proof. We make no novelty claim, no claim of a completed literal two visit universal tile and input loader, and no sharp one versus two visit threshold claim.

[write] *Note added with Parts II–IV (batch 88, 3 October 2026).* This is Report 38’s scope and stays its historical scope. The literal two-visit tile and input loader are now Theorem 40.1.1 of Part II, and the sharp one-versus-two-visit boundary, for the binary RL ant with a fixed residue-and-heading acceptance predicate and subject to the published U15 encoding, is its Corollary 40.1.2, whose lower side is the theorems of this Part. Report 40 itself says that Report 38 “correctly declined” the claim while the upper interface was absent.

2 Model and exact output contract

Let $m \geq 1$ and let the explicitly listed word $w \in \{L, R\}^m$ specify the turn for each colour $0, \dots, m-1$. Let $u, v \geq 1$ and let

$$b : \{0, \dots, u-1\} \times \{0, \dots, v-1\} \longrightarrow \{0, \dots, m-1\}$$

be an explicitly listed tile. The implementation stores rows indexed by y , then columns indexed by x . A finite map $D : \mathbb{Z}^2 \rightharpoonup \{0, \dots, m-1\}$ overrides the tile:

$$c_0(x, y) = \begin{cases} D(x, y), & (x, y) \in \text{dom}(D), \\ b(x \bmod u, y \bmod v), & \text{otherwise.} \end{cases} \quad (1)$$

Redundant defects, equal to the underlying tile colour, are allowed. Coordinates, periods, and integer parameters use ordinary binary notation; the colour rule and every tile entry are explicitly present. A succinct circuit description of an exponentially large tile is a different input model.

The starting head position is $p_0 \in \mathbb{Z}^2$ and its heading is $h_0 \in \{0, 1, 2, 3\}$. The headings are north, east, south, west, respectively; positive x points east and positive y north. Their unit vectors are $e(0) = (0, 1)$, $e(1) = (1, 0)$, $e(2) = (0, -1)$, and $e(3) = (-1, 0)$.

Time t denotes the configuration *before departure*. At time t , read $q = c_t(p_t)$, turn right or left according to w_q , increment the colour at p_t modulo m , and move one unit in the new heading. This convention agrees with the formal rotate, increment, move definition in the primary turmite paper [1].

Define the first repeated arrival time by

$$\tau = \min\{t \geq 1 : \exists s < t \ (p_s = p_t)\}, \quad (2)$$

with $\tau = \infty$ if the set is empty. Define the certified time domain

$$\mathcal{T} = \begin{cases} \{0, \dots, \tau\}, & \tau < \infty, \\ \mathbb{N}, & \tau = \infty. \end{cases}$$

The first repeated arrival belongs to $\mathcal{T}$. Its departure and later configurations do not belong to the claim.

Definition 2.1 (Affine lane). A lane $A = (a_A, d_A, t_A, \ell_A, N_A, g_A)$ represents

$$p_{t_A+n\ell_A} = a_A + nd_A, \quad h_{t_A+n\ell_A} = g_A, \quad 0 \leq n \leq N_A,$$

where $a_A, d_A \in \mathbb{Z}^2$, $\ell_A \geq 1$, and $N_A \in \mathbb{N} \cup \{\infty\}$. A singleton has $N_A = 0$. Lane endpoints are inclusive.

The returned lanes partition the times in $\mathcal{T}$, although on the revisit branch two times can have the same spatial position. Infinity is stored as a fixed symbol, not as an integer.

Theorem 2.2 (Total first revisit algorithm). *For every valid input, there is a total algorithm that returns either*

1. $\tau < \infty$, p_τ , an earlier $s < \tau$ with $p_s = p_\tau$, and an exact lane representation through time τ ; or
2. a certificate that $\tau = \infty$, an exact lane representation of every time, and a final translated periodic tail

$$p_{T_0+i+nL} = a_i + n\delta, \quad h_{T_0+i+nL} = g_i \quad (0 \leq i < L, n \geq 0), \quad (3)$$

where $1 \leq L \leq 4uv$ and $0 \neq \delta \in u\mathbb{Z} \times v\mathbb{Z}$.

The algorithm has polynomial bit complexity in the ordinary explicit input length and produces at most $(K+1)4uv$ lanes, where $K = |\text{dom}(D)|$.

Sections 3–5 prove totality and correctness. Section 6 supplies the bit complexity argument, which requires more than a count of large integer operations.

3 Static shadow and periodic excursions

Lemma 3.1 (Static shadow). *Consider the walk that always reads c_0 and never changes the board. Its positions and headings agree with the actual turmite through their common first repeated arrival, inclusive. Their first repeated arrival times are equal.*

Proof. Before a first repeated arrival, each departure is from a position never previously departed from. Its actual colour is therefore c_0 at that position. Starting with the same head, induction gives the same turn, unit step, and next heading for the two walks. The induction establishes the arrival at the first repeat as well; it does not require the next departure to agree. If either walk repeated earlier than the other, the induction up to that time would give the same earlier spatial equality in both, a contradiction. $\square$

Colour queries still require the evolving board and will be reconstructed separately.

Lemma 3.2 (The projected transition is a permutation). *On the defect free background, the transition of the projected state*

$$(x \bmod u, y \bmod v, h)$$

is a permutation of a set of size $S = 4uv$.

Proof. Given a successor position residue q and heading h' , subtract $e(h')$ to obtain the predecessor position residue. Its tile colour determines the turn that was applied there. Undo that quarter turn to recover the unique predecessor heading. Every successor thus has exactly one predecessor. $\square$

Consequently each projected orbit is a pure cycle, with no transient. From a restart at time r , compute the cycle's first L positions and headings, $1 \leq L \leq S$. If its displacement after L steps is δ , then

$$p_{r+i+nL} = a_i + n\delta, \quad h_{r+i+nL} = g_i, \quad 0 \leq i < L, \quad n \geq 0. \quad (4)$$

Its L phases are L lanes. Returning to the same residue and heading forces $\delta \in u\mathbb{Z} \times v\mathbb{Z}$. The reference implementation accommodates a generic finite state transient followed by a cycle, but Lemma 3.2 implies that its transient parameter is always zero on these inputs.

If $\delta = 0$, the hypothetical excursion repeats its start after L moves. Nonzero drift alone does not certify spatial self avoidance: distinct phases can still intersect. All phase pairs must be checked.

4 The exact lane intersection primitive

Given a prospective lane A and an old lane B , minimize the new time $t_A + \ell_A n$ subject to

$$\begin{aligned} a_A + nd_A &= a_B + kd_B, \\ 0 \leq n \leq N_A, \quad 0 \leq k \leq N_B, \\ t_B + \ell_B k &< t_A + \ell_A n. \end{aligned} \quad (5)$$

An infinite endpoint omits its upper inequality. When B is a static defect or target, omit the final chronological condition. Because $\ell_A > 0$, minimizing new time is the same as minimizing n .

4.1 Rank two

The spatial matrix has columns $d_A, -d_B$. If its determinant is nonzero, Cramer's rule gives the unique rational pair (n, k) . Integer divisibility checks decide whether that pair is integral. Check nonnegativity, inclusive endpoints, and, if requested, strict chronology. A passing pair is the unique feasible pair; otherwise no intersection exists.

4.2 Rank one

Choose a nonzero coordinate equation $an + bk = c$. Put $g = \gcd(a, b) > 0$. If $g \nmid c$, there is no solution. Otherwise extended Euclid gives $ax + by = g$ and the full integer family

$$n = n_0 + sz, \quad k = k_0 + rz, \quad n_0 = x(c/g), \quad k_0 = y(c/g), \quad s = b/g, \quad r = -a/g. \quad (6)$$

Substitution into the other spatial equation tests whether it holds identically or the rank one system is inconsistent. Each remaining bound is an integer inequality

$$C + Ez \geq 0.$$

For $E > 0$ it imposes $z \geq \lceil -C/E \rceil$; for $E < 0$ it imposes $z \leq \lfloor C/(-E) \rfloor$; for $E = 0$ it is either tautological or impossible. In particular strict chronology becomes

$$t_A + \ell_A n_0 - t_B - \ell_B k_0 - 1 + (\ell_A s - \ell_B r)z \geq 0. \quad (7)$$

The minus one is necessary: equality of arrival times must be excluded.

Intersecting the bounds produces one integer interval, possibly unbounded. If it is empty, return no hit. If $s > 0$, minimize at its lower endpoint; if $s < 0$, minimize at its upper endpoint. The bound $n \geq 0$ guarantees that the required endpoint is finite. If $s = 0$, any feasible parameter gives the same n ; use an available endpoint or zero. Reconstruct (n, k) and return the new time and earlier witness.

4.3 Rank zero

Both spatial steps vanish. Unequal bases make the relation empty. For equal bases, the earliest old index is $k = 0$, and it makes chronology easiest because $\ell_B > 0$. With no chronology the minimizing index is $n = 0$; with chronology it is

$$n = \max \left(0, \left\lceil \frac{t_B + 1 - t_A}{\ell_A} \right\rceil \right). \quad (8)$$

Check its endpoint. This case includes stationary lanes and zero drift cycles.

Proposition 4.1 (Exact primitive). *These three cases terminate using exact integer arithmetic and return the earliest feasible new time and an old witness, or correctly report emptiness. They handle negative coordinates and drifts, zero components, finite and infinite endpoints, same lane tests, and different headings at an equal position.*

Proof. The spatial system has exactly one of the three ranks. Cramer's rule is exhaustive in rank two. Equation (6) parameterizes all integer solutions in rank one, and every remaining constraint is included in the parameter interval, so minimizing its affine image is exact. In rank zero, existence of any chronological old index implies existence of $k = 0$, yielding (8). No branch tests heading equality, as required by (2). $\square$

5 Accelerating to the next decisive event

At a restart time r , maintain the current position and heading and a list H representing exactly the arrivals at times $s < r$. Thus the current arrival at r is excluded from history. Initially $r = 0$ and H is empty.

1. Construct the prospective defect free excursion E from the current head using (4).
2. Use strict chronology to find the earliest intersection of any lane of E with any lane of H or E . Call its time C , or ∞ if every pair is empty.
3. Test every lane of E against every defect, treated as a singleton spatial target without chronology. Let the earliest defect arrival be F , or ∞ .
4. If $C < \infty$ and $C \leq F$, return that repeat, its earlier witness, and H together with E truncated through C .
5. If $C = F = \infty$, return H and all of E as the complete fresh path and no revisit certificate.
6. Otherwise $F < C$. Append E through time F to H , perform the one departure prescribed by the input defect colour at that fresh position, and restart at $r = F + 1$.

Truncating a lane at time b replaces its endpoint by

$$\min \left(N_A, \left\lfloor \frac{b - t_A}{\ell_A} \right\rfloor \right)$$

and discards it if that endpoint is negative. The operation never enumerates its represented points.

Proposition 5.1 (Invariant and event ownership). *At every restart, the history is exact, its arrivals are pairwise spatially distinct, and all prior departures agree with the actual turmite. The first terminal event is the actual first repeated arrival, or a valid certificate that no repeat ever occurs.*

Proof. Assume the invariant at r . Before F , the prospective walk reads the correct initial board because it has not hit an overriding defect. Before C , all positions are fresh by the exhaustive prospective/history and prospective/prospective pair checks. Lemma 3.1 therefore validates these steps.

If $C \leq F$ is finite, the head reaches a repeated position at C and no earlier repeat exists. A defect tie does not change this: the arrival is already repeated, independently of what a subsequent departure would do. The truncated representation includes C .

If both event times are infinite, the entire prospective tail avoids all defects and has no collision with itself or with history. It therefore agrees with the actual walk forever and is globally fresh. Its drift must be nonzero, since zero drift would supply a same phase collision after one cycle.

If $F < C$, the defect position at time F is fresh. A prior occurrence in H or E would have yielded $C \leq F$. The actual colour there is exactly the override colour from D , so the special turn and unit move are correct. The appended history contains times through F and excludes the new current arrival at $F + 1$, preserving the invariant. If the move lands on an old point, the next iteration detects a collision at local time zero; it is not missed. $\square$

Proposition 5.2 (Termination and size). *There are at most K processed defect departures and $K + 1$ excursion constructions. The output contains at most $(K + 1)S$ lanes, where $S = 4uv$. A simple implementation uses at most*

$$P = S^2 \frac{(K + 1)(K + 2)}{2} + SK(K + 1) \tag{9}$$

pair intersection calls, plus $O((K + 1)S)$ projected successor steps.

Proof. Each processed defect position is fresh. A second arrival at a previously processed defect would be a historical collision at or before that arrival, and would terminate instead. Hence at most K such departures occur. At the stage with j previously processed defects, the history has at most jS lanes. Testing the at most S new lanes against history and all new lanes requires at most $(j + 1)S^2$ calls; testing all defects requires at most SK more. Sum over $j = 0, \dots, K$ to obtain (9). Each excursion adds at most S lanes. $\square$

What can be checked in a certificate. Retain the excursion start states, their projected cycles, their cut times, the fresh defect departures, and, when present, the final unbounded cycle. Every listed local transition is checkable. Comparing the excursion lanes with all defects verifies the first defect cut, and the same exact intersection primitive verifies all earlier disjointness conditions and the final tail's avoidance. Thus the no revisit claim rests on finite exact checks of infinite affine sets, rather than on an arbitrary simulation timeout.

6 Polynomial bit complexity

A polynomial number of arithmetic calls does not by itself prove membership in P: large temporary solutions could acquire too many bits. We bound both the candidate events and the arithmetic inside a call. The proof applies to the active first revisit engine and the same mathematical algorithm in the frozen original.

6.1 Parameters and a numerical first revisit bound

Let

$$\begin{aligned} S &= 4uv, & K &= |\text{dom}(D)|, \\ R &= \max\{|z_x|, |z_y| : z = p_0 \text{ or } z \in \text{dom}(D)\}, & B &= R + S + 1, \\ Q &= S(2B + 1), & A_0 &= Q + 1, \\ T_* &= (K + 1)A_0. \end{aligned} \tag{10}$$

R includes the starting position even when there are no defects. The symbols B, Q here are complexity bounds; the observation calculus below uses separate symbols for its eventual threshold and period.

Theorem 6.1 (Uniform numerical and representation bounds). *If a first revisit exists, $\tau \leq T_*$. Every restart time, stored lane base time, and finite new time returned by any pair call in the first revisit algorithm is at most T_* . All finite stored lane endpoints have index at most $2B$. Every lane base has each spatial coordinate of magnitude at most B . On the no revisit branch the tail starts no later than KA_0 .*

The numerical value of T_* can be exponential in the binary input length. Its bit length is polynomial, and no loop counts from zero to T_* .

6.2 Bounded bases and completed history

A restart position is either p_0 or one unit from a defect. Its coordinate magnitudes are at most $R + 1$. A phase base is reached in fewer than S further moves, so it lies in $[-B, B]^2$. The extra point used to compute the drift is also in that box. Each component of δ has magnitude at most $L \leq S$, and phase differences in the same excursion obey

$$|a_{i,q} - a_{j,q}| \leq |i - j| \leq L - 1 \tag{11}$$

for each coordinate q .

Consider a completed finite excursion ending at defect z at time F . In a retained lane its last point is $e = a_i + N\delta$. By the truncation formula,

$$0 \leq F - (r + i + NL) \leq L - 1.$$

The hypothetical walk needs fewer than S unit moves from e to z , so $e \in [-B, B]^2$. Every coordinate of an intermediate affine point lies between the corresponding coordinates of its endpoints. Thus *every historical point*, including those encoded by an extremely long lane, lies in this fixed box.

If $\delta \neq 0$, choose a coordinate q with $\delta_q \neq 0$. Then

$$N|\delta_q| = |e_q - a_{i,q}| \leq 2B,$$

so $N \leq 2B$. If $\delta = 0$, the excursion would revisit its start after L steps, while the processed defect satisfies $F < C \leq r + L$. Every retained lane then has endpoint zero. This controls all completed history before the final stage.

6.3 Local indices in every pair call

Historical times are strictly below the current restart r , whereas prospective times are at least r . Once index domains are imposed, chronology is automatic for a prospective/history pair.

Prospective lane versus history. Suppose $a_i + n\delta = y$ is a historical point. For nonzero drift, choose a nonzero coordinate of δ ; both the base and y lie in $[-B, B]^2$, so $n|\delta_q| \leq 2B$ and every feasible $n \leq 2B$. The old witness index is bounded by its historical endpoint. For zero drift, any matching historical point is already matched at new index zero.

Prospective lane versus a defect. The target's coordinates have magnitude at most $R \leq B$. The same argument gives $n \leq 2B$ for nonzero drift and minimizing $n = 0$ for zero drift. Thus a far away or unreachable defect cannot hide an uncontrolled intermediate event time.

Two phases with nonzero drift. Both phases have drift δ and period L . Their equality is

$$(n - k)\delta = a_j - a_i.$$

There is at most one integer $d = n - k$ satisfying it. Equation (11) gives $|d| \leq L - 1 \leq S - 1$. Chronology is exactly

$$i - j + Ld > 0,$$

independent of a common increment of the indices. If it holds, the minimizing pair is

$$n = \max(0, d), \quad k = \max(0, -d),$$

so both are at most $S - 1 \leq 2B$. If it fails, the ordered phase pair is empty. In particular, a same phase nonzero drift pair has $d = 0$ and is correctly rejected by strict chronology.

Two phases with zero drift. Unequal bases never match. For equal bases choose $k = 0$; the minimizing n is zero when $i > j$ and one otherwise. Thus $n \leq 1 \leq 2B$. The phase zero lane compared with itself guarantees a repeat at $r + L$.

Every finite new time returned by any of these calls therefore satisfies

$$r + i + Ln \leq r + (S - 1) + 2BS < r + Q. \tag{12}$$

This bound applies to discarded candidates as well as the winning event.

6.4 Absolute times and the terminal excursion

After a processed defect, $r' = F + 1$, and (12) implies $r' - r \leq Q \leq A_0$. After j defect departures, $r \leq jA_0$. Any candidate at that stage lies below $jA_0 + Q \leq T_*$. A stored phase base time is also at most T_* because $i < S \leq A_0$.

If the final event is a collision with winning index $n \leq 2B$, every other phase's endpoint when truncated at the same time is at most n : shifting its phase changes the quotient by zero or minus one. Thus every finite output endpoint is at most $2B$. Its points need not lie in the smaller historical box, but the sufficient bound

$$|\text{coordinate}| \leq B + 2BS = B(2S + 1) \quad (13)$$

holds. A final infinite tail is stored by at most S unbounded lanes; its future coordinates are symbolic. This proves Theorem 6.1.

6.5 Temporary arithmetic inside the primitive

Put

$$\begin{aligned} U &= 2T_* + 4BS^2 + 2B(S + 1) + 2S^2 + 4, \\ M &= 2T_* + 4(S + 1)^2U + 4(B + 1). \end{aligned} \quad (14)$$

These intentionally loose bounds cover the integer arithmetic arising from calls on the algorithm's lanes. They are not bounds for arbitrary externally supplied lanes in the general relation API.

1. Spatial matrix coefficients have magnitude at most S ; right hand sides, being base differences, have magnitude at most $2B$. Determinants have magnitude at most $2S^2$ and Cramer numerators at most $4BS$. Dividing by a nonzero integer cannot increase the numerator's magnitude when the quotient is integral.
2. Extended Euclid acts only on coefficients of magnitude at most S . Its stored Bezout coefficients have magnitude at most S , with zero input cases bounded by one; its quotient-coefficient products are bounded by S^2 . Scaling by the right hand side divided by the gcd gives $|n_0|, |k_0| \leq 2BS$. Parameter direction coefficients have magnitude at most S .
3. Testing the other spatial equation produces terms and sums bounded by $4BS^2$. Index bound constants have magnitude at most $2B(S + 1)$. The chronological constant in (7) has magnitude at most $2T_* + 4BS^2 + 1$ and its coefficient at most $2S^2$. Every constant and coefficient entering the interval calculation is below U in magnitude.
4. An interval endpoint is a floor or ceiling quotient by a nonzero integer, so its magnitude is at most U . The selected parameter is an endpoint or zero. Reconstructed indices therefore have magnitude at most $2BS + SU \leq 2SU$.
5. Multiplication by a time period or displacement and addition of a base produce magnitudes at most $T_* + 2S^2U + B < M$. Rank zero arithmetic, cycle construction, and truncation are smaller.

These estimates include inconsistent systems, nonintegral candidates, unsatisfiable intervals, discarded pair answers, and expressions formed before cancellation. They close the gap that would remain if only returned events had been bounded. Rule and colour indices are separately bounded by m .

6.6 Bit operation conclusion

A common bit width bound is

$$\begin{aligned}\beta &= 1 + \lceil \log_2(\max(M, m) + 1) \rceil \\ &= O(\log(K + 2) + \log(S + 1) + \log(R + S + 2) + \log(m + 1)).\end{aligned}\tag{15}$$

Each pair uses a bounded number of arithmetic operations besides extended Euclid, whose number of iterations is $O(\log(S + 1))$. Schoolbook integer arithmetic gives the coarse total bound $O(P\beta^3)$ for the pair calls. Input validation, finite state cycle construction, truncation, and writing the lane records are polynomial as well. Hash table average case assumptions are unnecessary: finite arrays or deterministic dictionaries suffice, and even linear scans add only polynomial overhead.

If N_{in} is the ordinary explicit binary input length, then

$$S = O(N_{\text{in}}), \quad K = O(N_{\text{in}}), \quad m = O(N_{\text{in}}), \quad \log(R + 1) = O(N_{\text{in}}).$$

Hence $P = O(N_{\text{in}}^4)$ and $\beta = O(N_{\text{in}})$. The arithmetic estimate is at most $O(N_{\text{in}}^7)$, a coarse upper bound rather than an optimized runtime characterization of Python. The compressed output occupies $O((K + 1)S\beta)$ bits, up to ordinary record overhead. This completes Theorem 2.2.

7 The exact observation language

A clause is a conjunction of the following optional restrictions:

1. The heading lies in a finite specified subset of $\{0, 1, 2, 3\}$.
2. Each specified integer linear coordinate congruence holds:

$$c_x(p_t)_x + c_y(p_t)_y \equiv r \pmod{q}, \quad q > 0.$$

3. The head lies in a finite specified set of absolute lattice positions.
4. For every stencil entry (f, α) , the pre-departure colour at offset $f \in \mathbb{Z}^2$ is exactly α :

$$c_t(p_t + f) = \alpha, \quad 0 \leq \alpha < m.$$

A query is a finite union of clauses. A missing heading or site restriction is unrestricted; an explicitly empty allowed set is impossible. Empty stencils and congruence lists are true. An empty clause union is false. Repeated stencil entries are interpreted conjunctively, so contradictory entries make a clause false. Palette size one is valid.

Theorem 7.1 (Exact first hit). *For every valid input and query of this language, occurrence in $\mathcal{T}$ is decidable. If it occurs, its exact least time, head position and heading, and a witnessing lane and clause can be returned. Otherwise the result is explicitly none. The procedure is implemented using only exact standard library integer arithmetic.*

7.1 Reconstructing the pre-departure board

For $t \in \mathcal{T}$ and $y \in \mathbb{Z}^2$, define

$$V(y, t) = \mathbf{1}\{\exists s < t : p_s = y\}.$$

Every cell has been departed from at most once before t . Even at $t = \tau$, the second departure from p_τ has not occurred. Therefore

$$c_t(y) = [c_0(y) + V(y, t)] \bmod m. \quad (16)$$

This identity is generally false after departure at τ . A future head point in the compressed representation must not count as visited yet; strict chronology is essential.

[write] The programme review [4] calls this the *strict first-arrival observation boundary*: every observation of Sections 7–9 is of the board before departure, on the domain $\mathcal{T}$, which ends with the first repeated arrival; nothing is asserted about the board after that site’s second departure.

7.2 The original Presburger formulation

Let the valid path lanes be $A_1, \dots, A_J$. For a fixed head lane A , set $x(n) = a_A + nd_A$ and $t(n) = t_A + n\ell_A$, and let $\text{Dom}_A(n)$ mean $n \geq 0$ together with its finite endpoint when present. At offset f write

$$\begin{aligned} \text{Visit}_{A,f}(n) \iff \bigvee_{j=1}^J \exists k [& \text{Dom}_{A_j}(k) \wedge a_{A_j} + kd_{A_j} = x(n) + f \\ & \wedge t_{A_j} + k\ell_{A_j} < t(n)]. \end{aligned} \quad (17)$$

The initial colour predicate is

$$\begin{aligned} \text{Initial}_q(y) \iff & \bigvee_{z \in \text{dom}(D): D(z)=q} (y = z) \\ & \vee \left[\bigwedge_{z \in \text{dom}(D)} (y \neq z) \wedge \bigvee_{(i,j): b(i,j)=q} (y_x \equiv i \pmod{u} \wedge y_y \equiv j \pmod{v}) \right]. \end{aligned} \quad (18)$$

At offset f , the desired colour α is exactly

$$\begin{aligned} & [\text{Initial}_\alpha(x(n) + f) \wedge \neg \text{Visit}_{A,f}(n)] \\ & \vee [\text{Initial}_{\alpha-1 \bmod m}(x(n) + f) \wedge \text{Visit}_{A,f}(n)]. \end{aligned} \quad (19)$$

Adding the lane domain, heading restriction, sites, congruences, and the equation $t = t(n)$ gives a first order Presburger formula $\text{Hit}(t)$. Vector equality means two linear equalities. All disjunctions outside the lane index quantifiers are finite and explicit.

A fixed constructive Presburger decision procedure, such as Cooper’s elimination procedure [2], decides $\exists t \geq 0 \text{ Hit}(t)$. If true, test existence in $[0, b]$ for $b = 0, 1, 2, 4, \dots$ until successful and then bisect. The bounded existence predicate is monotone, so this returns the exact minimum. This route is a complete effective proof, not a termination oracle. It was specified but not executed by the original program.

[write] Cooper’s procedure is formalized in this repository, in `Logic/PresburgerArithmetic` (Lean: `PresburgerArithmetic.Formula.presburgerArithmetic_decidable` and `PresburgerArithmetic.Formula.holds_iff_quantifierEliminate`, over the integers; an independent Rocq proof of the one-variable

step). The formula $\text{Hit}(t)$ above has not been written in that syntax, and no statement of this report is formalized.

The extension implements a specialized route instead. Its quantified relations involve only two lane indices, whose projection has a particularly simple form. The following construction avoids a general solver and avoids enumerating a huge modulus period.

8 The executable signed progression calculus

8.1 Atoms and exact intersection

An interval congruence atom is

$$\mathcal{A}(l, h, r, q) = \{n \in \mathbb{N} : l \leq n \leq h, n \equiv r \pmod{q}\}, \quad q > 0, \quad (20)$$

with $h = \infty$ allowed. Canonicalization clamps the lower bound to zero, advances to the first admissible integer, and, if bounded, retreats to the last admissible integer. Empty atoms disappear; a singleton uses step one. Thus stored endpoints are attained points.

To intersect residue classes $r_1 \pmod{q_1}$ and $r_2 \pmod{q_2}$, put $g = \gcd(q_1, q_2)$. If $g \nmid r_2 - r_1$, they are disjoint. Otherwise put $q'_2 = q_2/g$ and solve

$$k \equiv \frac{r_2 - r_1}{g} \left(\frac{q_1}{g}\right)^{-1} \pmod{q'_2}.$$

When $q'_2 = 1$, take $k = 0$. The combined residue is $r_1 + q_1 k$ modulo $\text{lcm}(q_1, q_2)$. Intersect the intervals and round inward to that residue. This proves closure under intersection using generalized CRT, including noncoprime moduli, modulus one, negative residues, and huge endpoints.

An atom's count in a finite interval $[L, H]$ is zero if there is no point. Otherwise, if f is its first surviving point and H' the smaller upper endpoint, it is

$$1 + \left\lfloor \frac{H' - f}{q} \right\rfloor. \quad (21)$$

Neither counting nor intersection enumerates the represented points.

8.2 Projection of a lane relation

Lemma 8.1 (One atom projection). *For lanes A, B , the set of new indices n for which there exists an old index k satisfying (5) is one interval congruence atom or empty. The same statement holds without chronology.*

Proof. In rank two, retain the unique integral feasible solution's n as a singleton. In rank one, use (6) and intersect every bound, including (7), to obtain an integer parameter interval. If $s = 0$, a nonempty interval projects to the singleton n_0 . Otherwise its affine image $n_0 + sz$ is an arithmetic progression with step $|s|$. For $s > 0$, its first point comes from the lower parameter endpoint and its last point from the upper endpoint; for $s < 0$ these roles reverse. The constraint $n \geq 0$ ensures a finite lower image endpoint even if the parameter interval is unbounded. In rank zero, unequal bases are empty; equal bases allow $k = 0$ and impose exactly the threshold (8), intersected with the new domain. This is an interval atom. $\square$

The implementation's `relation_indices` performs this projection rather than merely returning the first point. Shifting a head lane by a stencil offset changes its spatial base and leaves its time, period, heading, and endpoint unchanged. A static target is represented as a singleton lane and queried without chronology.

A coordinate congruence restricted to a lane has form $an \equiv b \pmod{q}$. It is empty unless $\gcd(a, q)$ divides b ; otherwise division by that gcd and a modular inverse give a single residue class. This also handles $a = 0$ and $q = 1$.

8.3 Boolean combinations as signed indicator sums

Represent a set of nonnegative integers by the finite signed sum

$$F(n) = \sum_{\mathcal{A}} c_{\mathcal{A}} \mathbf{1}_{\mathcal{A}}(n), \quad c_{\mathcal{A}} \in \mathbb{Z}. \quad (22)$$

The invariant is that $F(n) \in \{0, 1\}$ for every $n \in \mathbb{N}$. Individual coefficients can be negative. Public construction begins with the empty set, one validated atom, or the nonnegative universe. Boolean operations use

$$F \wedge G = FG, \quad \neg F = 1 - F, \quad F \vee G = F + G - FG. \quad (23)$$

Distribute products over the finite sums and use $\mathbf{1}_{\mathcal{A}}\mathbf{1}_{\mathcal{B}} = \mathbf{1}_{\mathcal{A} \cap \mathcal{B}}$. Equal canonical atoms are combined and zero coefficients removed.

Lemma 8.2 (Boolean invariant). *Every expression built by these public constructors and operations has a finite representation (22) and remains the exact Boolean indicator of the represented set.*

Proof. The assertion holds for all constructors. Intersection closure keeps each finite product finite. The three identities (23) are exactly the truth tables for conjunction, negation, and disjunction on zero and one. Structural induction proves the result. $\square$

Arbitrary injected signed coefficients are outside this contract. Merely counting a general signed function could give zero through cancellation while it was nonzero at some points; the Boolean invariant prevents that error.

8.4 Global emptiness and the exact minimum

Linearity and (21) give the exact number of represented points in any finite interval. By Lemma 8.2, this count is a nonnegative integer and is zero exactly when that interval contains no member.

For a finite signed representation define

$$\begin{aligned} B_F &= \max(\{0\} \cup \{\text{first}(\mathcal{A}) : \mathcal{A} \text{ unbounded}\} \cup \{\text{last}(\mathcal{A}) + 1 : \mathcal{A} \text{ bounded}\}), \\ Q_F &= \text{lcm}\{\text{step}(\mathcal{A}) : \mathcal{A} \text{ unbounded}\}, \end{aligned} \quad (24)$$

where the empty least common multiple is one. For $n \geq B_F$, every bounded atom has vanished and every unbounded atom is an unrestricted congruence class. Thus F is Q_F periodic there.

Proposition 8.3 (Finite exact minimum bracket). *The represented set is empty if and only if its count on $[0, B_F + Q_F - 1]$ is zero. Otherwise the exact minimum is found by binary search with predicate $\text{count}[0, h] > 0$ on that interval.*

Proof. The interval contains the entire exceptional prefix and one complete eventual period. A member beyond it can be reduced by multiples of Q_F to a member of its tail period. Hence zero count is equivalent to global emptiness. If the count is positive, the interval brackets the minimum, and prefix nonemptiness is a monotone predicate. Binary search returns its first true value. $\square$

This takes $O(\log(B_F + Q_F + 1))$ counting calls, each a finite signed sum over the current terms. A huge least common multiple is manipulated as an integer, not expanded into a residue array. The number of terms, coefficient sizes, and intermediate Boolean products may nevertheless be large. No polynomial bound for arbitrary observation formulas follows.

8.5 Compiling a complete clause and taking the first event

For a fixed head lane A and offset f , apply Lemma 8.1 to the shifted lane and each full path lane. Their finite union is exactly $\text{Visit}_{A,f}$, with strict chronology excluding present and future arrivals.

The initial colour set is compiled in two parts: union the matching defect singleton relations, then union the matching tile residue conjunctions after masking out *all* defect matches. This is exactly (18); defect overrides take precedence even if they are redundant. Compile (19) using the Boolean algebra and intersect over stencil entries. Intersect also the lane domain, constant heading test, allowed site union, and each linear congruence. These steps produce the exact set of satisfying indices for one lane and one clause.

Find its minimum by Proposition 8.3. Since time increases strictly with lane index, this yields that lane and clause's least event time. Minimize over the finitely many pairs of lanes and clauses. Every time in $\mathcal{T}$ appears in exactly one lane, and all predicates were compiled exactly, so the resulting hit or explicit none proves Theorem 7.1. Time zero, the first collision, and a collision tied with a defect remain included. Formula (19) is also correct when $m = 1$, because its two initial colour predicates coincide.

9 Eventual periodicity of finite observations

Assume $\tau = \infty$, and use the final tail (3). Fix phase i and offset f , so the observed position is

$$y(n) = a_i + n\delta + f.$$

The background colour at $y(n)$ is constant, since δ is a tile period. Because $\delta \neq 0$, this sequence meets each fixed defect and each point of the finite pre-tail history at most once. Those exceptions can be bounded without expanding that history.

Choose a coordinate q with $\delta_q \neq 0$. Compute m_0, M_0 , the minimum and maximum of that coordinate over all bounded pre-tail lanes and all defects. A bounded lane's extrema occur at its two endpoints. If that finite collection is empty, its exception threshold is zero. Otherwise write $z = (a_i + f)_q$ and take

$$n_{i,f}^{\text{exc}} = \begin{cases} \max(0, \lfloor (M_0 - z)/\delta_q \rfloor + 1), & \delta_q > 0, \\ \max(0, \lfloor (z - m_0)/(-\delta_q) \rfloor + 1), & \delta_q < 0. \end{cases} \quad (25)$$

Beyond that threshold, the observed point is strictly past every finite exceptional point in the chosen coordinate.

An earlier visit belonging to tail phase j must satisfy

$$a_j - a_i - f = d\delta, \quad k = n - d$$

for an integer d , unique if it exists. Its time is earlier precisely when

$$j - i - dL < 0. \quad (26)$$

If (26) holds, the visit exists exactly for $n \geq \max(0, d)$; otherwise it never contributes. Thus every tail contribution to the visited indicator is a threshold. Take the maximum of the applicable thresholds and (25), then the maximum over the finite set of phases and offsets. This yields a computable N_0 after which every requested colour is constant within each phase. The complete stencil and heading are therefore L periodic in absolute time from $T_0 + N_0L$ onward.

For a coordinate congruence $\ell(p_t) \equiv r \pmod{q}$, a phase index increment changes its left hand side by $\ell(\delta)$. A valid period in phase index is

$$\frac{q}{\gcd(q, \ell(\delta))}.$$

The least common multiple over all requested congruences gives a computable factor Q_{obs} , with empty least common multiple one. The resulting observations have eventual absolute period LQ_{obs} . A finite allowed absolute site set is eventually avoided by nonzero drift; its exception threshold is obtained by the same coordinate argument. Finite unions of clauses preserve the common eventual period.

Why the whole board is different. The head can drift while its visited trail grows and fixed initial defects remain behind. A finite window moving with the head eventually loses sight of these finite exceptions, which is all this proof needs. It does not show that the whole infinite board becomes a translate of itself after L steps.

10 Worked examples with enormous exact times

10.1 An affine staircase and an exponentially late revisit

Use rule $w = RL$, start at the origin heading north, and use the checkerboard

$$b = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad u = v = 2.$$

With no defects, induction gives

$$p_{2j} = (j, j), \quad h_{2j} = 0; \quad p_{2j+1} = (j+1, j), \quad h_{2j+1} = 1 \quad (j \geq 0). \quad (27)$$

At each even time the fresh colour zero turns north to east; at the next odd time colour one turns east to north. The head path never repeats. The projected cycle has $L = 4$ and $\delta = (2, 2)$, with four phase lanes:

Phase i	Base a_i	Heading g_i	Time	Point
0	(0, 0)	north	$4n$	$(2n, 2n)$
1	(1, 0)	east	$1 + 4n$	$(1 + 2n, 2n)$
2	(1, 1)	north	$2 + 4n$	$(1 + 2n, 1 + 2n)$
3	(2, 1)	east	$3 + 4n$	$(2 + 2n, 1 + 2n)$

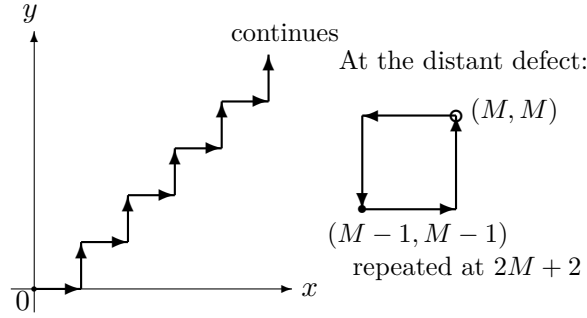


Figure 1: Exact staircase geometry. Left: the first ten unit steps of (27). Right: a translated local square at a defect of colour one at (M, M) . The bottom and right edges are traversed before the defect; its westward departure and the next southward step return to the lower left point. The right inset is schematic in location, with each edge representing one lattice step.

Now set $M = 10^{100}$ and add the single defect $D(M, M) = 1$. The initial colour at this even parity site is changed from zero to one. The head reaches it at time $2M$ while heading north. The defect turns it west, so at time $2M + 1$ it is at $(M - 1, M)$ heading west. That fresh odd parity site has colour one and turns the head south. Thus

$$\tau = 2M + 2, \quad p_\tau = (M - 1, M - 1), \quad s = 2M - 2. \quad (28)$$

The new point at $2M + 1$ is outside the earlier staircase; the next point is its first collision. At the earlier witness time the heading was north, while at the repeat it is south. A full head state repetition test would incorrectly miss this event.

The released test computes (28) using two excursions and six compressed lanes. With $S = 16$, $K = 1$, $R = M$, the coarse theorem gives $T_* = 64M + 1122$, safely above $2M + 2$. More generally, replacing M by 2^b produces a family whose first repeat is exponential in the defect coordinate's bit length. No path of that length is materialized.

10.2 A genuine two cell pattern whose first hit is huge

Return to the same staircase with *no defects*. Ask for heading north, $x \equiv 0 \pmod{M}$, and the exact two cell stencil

$$c_t(p_t) = 0, \quad c_t(p_t + (-1, -1)) = 1. \quad (29)$$

At a northward arrival $t = 2j$, the current fresh cell has initial colour zero. At $j = 0$, the offset cell $(-1, -1)$ has never been visited and has colour zero, so the stencil fails. At every $j \geq 1$, the offset cell is the former head point at time $2j - 2$; it has been incremented once from zero to one. Thus the stencil eliminates $j = 0$ and holds at every later northward arrival. The congruence makes j a multiple of M , giving the exact first hit

$$t_{\text{hit}} = 2 \cdot 10^{100}, \quad p_{t_{\text{hit}}} = (10^{100}, 10^{100}), \quad h_{t_{\text{hit}}} = 0. \quad (30)$$

Because M is even, the witnessing four phase lane is phase zero with index $M/2$. The packaged `example-result.json` records this result and the four lane count. This query is not equivalent merely to an input coordinate congruence: the evolving stencil is what excludes time zero.

The first revisit time bound in Section 6 does not bound arbitrary observation first hit times. This defect free run has no revisit at all, and its query's modulus can be arbitrarily large.

10.3 An unseen defect and a collision endpoint query

An off-path defect can matter to a stencil without ever changing the head’s motion. Put $N = 10^{100} + 7$ and override $(N, N + 1)$ to colour zero. That site is above the staircase and is never occupied. At a northward arrival, the offset $(0, 1)$ has background colour one and is unvisited, except at the override. The query “heading north and offset $(0, 1)$ has colour zero” therefore first holds at $2N$. The active tests compute exactly $2 \cdot 10^{100} + 14$.

In the collision example, a query requiring heading south, head site $(M - 1, M - 1)$, and current colour one holds at $\tau = 2M + 2$. The cell was changed on its first departure, and has not yet been changed a second time. This is a valid endpoint observation under (16). Attempting the same one visit colour reconstruction after this departure is outside the contract.

11 Implementation interfaces and assurance

The active standard library modules are `one_visit.py` and `observations.py`. The former implements the total first revisit decision and lane representation; the latter implements atoms, CRT intersections, relation projection, signed Boolean algebra, exact count/minimum, and the full clause compiler. No general Presburger package is installed or invoked.

End to end entry point. The recommended call is `observations.solve(rule, tile, defects, start, heading, clauses)`. It constructs the certified path and uses the same input board and rule for the query. The lower level `first_hit(result, rule, tile, defects, clauses)` has a substantive precondition: the supplied result must be the certified result for those same inputs. It does not authenticate a hand forged or mismatched result. The theorem is not a claim about arbitrary mutated Python objects.

Input and optimization guards. The active copy of the boundary engine replaced removable assertions with explicit exceptions. The active libraries contain no `assert` statements; their public validation and internal consistency guards still execute under `python -O`. Tests use `unittest` assertion methods, which are not removed by optimization.

The reviewed iterator defect. Independent review found that an earlier version consumed one shot clause iterators while validating them, silently changing later execution. The final `Clause` constructor materializes each field, including nested entries, into immutable tuples exactly once. Regression tests compare tuple and iterator forms, nested iterators, repeated queries, and mutation of the original input list. The preserved review records the initial finding and its resolution; it does not conceal the superseded failure.

11.1 Author tests and independent checks

Both the 16 author test groups and the seven independent groups pass in ordinary and optimized Python. The author regressions include:

- 1,344 exhaustive small binary instances, 3,000 randomized turmite instances, 60,000 exact lane comparisons, and 1,000 projected permutation checks

- 1,000 random Boolean expressions, 10,000 projected lane relations with 400,000 point checks, and 350 complete query comparisons with an evolving board
- enormous exact times, zero drift, unreachable defects, defect and collision ties, time zero and collision endpoints, invalid inputs, and iterator regression cases

The independent checker uses direct congruence evaluation, separately compiled Boolean truth tables, direct spatial membership, and a separately written changing board simulator. It does not use the release collision or colour reconstruction functions as its correctness oracles. Its seven groups cover:

- 1,470 small atom inputs, 8,000 CRT intersections, and 350 arbitrary four predicate Boolean truth tables
- 18,000 lane pairs, balanced as 6,000 of each rank, with 2,448,000 membership comparisons and both chronology modes
- 356 separate evolving board runs, 251 digit endpoints, a 501 digit CRT product, equal time exclusions, first collision colour observations, 17 invalid input checks, and iterator normalization

Finite revisit simulator comparisons cover their entire certified domains. Infinite run comparisons inspect finite prefixes and manageable reported hits; those experiments cannot alone prove absence of a later event. Global nonoccurrence is proved by the exact minimum bracket and is also tested on analytic impossible cases. Tests supplement the proofs.

The separate complexity review additionally records 10,000 deterministic random bound checks with seed 301003. That historical supplementary check is not a new executable test in this release and is not a premise of the mathematical proof. Its recorded result is retained in the frozen note without inventing a replayable checker for it.

11.2 Frozen evidence and the enclosing release

The complete 31 file source packet is retained under `evidence/source-packet/`. Its manifest file is pinned by SHA256

7dfd4f270ad7bf667bc123aace7bd97fd13b992949a9dfc8e3dca3f8105cea62.

That packet includes the historical `boundary-context/`, the active implementation and tests, the original normal and optimized logs, the observation review, and the complexity note. Every source byte is preserved. Original elapsed times remain part of that historical evidence and are not presented as deterministic output.

The source packet's own `verify_release.py` is a testing gate rather than a complete enclosing authenticity gate: it can use current hashes as its preservation baseline and allows caller selected output under the packet. The report therefore supplies a separate enclosing verifier. It checks the pinned source identity and every approved executable, rejects an output directory within the release, and stages only the active modules and author/independent tests. Historical boundary Python and the packet's own gate remain inert; no upstream repository code is executed.

Fresh logs, compiler products, temporary extraction, visual review images, and test receipts belong outside the frozen release. New test processes are run in ordinary and optimized modes with explicit exit checks. Reproducible scientific stdout and outcomes are compared after excluding

only the nondeterministic unittest elapsed-time field. Raw original logs remain unchanged. Hash verification, test replay, and proof validity are separate claims: none substitutes for the others.

The enclosing tools also support external no-shell-escape PDF rebuilding, deterministic ZIP construction, safe moved archive extraction and replay, and regression cases for tampering and hostile archive paths. The external QA receipt records the actual executed checks and final artifact digests. These mechanisms detect alteration relative to the approved local trust anchor; a hash manifest is not a digital signature or an identity proof for an unknown distributor.

[write] *In this repository.* The packet is shipped flattened beside this article: `evidence/source-packet/X` became `source-packet-X` (Markdown, at the report root), `code/source-packet-X` (Python) or `data/source-packet-X` (logs, receipts, JSON), with `boundary-context/` kept as a name segment; `verification/X` became `data/verification-X`; the release tools are in `code/`, and `MANIFEST.json` in `data/`. Not shipped: the delivered PDF, `SHA256SUMS`, both `manifest-sha256.json` files (including the one whose SHA-256 is printed above) and the byte-identical duplicate `boundary-context/test_one_visit.py` of `test_boundary_regression.py`; they survive in the archive of the arrival commit (Appendix C). The release verifier checks POSIX file modes and the delivered layout and therefore does not run on the shipped tree; the report README gives a rerun route for the six active programs on a copy. At placement and at the write, the 16 author and seven independent test groups and the example passed in normal and optimized Python, with stable outputs equal to the frozen receipts. The programme review did not execute the archived code [4].

12 Consequences for universality and further research

Corollary 12.1 (No reduction through globally fresh runs). *There is no computable reduction of an undecidable language that maps every input to a globally one visit finite defect periodic turmite run and characterizes acceptance by a query in the finite observation language of Section 7.*

Proof. Apply the reduction, run the total first revisit algorithm, and evaluate the exact query on its certified all-time domain. Under the stated reduction contract every produced run is globally fresh, so Theorem 7.1 decides acceptance. This would decide the undecidable language. The one visit class itself is decidable; no undecidable promise is hidden here. Fixing the rule or tile only restricts the input family and does not invalidate the argument. $\square$

Maldonado, Gajardo, Hellouin de Menibus, and Moreira prove universality for nontrivial cyclic turmites with periodic backgrounds and finite perturbations; their Section 5 explicitly describes an at-most-two-visits property of their constructions [1]. This is relevant upper-side context. The present report does not turn those diagrams into a literal coordinate atlas, input loader, and guarded accepting-port equivalence. Such a physical interface must specify exactly which head or stencil event means acceptance and prove both directions. Physical halting or return to a chosen absolute site does not follow merely from universality.

[write] **Attribution, checked.** The programme review [4] checked the primary paper [1] directly: its Theorems 2.1 and 3.1 establish simulation on periodic backgrounds with finite input perturbations, and its Section 5 states the at-most-two-visits property of its constructions. That is the upper side of the picture, and it is third-party work, cited and not reproved here. The paper does not instantiate a literal periodic colour atlas, an ordinary-input arithmetic loader or an iff accepting port for this repository. Ordinary turmites do not physically halt, and no sharp one/two-visit threshold is newly established: Corollary 12.1 is the lower side only.

[write] *Note added with Parts II–IV (batch 88, 3 October 2026).* For Report 38 this remains true. Part II instantiates the literal periodic colour atlas, the finite loader and the iff accepting port for one fixed construction in the spirit of [1] (whose qualitative universality and two-visit principle it credits), and its Corollary 40.1.2 establishes the sharp threshold in the stated model; Part III adds a raw-input arithmetic loader for the fixed U15 sentinel-pair language only, not an ordinary-input loader for the programme’s input convention (preface to the four Parts, item (a)).

[write] **Relation to the programme’s interface obligations.** The research-tree notes `EXPLORATION_ANT_CHECKERBOARD_HISTORY.md` (line 394) [5] and `EXPLORATION_TOGGLE_ROUTER_UNIVERSALITY.md` (line 128) [6], both in `Computability/HilbertTenthProblem/Papers/1980/` and unchanged since the pin 5883b08b7, list among the unpaid obligations of a Langton-ant or periodic-turmite Diophantine interface the choice of a macro-pattern whose occurrence is equivalent to simulated halting, and the identification of an actual simulated halt event. Corollary 12.1 answers these from the negative side: on runs that are globally one-visit, no acceptance event expressible in the observation language of Section 7 can carry an undecidable language, so such an event must be read on runs with revisits, as in the two-visit constructions of [1]. The corollary does not construct such an event, and it does not bear on the notes’ operation counts (the 174 operations of the bounded-history interface in the first note). The programme’s universal-operation frontier is unchanged by this report; the review recorded it unchanged, and the later record of 84 operations (commit 20aafb9a5) does not use it. The programme’s index still lists first-hit minimality and an ordinary-input universal loader as unpaid for the turmite route (`Computability/HilbertTenthProblem/README.md`).

[write] *Note added with Parts II–IV (batch 88, 3 October 2026).* That index sentence concerns Report 38 and its Boolean-DAG lead, and it still stands there. For the literal ant of Parts II–IV the programme’s later review [W1] records a narrower statement, not that the loader is paid: Part III’s polynomial recognizes the fixed U15 sentinel-pair halting language, an ordinary-integer recognizer of a fixed recursively enumerable complete language, but it is not an emitted per-program map from the research tree’s ordinary input into the U15 tape encoding. In the programme’s sense an ordinary-input universal loader therefore remains unpaid. That certificate quantifies over some finite accepting history, so it does not need first-hit minimality. The universal bound of 84 operations is unchanged by Parts II–IV as well.

[write] **A component lead from the review.** The review proposes to keep the observation formula of Section 8, after exact lane projection, as a shared Boolean DAG and to compile *membership at a supplied lane index* without expanding signed indicator products. Its atoms are affine inequalities and fixed-modulus congruences; the programme’s five-witness congruence component (`presburger_congruence_five.md` in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`, whose standalone 21-operation sum of squares excludes evaluation of the affine input, truth-output use and outer composition) supplies canonical truth and falsehood for congruences. An inequality truth bit $\chi = [\Lambda \geq 0]$ of an affine form Λ takes two natural coordinates χ, η and the residuals

$$\chi(\chi - 1), \quad \Lambda - (2\chi - 1)\eta - \chi + 1,$$

which force $\eta = \Lambda$ when $\chi = 1$ and $\eta = -\Lambda - 1$ otherwise; AND/OR/NOT gate values then have quadratic or affine residuals with unique values. (The review writes $b = [L \geq 0]$ with coordinates b, s ; the letters are changed here because b, L and s already have meanings in this report.) This is an *unimplemented* paid-component lead, not a saving or a bound: a paid compiler must still charge every affine form, atom, gate, truth-output use, square and final sum. It does *not* certify first-hit minimality (the global count and minimum of Proposition 8.3 would need a separate certificate); it does not certify arbitrary supplied lane tables (the cycle, defect-ownership and no-earlier-collision checks of Section 5 remain necessary); and it is not an ordinary-input universal loader. The review’s suggested next task is a bounded literal DAG compiler for one nontrivial clause of this report, with a full natural-fibre proof and a paid finalizer, compared against its expanded-atom form.

[write] **Neighbouring reports.** No other report of the collection treats turmites, Langton’s ant or first revisits. Research Report 35 (batch-83 manuscript 07), placed as source 22 of Part XX of the collection’s

report *canonical-diophantine-certificates* [7], supplies for abelian sandpiles what question 3 below asks for turmites: a literal periodic background, a finite input loader and an acceptance event (finite total activity) equivalent to halting of a fixed universal machine. It shares no theorem with this report.

[write] *Note added with Parts II–IV (batch 88, 3 October 2026)*. The first sentence remains true of the *other* reports of the collection, but this report now treats Langton’s ant itself: Parts II–IV (Reports 40, 42, 44, 47 and 48) give the literal periodic Langton ant for U15 and its Diophantine certificate. They are the turmite counterpart of Report 35 and of the certificate built on it in the same Part XX: in both, a literal periodic substrate, a finite loader for the same fixed machine U15 and an acceptance event equivalent to its halting. The constructions share no theorem.

Several focused next questions remain.

1. **Observation complexity.** Bound term growth for restricted stencils or congruence families, or devise a more compact representation that avoids distributive blowup while preserving exact count and minimum. A polynomial observation result would require a separate proof.
2. **Sharper first revisit bounds.** Improve the deliberately loose T_* and $O(N_{\text{in}}^7)$ estimates, determine tight parameter dependencies, and separate decision from certificate construction complexity.
3. **A literal two visit interface.** Supply explicit periodic colour coordinates, a finite input loader, exhaustive gadget visit accounting, and an iff acceptance port in the stated observation class. This is needed before asserting a sharp threshold in a common formal model.
4. **Independently checkable certificates.** Formalize the cycle, truncation, intersection, and colour reconstruction claims in a small proof checker or proof assistant. The current standard library replay checks are strong finite evidence, not machine checked proofs of all inputs.
5. **Model changes.** Investigate succinct tiles, other lattices or turning rules, multiple heads, and restricted post-repeat dynamics separately. The permutation, fresh-cell, and bounded-defect arguments must be re-established rather than assumed to transfer.

[write] All five questions are open at the write. For question 3, the sandpile analogue of Report 35 is cited in the remark on neighbouring reports above; it does not transfer to turmites. For question 4, the repository formalizes none of the cycle, truncation, intersection or colour reconstruction claims; the review’s Boolean-DAG lead above concerns Diophantine occurrence certificates, not first-hit minimality.

[write] *Note added with Parts II–IV (batch 88, 3 October 2026)*. **Question 3 is answered** by Part II, for the binary RL ant (Langton’s ant) and the fixed machine U15: Theorem 40.1.1 supplies explicit periodic colour coordinates (periods 576000 and 481238074400, a total colour query), a finite input loader with exactly $2812 + 4(\text{popcount } \ell + \text{popcount } r)$ overrides, exhaustive gadget visit accounting (Lemma 40.3.1, Proposition 40.4.1, Theorem 40.10.2: at most two visits per cell over the whole infinite run) and an iff acceptance port in this report’s observation class, a heading and two coordinate congruences (Theorem 40.12.2). Its Corollary 40.1.2 then asserts the sharp threshold in the common model, and Section 40.14 checks that the models match. The answer depends on the published U15 encoding of Neary and Woods for undecidability, and it does not cover other rules, blank backgrounds or other observation classes. Parts III and IV carry it into arithmetic, with the scope of item (a) of the preface to the four Parts. Questions 1, 2, 4 and 5 remain open; Report 40 also names a proof-assistant formalization of its finite-history composition as a next step (Section 40.17).

A Reader guide to the reproducible materials

The report source and PDF are `Research_Report38.tex` and `Research_Report38.pdf`. The frozen packet’s main scientific entry points are:

- `boundary-context/proof.md`: original total decision, Presburger formulation, eventual periodicity, and universality scope
- `PROOF.md`: the complete specialized first hit set calculus and colour compiler
- `complexity-review.md`: the numerical and intermediate bit bounds for first revisit
- `review.md`: independent implementation and mathematics audit, including the resolved iterator hazard
- `example.py` and `example-result.json`: the exact first hit at $2 \cdot 10^{100}$

Use the enclosing release’s instructions and verifier rather than executing historical context scripts. Python 3.10 or later is required by the source syntax; a local pdfLaTeX installation and standard TeX packages are needed only to rebuild the PDF. The scientific algorithms themselves use the Python standard library. No online service is required for replay.

[write] In this repository the report source is `article.tex` and its PDF `article.pdf`, built from this text (the delivered `Research_Report38.pdf` is not shipped). The entry points above are shipped as `source-packet-boundary-context-proof.md`, `source-packet-PROOF.md`, `source-packet-complexity-review.md`, `source-packet-review.md`, `code/source-packet-example.py` and `data/source-packet-example-result.json`. The enclosing release’s verifier does not run on the shipped tree (it checks POSIX file modes and the delivered layout); the report README gives the rerun route.

B Selected boundary cases and why they are covered

Case	Reason the algorithm remains exact
Repeated position with a new heading	Spatial equality is the only revisit criterion; headings are used for dynamics and optional query restrictions
Defect at the starting point	Its first arrival has $F = 0$ and is processed before its departure unless a genuine earlier collision exists, which is impossible at time zero
Defect tied with a repeat	Collision wins at $C = F$; its repeated arrival is included, and the subsequent turn is not used
Move from a defect directly to history	The next restart’s local index zero is checked against all history
Redundant or unreachable defect	Redundant overrides are harmless fresh events; unreachable defects generate no candidate and do not obstruct the final certificate
Zero drift or stationary relation	Rank zero handles stationary matches; a zero drift cycle necessarily supplies a repeat
Negative drift and parameter direction	Floor and ceiling bounds retain signs; negative parameter step reverses which endpoint gives the first projected index
Equal-time lane matches	The integer minus one in strict chronology excludes the current arrival and future arrivals

Case	Reason the algorithm remains exact
One colour palette	Increment is modulo one; the two initial colour alternatives coincide, and the Boolean visited split remains valid
Empty or contradictory query	Empty clause union is false, an unrestricted empty clause is true, and contradictory stencil entries have empty intersection
Huge first hit or large CRT product	Endpoints, periods and counts are exact integers; neither the trajectory nor the complete residue period is expanded
A query beyond first revisit	The certified domain ends at the repeated arrival. No claim about a later board is supplied

C [write] Provenance of this report

[write] *Note added with Parts II–IV (batch 88, 3 October 2026).* This appendix records Part I as written in batch 83. The report now has four Parts built from six manuscripts; the provenance of Parts II–IV (batch-88 manuscripts 03, 04, 07, 09 and 10) is in the provenance section at the end of the report.

Source. The report is built from one manuscript.

Manuscript	Research Report 38 of the programme’s report pipeline; batch-83 manuscript 10 of ProveIt’s incoming reports
Archive	<code>Polynomial_First_Revisit_and_Exact_Pattern_Queries_for_Turmites_Package.zip</code> , 504,531 bytes, SHA-256 <code>e5abfdbfbfcc9c203bdfafb040cba7f8c280af8b031b02103c989aeb97e085277</code> ; 47 files under <code>Research_Report38/</code> , main file <code>Research_Report38.tex</code> (703 lines), 21-page PDF
Arrival Pin	commit <code>3051d1446</code> <code>5883b08b7</code> (3 October 2026): the commit to which the frozen source audit (<code>source-packet-boundary-context-source-audit.md</code>) pins its repository searches, after reading <code>EXPLORATION_ANT_CHECKERBOARD_HISTORY.md</code> (blob <code>2e762cc5</code> , lines 305–401) and <code>EXPLORATION_TOGGLE_ROUTER_UNIVERSALITY.md</code> (blob <code>a70af5cc</code> , lines 1–95); the manuscript pins no commit for its mathematics. Both blobs are unchanged at the write.
Placement	commit <code>a51a439cd</code> (batch 83, cluster H2)
Review	commit <code>308f14072</code> , <code>review_turmite_first_revisit38_intake.md</code> [4]
Printed as	the whole report, with the manuscript’s numbering

What the write changed. Every label gained the prefix `ptr:` (59 delivered labels; the write added `ptr:sec:preface` and `ptr:app:provenance`). The write added the status box after the abstract, the editorial preface, the notes marked [write] (Sections 7, 11 and 12, Appendix A), this appendix and the bibliography entries marked [write]. No statement, proof, example, non-claim or question of the manuscript was removed or changed, and no symbol was renamed. The bibliography entry for the frozen packet was cited nowhere in the delivered text; the preface now cites it.

Where the placement chose. (1) Category: the report is filed under *hilbert-tenth-problem*, not *automata-and-formal-languages*, because its comparison targets are the programme’s turmite-interface notes, the programme reviewed it and derived a Diophantine component lead from it,

and the category already holds the collection’s dynamical-substrate reports. (2) Layout: as a new single-source report its files were staged unprefix; the nested `evidence/source-packet/` and `verification/` paths were flattened with “-” (the segment `evidence/` dropped). (3) Not staged: the delivered PDF, SHA256SUMS, the two `manifest-sha256.json` files and one byte-identical duplicate test file. All checksum manifests were verified at placement, and the archive is recoverable with `git show 3051d1446:docs/incoming/Polynomial_First_Revisit_and_Exact_Pattern_Queries_for_Turmites_Package.zip`.

References

- [1] Diego Maldonado, Anahí Gajardo, Benjamin Hellouin de Menibus, and Andrés Moreira. *Non-trivial Turmites are Turing-universal*. arXiv:1702.05547v1, 18 February 2017. Formal model in Section 1, simulation in Theorems 2.1 and 3.1, and visit accounting in Section 5. <https://arxiv.org/abs/1702.05547>; primary full text at <https://arxiv.org/html/1702.05547>. Inspected 3 October 2026.
- [2] D. C. Cooper. *Theorem Proving in Arithmetic without Multiplication*. In *Machine Intelligence* 7, 1972, pp. 91–99. Primary scan cited in the frozen boundary source audit: <https://www21.in.tum.de/teaching/logik/SS16/Exercises/Cooper.pdf>. Used for the original constructive Presburger route; the executable extension here uses the fully specified elementary calculus in Section 8.
- [3] *Effective one-visit turmite boundary and complete exact first-hit implementation packet*. Frozen local research sources dated 3 October 2026. Complete contents and SHA256 manifest are retained in `evidence/source-packet/`; its fixed manifest identity is stated in Section 11. Includes separate original proof, extension proof, complexity review, independent review, active source, test evidence, and the unchanged historical boundary context. This is the source record for the present article, not an external publication or priority assertion. [write] Shipped flattened beside this article (prefix `source-packet-`); its two `manifest-sha256.json` files are not shipped (Appendix C).
- [4] [write] Hilbert’s-tenth research programme, *Report38: first revisits and exact finite observations*, review note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_turmite_first_revisit38_intake.md`, commit 308f14072, 3 October 2026.
- [5] [write] *A counted 174-operation bounded ant history*, research-tree note `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md`, blob 2e762cc5.
- [6] [write] *Toggle/router models: three primary-source interface checks*, research-tree note `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_TOGGLE_ROUTER_UNIVERSALITY.md`, blob a70af5cc.
- [7] [write] *canonical-diophantine-certificates*, ProveIt research-report collection, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/canonical-diophantine-certificates/`; Part XX, source 22: Research Report 35 (batch-83 manuscript 07), placed by commit 216bd81e1.

Part II

A literal periodic Langton ant and the one/two-visit boundary (Report 40)

[write] **Source.** *A Literal Periodic Langton Ant Interface and the One Visit Two Visit Boundary. Exact finite templates and a fixed U15 accepting port*, Research Report 40, dated 3 October 2026, author line “Mathematical research report”; batch-88 manuscript 03, archive `Literal_Periodic_Langton_Ant_and_Visit_Boundary_Package.zip`; pin 5883b08b7 (its bibliography entry [5]), the same as Part I’s. Its sections, statements and equations are numbered 40.*k* here; its labels carry `ptr:la:` and its shipped files the prefix `02-literal-ant-`. It answers question 3 of Part I (Section 12) and proves the threshold Report 38 declined to claim. Report 40’s archive embeds Report 38 unchanged as its lower-side companion; that companion is Part I of this report, and its files are shipped once, under Part I’s names. The programme reviewed the report in commit `cfbe37de7` [W1] and found no defect in what it read; it did not regenerate the physical templates or the atlas. The scope of the arithmetic that later Parts build on this ant is item (a) of the preface to the four Parts.

Abstract (Report 40). We specify a single doubly periodic binary Langton ant background, a fixed head start, and a finite input perturbation that simulates the fixed fifteen-state two-symbol machine U15. The background has rectangular periods 576000 and 481238074400. For finite nearest-head-first tape words ℓ, r , the loader changes exactly $2812 + 4(\text{popcount } \ell + \text{popcount } r)$ cells, including a fixed 2806-cell startup patch. A fixed pre-departure residue-and-heading event occurs exactly when the simulated machine reaches its undefined transition J1. Every cell is visited at most twice over the entire infinite ant trajectory, including after simulated halting. We give the finite-history composition argument, a complete 862-NAND cellular rule, an exact arithmetic grammar for 601547591 physical instruction rounds, the growing-layer and long-wire proofs, and the accepting-port exclusivity argument. Together with the complete one-visit decision and observation theorem in Report 38, this yields a sharp one-versus-two-visit boundary for reachability in the precisely stated fixed-periodic-background model. The qualitative universality and two-visit principle are due to Maldonado and coauthors; the arbitrary-program-to-U15 encoding remains a published dependency. Neither a dense giant board nor the expanded instruction list is claimed to have been constructed. The accompanying archive includes all literal numerical maps, own-code generators, proofs, and receipts, with authenticated isolated replay and explicit limits on what those checks establish.

40.1 The result and its exact scope

The object of this report is a complete interface between a fixed source computation and one fixed physical ant, rather than a collection of truth-table gadgets without a controller. The interface includes the initial head, a finite list of changed colours, an effective colour query at every lattice site, and a specified physical event whose occurrence has both a soundness and a completeness proof. The controller keeps running after the simulated halt.

Theorem 40.1.1 (Literal periodic interface). *There is a fixed binary colouring $b : \mathbb{Z}^2 \rightarrow \{0, 1\}$ with periods*

$$S = 576000, \quad 2V = 481238074400,$$

and an explicit computable map I from pairs of finite binary words to finite colour overrides, with the following properties. Start the colour-zero-right, colour-one-left ant at

$$p_0 = (288650, 75), \quad h_0 = E$$

on b modified by $I(\ell, r)$. The source machine starts in state A reading zero, with nearest-head-first words ℓ and r on its two sides and zero elsewhere.

1. The override support has exactly $2812 + 4(\text{popcount } \ell + \text{popcount } r)$ sites
2. At every site there are at most two visits in the whole infinite ant run
3. $U15$ reaches $J1$ if and only if a pre-departure head state has heading S and residue, modulo $(S, 2V)$, in

$$\{(546702, 240606225650), (258702, 481225262850)\} \quad (40.1)$$

4. On the valid primary-loaded runs just specified, the first residue clause never occurs; the second clause alone gives the same equivalence

No colour stencil is part of the accepting event. The colouring, all overrides, and both residue clauses are independent of a simulated running time bound.

The proof is distributed across the construction, composition, initialization, and observation sections below. In particular, item 2 is an all-history theorem; it is not inferred from bounded simulation.

The distinction between the implemented $U15$ interface and its universality dependency matters. The included table is an exact fixed source machine. Its 30 table entries, one undefined, are fully simulated. The reduction from an arbitrary program to the appropriate finite $U15$ tape encoding is the published construction of Neary and Woods [2]; it is not a second compiler implemented by this packet. Invoking that dependency gives undecidability of the indicated accepting-port problem. The literal pair-to-ant loader is implemented here.

Corollary 40.1.2 (Precisely aligned visit boundary). *For the binary RL ant on fixed doubly periodic backgrounds with finite perturbations and a fixed residue-and-heading acceptance predicate, globally at-most-one-visit reachability is decidable. There is a fixed background, fixed start, fixed predicate, and computable family of finite perturbations whose trajectories are globally at-most-two-visit and whose reachability is halting-equivalent. Thus one visit cannot support a halting reduction with this observable, whereas two visits suffice, subject to the stated published $U15$ encoding dependency.*

Section 40.14 checks the model match explicitly. This is a sharp boundary for this decision model, not a statement that every two-visit run is universal, nor a threshold for every conceivable output interpretation.

Prior result and contribution. Maldonado, Gajardo, Hellouin de Menibus, and Moreira give periodic-background universality and the two-visit construction principle [1]. These qualitative results are credited to them. This report makes one fixed implementation literal through numerical cell maps, history-aware interfaces, source-machine compilation, global geometry, input anchoring, and exclusive observation. No novelty or priority theorem is asserted. The construction uses an infinite periodic background; it gives no finite blank-background universality, physical stopping event, or return-to-a-fixed-absolute-site detector.

40.2 Dynamics and evidence conventions

Coordinates increase east and south. Headings N, E, S, W have codes 0, 1, 2, 3 and unit vectors

$$d_0 = (0, -1), \quad d_1 = (1, 0), \quad d_2 = (0, 1), \quad d_3 = (-1, 0).$$

At time t , the snapshot (p_t, h_t, c_t) is taken before departure. If $a = c_t(p_t)$, set

$$h_{t+1} = h_t + 1 - 2a \pmod{4}, \quad c_{t+1}(p_t) = 1 - a, \quad p_{t+1} = p_t + d_{h_{t+1}},$$

leaving all other colours unchanged. A visit means an arrival snapshot, including time zero. Because the physical dynamics never stop, every visit has one departure; an all-time departure bound is therefore an all-time visit bound.

An owned support is the finite set of cells that a template supplies, including supplied colour-zero cells. Unowned cells are not automatically forbidden terrain; the global default is zero. A segment ends on its specified terminal head state *before processing that cell*. Consequently the terminal cell is owned and processed by the next segment when it starts. Counting both the upstream endpoint and the downstream entry as processed cells would be an error.

There are four kinds of evidence, with different logical force:

1. A finite cell map, explicit history, and step-by-step interpreter establish a finite template fact
2. Complete truth-table evaluation establishes equality of two finite Boolean functions
3. Arithmetic interval and ownership proofs cover all rows, coordinates, and histories, including infinite families
4. Hashes and replay receipts establish byte identity and repeatability of specified computations, not truth of a theorem by themselves

The article supplies ordinary mathematical proofs with independently replayed finite certificates. It is not a proof-assistant formalization. Historical component status strings are retained byte-for-byte: a record saying “Boolean cell only” still has that local scope even though the enclosing construction now supplies the missing geometry.

40.3 Literal primitive maps and legal histories

Seven finite numerical maps form the source basis: a straight cable, two turns, one bit box, two ordered crossings, and a union. The maps were extracted as numerical unit-square fills from the seven named MetaPost assets of [1]; no upstream PostScript, MetaPost, TeX, or other source program was executed. The complete extracted maps and their independent histories are retained as pure data. Appendix 40.A prints every primitive row, and the archive retains the exact coordinates, halo cells, intermediate boards, traces, and asset hashes.

[write] *Third-party data*. The seven maps are shipped as `data/02-literal-ant-literal-interface-common-primitive_maps.json` and `data/02-literal-ant-literal-interface-common-primitive_catalog.json`. They are numerical data derived from the MetaPost figure sources of the arXiv preprint [1], whose licence the archive does not state; the source-asset hashes are recorded in `02-literal-ant-literal-interface-SOURCE_PROVENANCE.md`. They are credited to Maldonado, Gajardo, Hellouin de Menibus and Moreira and are not covered by the repository’s MIT-0 licence (report README).

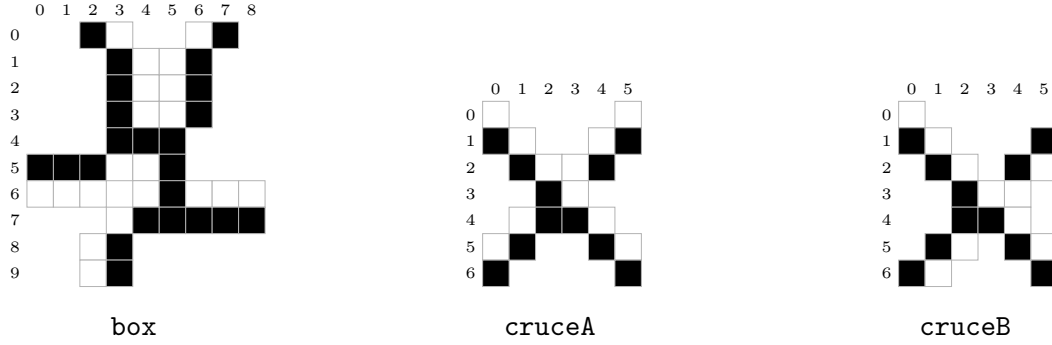


Figure 2: Literal numerical maps for the 43-cell bit box and the two ordered crossings. Black is colour one, white is a supplied colour-zero cell, and unfilled positions are outside the owned support. Coordinates increase right and down. The grids are generated directly from the archived numerical maps; they are not screenshots of a third-party article.

40.3.1 Ports and storage state

Template	Port	Entry to exit	Cells	Histories
cable_a	wire	$(0, 1, E) \rightarrow (9, 0, E)$	18	2
cable_b	wire	$(0, 1, E) \rightarrow (4, 6, S)$	17	2
cable_c	wire	$(0, 5, E) \rightarrow (5, -1, N)$	19	2
box	write	$(2, 0, S) \rightarrow (7, -1, N)$	43	6
box	read0	$(3, 9, N) \rightarrow (-1, 6, W)$		
box	read1	$(3, 9, N) \rightarrow (9, 6, E)$		
cruceA	first	$(0, 0, E) \rightarrow (6, 6, E)$	22	3
cruceA	second	$(0, 6, E) \rightarrow (6, 0, E)$		
cruceB	first	$(0, 0, E) \rightarrow (6, 6, E)$	21	3
cruceB	second	$(5, 1, W) \rightarrow (0, 7, S)$		
union	left	$(0, 3, E) \rightarrow (3, -1, N)$	25	3
union	right	$(8, 3, W) \rightarrow (3, -1, N)$		

The port coordinates are relative to each primitive’s normalized origin. In the box, INIT0 is the printed map. INIT1 changes exactly $(4, 4)$ and $(5, 4)$ from colour one to zero. READ enters at $(3, 9, N)$; its exit is $(-1, 6, W)$ for zero and $(9, 6, E)$ for one. WRITE enters at $(2, 0, S)$ and exits at $(7, -1, N)$. WRITE is the operation that changes an INIT0 logical box to the written-one form. A logical zero output is realized by omitting WRITE on a fresh INIT0 box.

The legal box histories are

$$\epsilon, \text{ WRITE}, \text{ READ0}, \text{ WRITE}; \text{ READ1}$$

from INIT0, and $\epsilon, \text{ READ1}$ from INIT1. The total departure counts in the four nonempty cases are respectively 18, 19, 41, 23. READ1 from INIT1 already has cells departed from twice. Thus the proof never relies on every individual primitive traversal being self-avoiding.

For each crossing the allowed history is ϵ , FIRST, or FIRST followed by SECOND. The first and second ports are the exact ones in the table. A union takes exactly one of its two alternatives, not both sequentially. Straight and turning cable pieces are used once. The catalogue includes explicit

failed histories: READ0 followed by WRITE does not yield the specified write exit, and SECOND before FIRST exits incorrectly. Those are actual interface restrictions, not interchangeable Boolean operations.

Lemma 40.3.1 (Finite history verification). *For the supplied seven maps, each legal history reaches every indicated terminal head state, has the recorded intermediate board, and departs from no cell more than twice in aggregate. INIT0/INIT1 changes and each optional read are included in that aggregate. Cable pieces have maximum one.*

Proof. For each history start with its explicit map and initialization, and apply the recurrence of Section 40.2 until the next specified terminal state. At each step the current coordinate must be in the owned support, its listed colour determines a unique new heading, and the colour is flipped once. Concatenate the operations on the *same* resulting board. Compare each generated transition, endpoint, final board, and per-cell count with the catalogue. There are two histories for each cable or corner, six for the box, three for each crossing, and three for the union. These finite lists are exhaustive for the stated interface languages. Taking the maximum over each listed count map gives one for cables and corners and two for the other resources. This finite computation is fully specified by the displayed maps and recurrence; complete traces are included for independent inspection. $\square$

40.3.2 Arbitrary even cable lengths

A horizontal cable of even longitudinal length L uses two lanes, colours zero above and one below, translated so its entry heading is east. Its local two-column trajectory is a four-step zigzag. After those four steps the head has moved two cells east, regained heading E and the same lane, and used each of the four cells once. Induction on $L/2$ proves correct exit and one departure per cell. Rotation by a quarter-turn preserves the rule and gives vertical and westward cables. This lemma is used for enormous but exactly specified long strips; it does not require materializing each site.

40.3.3 Shared box ownership

An output box of one stage can coincide with an input box of the next. The allowed intersection is the *whole 43-cell box with agreeing initial colours*. It has one physical owner and a single chronological history. Naming the box in two templates does not create two boxes. Its projected history is at most one optional WRITE and then one READ. A proof that added “two visits in the upstream gadget” to “two visits in the downstream gadget” would only give four and would lose the real invariant. Instead the merged box history is checked directly by Lemma 40.3.1. No triple painter or partial-box alias is admitted.

40.4 A complete physical instruction basis

Columns have pitch 600. A full physical round has old boxes near $y = 50$, new boxes near $y = 450$, an eastward control sweep at $y = 75$, and a westward sweep at $y = 305$. All output boxes begin as fresh INIT0 resources. The four adjacent-pair operations are

$$\begin{aligned} \text{NAND}(p, q) &= (1 - pq, 0), & \text{DUP}(p, q) &= (p, p), \\ \text{MOVE_LEFT}(p, q) &= (q, 0), & \text{MOVE_RIGHT}(p, q) &= (0, p). \end{aligned} \quad (40.2)$$

A delayed-COPY tile handles every unselected column and preserves its input. In DUP and the two moves, the ignored input may be either bit and may already have received a WRITE; it is not read. This is why unknown workspace contents do not invalidate the later word compiler.

The physical NAND template is a NAND macro followed by two normalized lower COPY pieces. NOT and ordinary COPY use a 200-unit half-round; delayed COPY and all pair instructions use a 400-unit complete round. The pair instruction replaces both consumed delayed-COPY slots exactly once. Its input/output ports and all translated primitive placements and routes are literal numerical data.

Resource	Owned cells	Height convention
NOT	3064	half-round
COPY	3134	half-round
Delayed COPY	5664	full round
DUP	10604	full round
MOVE LEFT	10550	full round
MOVE RIGHT	10318	full round
NAND with lower copies	13197	full round
Right marker joint	6269	exceptional joint
Left marker joint	6193	exceptional joint
Right and left row turns	684 and 564	full-round connectors

These are support sizes, not independent allowances that may be summed at shared cells.

Proposition 40.4.1 (Physical instruction contract). *With the declared chronology and fresh outputs, each complete round realizes (40.2) on its selected pair and preserves all other active columns. Its horizontal and vertical neighbors agree on initial colours and intersect only at declared whole-box aliases. Every cell in the complete legal resource history has at most two departures.*

Proof. For each tile, expand its listed primitive placements and cable routes by exact translation and rotation. Their maps agree on every shared owner. Enumerate the input bit assignments, prior input WRITE possibilities, both crossing orders where relevant, and all optional later reads of outputs. Project each composite trace to each owner and check it against the legal histories of Section 40.3. The resulting output state fields give (40.2); an untouched fresh output has value zero. The independent finite atlas then checks both horizontal neighbor halos, every vertically meeting translate, all pair types and both edge-turn continuations. Non-box supports are disjoint. Whole-box intersections are merged before counting. Hence local maxima apply to one real history at every physical cell. The archived independent interpreters reproduce the generated histories rather than trusting a generator's status field. $\square$

Two partial cases are needed in addition to full rounds. A skipped upper NOT pass may be followed only by the lower westward COPY. Its intermediate box remains INIT0, so its output is zero without reading the old input. Also the modified left marker/header joint can receive a normal east-only zero input when its earlier outer COPY was never used. The certificate for this second case covers unused old-source INIT0, INIT1, or prior WRITE0, followed by stopping, an output READ1, or the next header COPY and READ1: nine histories, all bounded by two. These cases close the initial partial header and every newly exposed outside edge; they cannot be assumed from the ordinary full-round cases.

40.5 The fixed machine and the total cellular rule

40.5.1 The exact U15 table

The fixed machine has states A through O, symbols 0 and 1, and the following transition table. An entry is written symbol, direction, next state. The undefined J1 transition is the simulated halting condition.

State	On 0	On 1	State	On 0	On 1
A	0RB	1RA	I	0RA	1LJ
B	1RC	1RA	J	1LK	halt
C	0LG	0LE	K	0RL	1RN
D	0LF	1LE	L	0RM	1RL
E	1RA	1LD	M	0LB	1RL
F	1LD	1LD	N	0LC	0RO
G	0LH	1LG	O	0RN	1RN
H	1LI	1LG			

The table's archived JSON hash is

0c6d8ae506f503e2783f6ed2ee68db4cde2f501b9864f376ab3773f86b59428a

The interface uses the 32-state encoding below. A separate 34-state sandpile convention is not imported.

[write] The table JSON (`ca/u15_table.json`) is not shipped in this report: it is byte-identical to `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/quadratic-orthant-certificates/data/16-universal-membrane-tm_table.json`. The table agrees with Table 16 of Neary and Woods [2] under $u_1, \dots, u_{15} \mapsto A, \dots, O$, $c \mapsto 0$, $b \mapsto 1$; its single undefined pair is (u_{10}, b) , here J1. The last sentence of that paper's Section 3.5 names c instead in the halting transition, which contradicts the table; Report 40 (its enclosing review, `02-literal-ant-enclosing-review-REVIEW.md`), Report 44 (Section 44.1) and the programme's review [W1] all note this and follow the table. The same table is the machine U15 of Part XX of *canonical-diophantine-certificates*, whose 34-state sandpile convention is the one not imported here.

40.5.2 Radius one with 32 primary states

A headless tape bit has code 0 or 1. If q is the zero-based index of A through O and a is the scanned symbol, the headed code is

$$H(q, a) = 2 + 2q + a \in \{2, \dots, 31\}.$$

Thus J1 has code 21. Define the total radius-one map $g(l, c, r)$ as follows. If c is headed, write its transition's tape bit; if $c = 21$, retain 21. If c is headless, an incoming right-moving head from l or left-moving head from r gives the corresponding next state combined with the current bit c . If both arrive, give the left arrival priority. If neither arrives, retain c .

The center-headed case takes priority over arrivals. These rules define g for all $32^3 = 32768$ triples, including malformed multi-head configurations. The priority on malformed double arrivals is a total extension; it is not an assertion that such configurations simulate a Turing machine.

Lemma 40.5.1 (Valid primary simulation). *On a configuration with one headed tape cell, one application of $G(X)_i = g(X_{i-1}, X_i, X_{i+1})$ gives the next machine configuration when the current*

transition is defined. A configuration with the unique head $J1$ is fixed. In particular, code 21 occurs in the valid evolution exactly when the machine reaches $J1$.

Proof. At the current head, the center-headed branch writes the prescribed tape bit. Exactly its destination neighbor sees an incoming head in the prescribed direction and constructs the prescribed next state over that neighbor's old symbol. The other neighbor sees no incoming head in its direction, and every other headless cell remains unchanged. Hence there is again exactly one head and the tape and state are correct. For $J1$ the center is retained, its undefined transition causes no arrival in a neighbor, and all other cells remain unchanged. Induction gives the assertion. $\square$

40.5.3 Radius half with 11-bit states

Use codes $0, \dots, 2047$. Primary codes remain $0, \dots, 31$. Define

$$\text{Pair}(a, b) = \begin{cases} 0, & a = b = 0, \\ 1024 + 32a + b, & \text{otherwise,} \end{cases} \quad 0 \leq a, b < 32.$$

The valid pair codes are 0 and 1025 through 2047. Code zero serves both phases. Define the two-input rule f in the following order:

1. If both inputs are primary, set $f(a, b) = \text{Pair}(a, b)$
2. Otherwise, if they are valid and consistent pair codes $\text{Pair}(a, b), \text{Pair}(b, c)$, set the output to $g(a, b, c)$
3. In every remaining case set the output to zero

There is no ambiguity at $(0, 0)$ because both interpretations give zero. The complete table has $2048^2 = 4194304$ entries, stored as little-endian two-byte integers.

Let $F(X)_i = f(X_{i-1}, X_i)$, and define $(\sigma X)_i = X_{i-1}$. For primary configurations,

$$F^2(X) = \sigma G(X), \quad F^{2t}(X) = \sigma^t G^t(X). \quad (40.3)$$

Indeed $F(X)_i = \text{Pair}(X_{i-1}, X_i)$, so the next step at i uses the consistent pair of pairs and returns $g(X_{i-2}, X_{i-1}, X_i)$. The second identity follows by induction and translation equivariance. Odd times contain only pair codes or zero, so cannot contain primary 21. Existence of primary 21 is invariant under the shift in (40.3).

40.5.4 Markers and the Boolean cell

The actual cell takes 24 bits, grouped as (x_L, s_0, s_1, x_R) with 11 bits for each state, and computes

$$\Phi(x_L, s_0, s_1, x_R) = (s_1, f((1 - s_1)x_L, x_R), f((1 - s_1)x_L, x_R), s_0(1 - s_1)). \quad (40.4)$$

The scalar multiplication in this notation means mask every bit of x_L by $1 - s_1$. In particular the left-edge branch $s_1 = 1$ ignores both x_L and s_0 for the state computation. This validates skipping their initial physical header prefix. There are again 24 output bits: one left marker, two copies of the 11-bit state, and one right marker. The separate halt bit is

$$h = \mathbf{1}\{f((1 - s_1)x_L, x_R) = 21\}.$$

40.6 The complete 862 NAND proof

The Boolean DAG has 24 inputs and 862 NAND gates, hence wire indices 0 through 885. Each gate lists two strictly earlier wires. The physical header first complements the 24 real input bits, and the footer later complements the 24 circuit outputs. Accordingly, the DAG receives complemented inputs and emits complemented Φ outputs; its halt wire is deliberately *uncomplemented*.

For clarity, the construction uses only identities reducible to NAND:

$$\neg a = \text{NAND}(a, a), \quad a \wedge b = \neg \text{NAND}(a, b), \quad a \vee b = \text{NAND}(\neg a, \neg b),$$

$$\text{MUX}(c, a, b) = (c \wedge a) \vee (\neg c \wedge b).$$

Equality to a fixed binary code is a conjunction of either each input bit or its negation. Finite conjunctions and disjunctions start from constructed constant-one and constant-zero wires. Identical unordered NAND pairs are memoized; no non-NAND primitive is left in the serialized circuit.

The circuit first negates each complemented input and masks the left state. A primary validity predicate tests that bits 5 through 10 are zero. Pair validity tests zero or bit 10 together with a nonzero lower ten-bit field. In the paired branch, the five low bits of the left pair equal the five middle bits of the right pair. The local radius-one circuit tests each machine-head code, constructs the incoming direction and next-state bits, and implements the center-head, halt, and left-arrival priorities exactly as in Section 40.5. The selected output bits combine the primary and consistent-pair branches; all invalid branch combinations vanish to zero. Duplicating those state bits and appending the markers gives (40.4). The 24 final negations supply the complemented footer inputs. A code-equality test produces h .

Theorem 40.6.1 (Complete Boolean agreement). *For every one of the $2^{24} = 16777216$ Boolean inputs, the 862-gate DAG on complemented inputs has complemented outputs Φ and halt wire exactly h . Its comparison is with the independently enumerated total tables, including malformed codes and all four marker assignments.*

Proof. The NAND identities prove every helper expansion. Code comparisons and branch selection implement the preceding total definition; the encoding calculation establishes its valid-machine meaning. To eliminate any ambiguity in the concrete compiled DAG, the independent checker also enumerates all 32768 primary triples to build g , all 4194304 state pairs to build f , and all four assignments (s_0, s_1) . For each marker assignment it forms one 4194304-bit integer for every input bit: bit position (l, r) stores that input on that state pair. A NAND is evaluated as bitwise complement of conjunction, masked to those 4194304 positions. Induction on the DAG rows proves that each live integer is the complete truth plane for that wire. The independent table-derived output truth planes are compared bit-for-bit with all 24 complemented outputs and h . Zero mismatch over four planes of assignments covers all 2^{24} inputs, not a sample of valid states. The serialized DAG and both complete table files are pinned in the receipt. Its peak of 63 live bitplanes is a memory implementation detail and does not reduce the exhaustive domain. $\square$

The short circuit-builder's randomized checks are ancillary. They are not used in place of the independent complete-table proof. Conversely, exact Boolean agreement alone establishes no physical tiling or visit bound; those are supplied elsewhere in this report.

40.7 An exact compressed physical program

The physical columns do not move. The logical occupied word is a prefix of those columns, with arbitrary contents allowed in unused slack. A logical WORD operation is implemented through (40.2); compilation never assumes fresh zero workspace unless a physical instruction itself creates it.

40.7.1 WORD insertion and contraction

For a word of length m and index i , put $r = m - i$.

- WORD(DUP, m, i) moves the suffix one position right, using MOVE_RIGHT at indices $m - 1, m - 2, \dots, i + 1$, then applies DUP at i . Its length is r and pair-left range $[i, m - 1]$
- WORD(NAND, m, i) applies NAND at i , then uses MOVE_LEFT at $i + 1, \dots, m - 2$. Its length is $r - 1$ and range $[i, m - 2]$

Descending right moves preserve every suffix bit before overwriting its former cell; ascending left moves likewise preserve each source until used. Hence the first operation inserts a copy immediately after the chosen bit, and the second replaces adjacent bits by their NAND and contracts the suffix. These proofs hold for arbitrary preceding, following, and slack contents.

40.7.2 SWAP and COPY to the end

The explicit logical XOR macro has three duplications and four NANDs. Three XORs and three more duplications give the standard adjacent swap identity. The resulting archived SWAP grammar contains 24 WORD operations, exactly 12 DUP and 12 NAND. Its truth table on the two selected bits is $(a, b) \mapsto (b, a)$, with every other logical bit untouched. The logical width increases by at most four and returns to its original value.

For a SWAP at index i in width m , the lengths of its 24 WORD terms are $r + c_j$, where

$$(c_j) = (0, -1, 1, 0, 1, 1, 2, 0, 0, 0, 2, 1, \\ 2, 2, 3, 1, 1, 0, -1, 0, 0, 1, -1, -1).$$

All terms are positive when $r \geq 2$, and $\sum c_j = 14$. Thus

$$L_{\text{SWAP}}(r) = 24r + 14.$$

Its physical counts are DUP 12, NAND 12, right moves $12r - 7$, and left moves $12r - 3$. Applying the WORD ranges gives exact pair-left range $[i, m + 2]$ and exact peak logical width $m + 4$.

To COPY(m, i) a source to the end, first insert a duplicate and then move that duplicate through the old suffix by $r - 1$ adjacent SWAPs. The original word is restored and one copy is appended at position m . The prefix length of the first t swaps is

$$P_r(t) = 12t(2r - t) + 26t, \quad 0 \leq t \leq r - 1. \quad (40.5)$$

Its increment is $24(r - t) + 14$, exactly the next SWAP length and strictly positive. Consequently

$$\begin{aligned} L_{\text{COPY}}(r) &= r + P_r(r - 1) = 12r^2 + 27r - 38, \\ n_{\text{DUP}} &= 12r - 11, \quad n_{\text{NAND}} = 12r - 12, \\ n_{\text{MOVE_RIGHT}} &= 6r^2 - 6, \quad n_{\text{MOVE_LEFT}} = 6r^2 + 3r - 9. \end{aligned} \quad (40.6)$$

For $r = 1$, COPY is the single final DUP; there is no empty-suffix exception to decode. For $r > 1$ its exact pair-left range is $[i, m + 3]$ and its exact peak width $m + 5$.

40.7.3 Appending a circuit gate

At logical width $m = 24 + j$, the j th DAG gate references wires $u, v < m$. Apply COPY(m, u), then COPY($m + 1, v$), and finally NAND at pair index m . The first two operations preserve every old wire and append the referenced bits at $m, m + 1$; the NAND stores their result at m and zero at $m + 1$. The new logical width is $m + 1$. This remains valid when $u = v$. Induction over all 862 gates proves circuit evaluation. Each GATE has three contiguous nonempty row intervals; its exact peak width is $m + 6$ and its exact pair-left range $[\min(u, v), m + 4]$.

The program gathers its 24 uniformly spaced real inputs, evaluates the DAG, copies the 24 selected complemented outputs and then h to the end, performs one dedicated observer DUP on h , gathers the outputs, and scatters them back to the uniform positions. The 958 positive-length top-level grammar records define exactly

$$R = 601547591$$

physical pair rounds. Their exact counts are

Instruction	Rows
NAND	1365370
DUP	1366258
MOVE LEFT	299589177
MOVE RIGHT	299226786
Total	601547591

No full 601-million-row list is included or needed.

40.7.4 Gather and scatter correctness

The selected active columns are $40j$, $0 \leq j < 24$. Input zero already lies at column zero. For $j = 1, \dots, 23$, left moves at descending pair indices $40j - 1, \dots, j$ move input $40j$ into j . Earlier gathered inputs lie below j and future sources above $40j$, so neither is touched. There are $39(1 + \dots + 23) = 10764$ rows.

The completed DAG occupies 886 wire positions. Copying its 24 outputs and h places them at 886 through 910. The observer DUP at 910 copies h to 911. For output $j = 0, \dots, 23$, descending left-move indices $886 + j - 1, \dots, j$ move that source into j , requiring 886 rows each, or 21264 total. Earlier gathered outputs and later sources are untouched. Finally, for $j = 23, \dots, 1$, ascending right-move indices $j, \dots, 40j - 1$ move output j to $40j$, for another 10764 rows. Descending order protects already scattered outputs. Any later overwrite of halt workspace happens after its dedicated observation and cannot alter the selected outputs.

40.7.5 Random access for every row

The top-level prefix array has 959 entries, starting at zero and ending at R , strictly increasing. For $0 \leq k < R$, binary search finds the unique interval containing k . The selected node's decoder

chooses a WORD, SWAP, COPY, or GATE branch. WORD has affine row descriptions. SWAP sequentially subtracts the 24 positive WORD lengths. GATE subtracts the two COPY lengths before its final NAND.

For COPY, after excluding its first r rows, use the invariant

$$P_r(\text{low}) \leq k < P_r(\text{high}), \quad 0 \leq \text{low} < \text{high} \leq r - 1.$$

Each binary branch preserves it and strictly shrinks the interval. On termination high equals low plus one; strict increase in (40.5) identifies exactly the next SWAP and a valid residual row inside it. At $r = 1$ this search is never entered. Thus the decoder terminates and returns the correct pair operation for *every* legal row.

The independent audit reconstructs all 958 nodes and every prefix entry from the raw DAG and the gather/scatter definitions. Its interval certificate covers 958 top-level intervals, 2586 GATE intervals, 83783 COPY intervals over all occurring residual widths, 20112 SWAP WORD intervals, and 40220 affine WORD subintervals. This is not random-row sampling: within each half-open arithmetic interval the branch decisions are constant and the residual index is affine, so endpoint checks together with the algebra cover its interior. Translation of both m and i shifts emitted column indices without changing lengths, allowing normalization by residual width.

The exact maximum pair-left index is 919, and the right index is at most 920. Active columns 0 through 957 therefore suffice. The exact compiler prefix peak is 915; the earlier conservative bound 917 is valid but not the exact peak. Physical scattering, rather than compiler scratch, accounts for the larger used destination 920. There are two surrounding routing columns, giving 960 total column slots.

40.8 The literal fixed colouring

A computation block C has an all-NOT/header-COPY round, R instruction rounds, and a NOT-footer/footer-COPY round. Put

$$K = 958, \quad W = 960, \quad S = 600W = 576000,$$

$$F = 400(R + 1) = 240619036800, \quad V = 400(R + 2) = 240619037200.$$

Layer $j \in \mathbb{Z}$ has vertical origin jV and horizontal shift $(j \bmod 2)S/2$. Its block i has origin

$$(iS + (j \bmod 2)S/2, jV).$$

All such blocks, including unused negative-index blocks and layers, belong to the background. Finite input perturbations select a growing portion of this already fixed hardware.

The literal generator is a total colour query. Given (x, y) , let $j_0 = \lfloor y/V \rfloor$ and inspect $j_0 - 1, j_0, j_0 + 1$. For each, translate to local $Y = y - jV$ and retain only $50 \leq Y \leq V + 259$. Let i_0 be the floor of its shifted x/S and inspect $i_0 - 1, i_0, i_0 + 1$, retaining the block horizontal support. For header and body positions inspect the three possible rounds around $\lfloor (Y - 50)/400 \rfloor$; only the legal range $0 \leq r \leq R$ survives. In round zero place the NOT and lower COPY resources, except for the specially supplied marker header. For $r \geq 1$, decode instruction $r - 1$, place that pair once, and place delayed COPY in all other active slots that can meet the coordinate. Add the exact left and right turns. A cell near a 600-wide column boundary needs only columns $k_0 - 1, k_0$.

The footer chooses the right-marker joint at absolute column 921 and ordinary NOT elsewhere. Its outer sweep includes all 960 storage columns; the left-marker joint is at absolute column 1. The normal C-to-C rising channel is supplied separately. Every nonempty painter list must agree in colour; an empty list returns zero.

Proposition 40.8.1 (A well-defined periodic board). *The selector terminates at every integer coordinate, finds every possible painter, and gives a single colour. Its rectangular periods are $(S, 0)$ and $(0, 2V)$.*

Proof. Each resource lies within $0 \leq X \leq S + 599$ and $50 \leq Y \leq V + 259$, so its true layer and block lie among the three candidates just described, including at negative coordinates because integer floor is used. A full-round support reaches at most one neighboring 400-unit slab; a column support reaches at most its neighboring 600-unit slot. Thus the finite round/column candidates are exhaustive. Pair placements replace both selected delayed COPY slots, preventing duplicate unrelated owners. The independent finite compatibility atlas checks all template classes, boundary classes, marker exceptions, and both stagger parities; its 1974 multiply-painted groups are complete agreeing 43-cell boxes and contain no triple painter. The long pieces are covered by the uniform inequalities proved in Section 40.9. Hence no missed painter or conflicting colour remains. Finitely many comparisons, lookups, and terminating row decoders make the query total. Increasing x by S increments the block index, while increasing y by $2V$ increments the layer by two without changing its horizontal phase. All local coordinates and maps are unchanged. $\square$

The finite review evaluated 263525 template, box, and anchor cells. That is finite evidence for the finite cases. The selector-range argument and arbitrary-even-wire proof cover the infinite translations and long cable lengths; the finite count is not represented as an exhaustive enumeration of the huge period rectangle.

40.9 Turnarounds and the long return channel

Each complete internal round has the verified east-to-west and west-to-east connectors. The exceptional footer and header controllers are chronological joints, rather than a Boolean marker wired onto a path of unknown history.

40.9.1 The right marker

The right marker is selected output slot 23, active column 920, absolute column 921. Its joint computes the marker before selecting a branch. On zero it continues the eastward footer and later accepts the ordinary westward COPY visit. On one it performs the output WRITE, joins its own outer COPY, and turns into the leftward sweep immediately. Its route never requires a crossover SECOND whose FIRST was skipped. This is why the marker gadget is verified as a combined NOT/COPY/branch resource, not as separate maxima.

40.9.2 The left marker

The left marker is selected output slot zero, absolute column one. Its joint either copies zero and continues west, or copies one and immediately enters the next layer's s_1 header NOT. The next

layer is shifted by half a block. Since $S/2 = 12 \cdot 40 \cdot 600$, selected output slot t aligns with selected input slot $t + 12$ modulo 24:

$$0 \mapsto 12, \quad 1 \dots 11 \mapsto 13 \dots 23, \quad 12 \dots 22 \mapsto 0 \dots 10, \quad 23 \mapsto 11.$$

Vertically, ordinary outer-COPY output is at $F + 450 = V + 50$, exactly the next header input. The left-marker header output is at $F + 650 = V + 250$, exactly the next header COPY input. These identities match full box maps, not just representative center points.

For fixed physical block indices, even-to-odd outputs 0 through 11 go to block $i - 1$ and 12 through 23 to block i . Odd-to-even they go to i and $i + 1$. The resulting phase-dependent spatial reindexing must be distinguished from the fixed logical index in (40.3). Both preserve the existence of state 21, which is the observation used here.

40.9.3 Normal completion into the next block

Let $e = 575400$. A normally completed C rises through its far-right margin and enters the next C header. Its exact successive head states are

$$\begin{aligned} (e, F + 75, E) &\longrightarrow (e + 170, F + 75, E) \\ &\longrightarrow (e + 175, F + 69, N) \\ &\longrightarrow (e + 175, 79, N) \\ &\longrightarrow (e + 180, 75, E) \\ &\longrightarrow (S + 600, 75, E). \end{aligned}$$

The pieces are an east strip of length 170, the 19-cell E-to-N corner, a north strip of even length $F - 10$, the rotated 17-cell N-to-E corner, and an east strip of length 1020. Their owned rectangles are

Piece	x interval	y interval
Bottom east	$e \dots e + 169$	$F + 74 \dots F + 75$
E-to-N corner	$e + 170 \dots e + 176$	$F + 70 \dots F + 75$
North strip	$e + 174 \dots e + 175$	$80 \dots F + 69$
N-to-E corner	$e + 174 \dots e + 179$	$74 \dots 79$
Top east	$e + 180 \dots S + 599$	$74 \dots 75$

They are pairwise disjoint; shared head seams are terminal states excluded from the earlier piece. Their total support and exact departure count is

$$2 \cdot 170 + 19 + 2(F - 10) + 17 + 2 \cdot 1020 = 2F + 2396 = 481238075996. \quad (40.7)$$

Lemma 40.9.1 (Long-channel isolation). *Every cell of this route has the specified cable or corner colour and is owned by no unrelated active resource. The route visits each owned cell once.*

Proof. The two finite strips and corners have explicit map checks and exact ports. On the north strip, for every $80 \leq Y \leq F + 69$, the $x = e + 174$ cell is colour zero and the $x = e + 175$ cell is colour one. Substitution into the generator's rotation formula gives longitudinal coordinate $a = F + 69 - Y \in [0, F - 11]$ and lane $b = x - (e + 175) + 1 \in \{0, 1\}$. This proves every long-wire cell, not selected sample heights.

The active body is at $x \leq e$ and internal right turns at $x \leq e + 59$, strictly left of the north strip. The outer right-guard COPY starts at $F + 250$, below its last owned row $F + 69$, and the next layer starts at $V + 50 = F + 450$. Prior ordinary outer-COPY and right-marker output footprints end at current height 59. The only previous-layer extension reaching current heights 80 through 259 is the left marker, confined modulo S to $[288600, 289200]$, far left of this channel. A previous same-layer block reaches at most current $x = 600$, while the next starts at S ; the north strip is strictly between them. The top strip ends at $S + 599$ and excludes its next header entry $(S + 600, 75, E)$. These uniform inequalities exclude all other painters. The even-cable induction and finite corner traces give exactly one departure per owned cell. $\square$

40.10 The growing-layer controller and global visit proof

At a layer boundary let the represented state word have length $m + j$. Its unique left marker is $s_1 = 1$ at block L and its unique right marker is $s_0 = 1$ at block R , with $R - L = m + j$. All intermediate marker bits are zero. One layer computes the $m + j + 1$ pairs spanning the word and a quiescent neighbor at each end.

Lemma 40.10.1 (One complete layer). *Starting in the marked left header, the ant computes the correct next radius-half layer, returns through the outer westward COPY sweep, and enters the next marked header. The next word has one additional state; its physical left and right edges expand by $S/2$ in the two directions. Every newly exposed input is quiescent.*

Proof. The first C is entered at its s_1 header, skipping the x_L, s_0 prefix. Equation (40.4) ignores the skipped left state under $s_1 = 1$, and the certified W-only header histories account for all skipped columns. Its complete instruction program is then executed. Subsequent blocks are entered normally by the isolated rising route of Lemma 40.9.1. The right-marker joint at the unique $s_0 = 1$ computes the last meaningful output and diverts before escaping into uncomputed eastward hardware.

The outer westward sweep copies every computed output box in the interval. Zero-valued intermediate left-marker tests continue west. The unique copied $s_1 = 1$ enters the next layer. The marker equations in (40.4) retain one left endpoint and propagate one right endpoint; the duplicated state and stagger identities of Section 40.9 give the correct adjacent input pairs. The next interval has one extra state and edges shifted by half a block. Fresh outside boxes were never activated, so they are INIT0. The certified normal header E-only zero case covers the boundary whose preceding outer COPY never ran. Thus the same invariant is established for the next layer. $\square$

Theorem 40.10.2 (Infinite global composition). *For every loaded finite pair, the unique physical trajectory is an infinite concatenation of the certified resource histories. No lattice site is visited more than twice, for the whole trajectory rather than only through acceptance.*

Proof. Section 40.11 establishes the initial marker and board invariant at the actual fixed start. Assume it at one layer. Every selected instruction index is in range by Section 40.7; each local operation has Proposition 40.4.1's history; Lemma 40.10.1 supplies the whole controller sequence and a next layer. Each layer is a finite number of finite instructions and finite cables, so its endpoint is reached in finite time. Induction produces arbitrarily many layers. The rule itself never lacks a transition.

It remains to bound all-time reuse rather than merely to validate endpoints. Nonalias resources have disjoint owned supports by the finite neighbor atlas, selector proof, exceptional-marker bounds,

and long-channel inequalities. Each such resource is scheduled once. The only temporal support overlaps are complete storage-box aliases across stages. Projecting the global schedule onto one such box gives exactly its optional-WRITE-before-READ legal history. Every crossing receives FIRST before SECOND and every union receives one alternative, as checked within the controller's local schedule. Thus each physical cell belongs either to one two-bounded complete resource history or to one merged two-bounded storage history. No separate macro maxima are summed at aliases.

Any purported third departure has a finite time and therefore belongs to a finite number of completed layers plus one finite prefix. The finite-prefix induction just proved excludes it. Hence all infinite-history departure counts are at most two. Every arrival, including the starting site, is followed by departure in this infinite run, so the same bound holds for visits. A frozen simulated J1 still undergoes new layer computation on fresh hardware; no later post-halt reuse is omitted from the argument. $\square$

40.11 The exact finite loader and actual startup

The input consists of finite binary lists ℓ, r , each with its first element nearest the machine head. Form the primary state word

$$(a_0, \dots, a_{m-1}) = (\text{reverse}(\ell), H(A, 0), r), \quad m = |\ell| + |r| + 1 \geq 1.$$

The head code $H(A, 0) = 2$ has exactly one set bit. For each set bit b of state a_i , set two source fields: block i , selected slot $13 + b$, and block $i + 1$, selected slot b . A slot k in block i has source-box origin

$$(iS + 600 + 24000k + 100, 50).$$

A logical one sets its two state cells $(x + 4, 54)$ and $(x + 5, 54)$ to colour zero; their background colour is one. State zero changes no state field. Set the left marker at block 0, slot 12, and the right marker at block m , slot 11.

40.11.1 The fixed 2806-cell anchor

The startup patch is the exact net board change left by a certified virtual COPY1-to-header prefix, translated by $(288600, -400)$. The virtual prefix is a way of deriving a finite initial state, not an actual uncounted physical run. Its unprocessed endpoint is

$$(288650, 75, E),$$

which is the real fixed starting head. The ordinary unused slot entry $(288600, 75, E)$ is not the start. The two virtual INIT1 setup flips cancel in the virtual read and are not applied a second time.

All 6193 cells in the translated marker's baseline agree with the global colour query, and all 2806 expected old colours in the net patch agree. The real sparse patch plus the global background reproduces the exact certified 1276-departure suffix to $(289200, 75, E)$, with maximum two departures at a cell. More generally the real startup is a suffix of a legal complete history, so deleting its earlier virtual prefix cannot increase a physical count. The anchor lies in

$$[288617, 289200] \times [-144, 76].$$

It meets raw input fields only at the left marker's two state cells, already set to the same zero colour.

Proposition 40.11.1 (Exact support and initial invariant). *The loader supplies a finite perturbation of the fixed board, establishes the initial valid layer invariant at the actual start, and changes exactly*

$$|I(\ell, r)| = 2812 + 4(\text{popcount } \ell + \text{popcount } r) \quad (40.8)$$

cells. Its support is contained in

$$[288617, 576000m + 264705] \times [-144, 76].$$

Proof. Every primary set bit is copied into two different selected source fields, each changing two cells. Different state copies lie in disjoint fields. The head contributes one set bit. The right marker is at block $m \geq 1$, disjoint from the anchor and state fields; it changes two cells. The left marker is already part of the anchor and contributes no additional support. Therefore

$$2806 + 4(\text{popcount } \ell + \text{popcount } r + 1) + 2 = 2812 + 4(\text{popcount } \ell + \text{popcount } r).$$

Every assigned colour differs from the background, as checked by the literal query; no redundant override is counted. The anchor's sole overlapping marker assignment agrees in colour. Its suffix places the controller at the active left header, while the duplicated source bits and two markers give exactly the initial state word and boundaries. The source-field formula and the anchor bounds give the displayed support rectangle. $\square$

Zero-only words retain their lengths. For example (ϵ, ϵ) has $m = 1$, while $(000, 00)$ has $m = 6$. Both change exactly 2812 cells, but the right marker is in a different block. Removing zero prefixes or suffixes as if the loader depended only on popcount would silently alter its declared finite input representation.

The public loader accepts the two words, not caller-chosen hardware, anchor, or program. A concrete output contains the fixed start, fixed periods, accepting clauses, and sorted changed-cell list. Its defensive-return checks ensure that a caller's mutation of returned metadata does not alter future compilations; such API isolation is separate from the physical simulation theorem.

40.12 An exclusive accepting port with the correct polarity

The accepting instruction is zero-based program row

$$k_* = 601515562,$$

and is a DUP on active column 910. At its entry, that input is the *uncomplemented* bit $h = [f = 21]$. The associated first output box's WRITE entry is the physical observation cell. An incoming heading S there means that this DUP is conditionally writing one.

40.12.1 Why the bit at that row is really h

The DAG halt wire is 885, the output of zero-based gate 861, namely NAND(884, 884). After all 862 gates and 24 output copies, the final copy-to-end has width 910 and source 885. Here $r = 25$, so (40.6) gives

$$12 \cdot 25^2 + 27 \cdot 25 - 38 = 8137$$

physical rows, transporting h to column 910. The independent polarity checker eagerly expands these 8137 rows, compares each with the random-access decoder, verifies that no unused workspace bit is read, and checks the resulting transported value on all 16777216 Boolean cell inputs. No footer NOT lies between this copy and the observer DUP. Thus “the DAG computes a halt bit” is strengthened to the exact polarity at the actual physical observation instruction.

Lemma 40.12.1 (Local observer exclusivity). *In every legal history of the observer DUP, its first output WRITE-entry cell is visited with heading S exactly once when $h = 1$, and never when $h = 0$. The eastward pass, later output READ, and next-row input alias cannot produce this head state.*

Proof. The DUP input read branches on the selected bit. For zero it omits both output writes, leaving fresh zero boxes; for one it follows the certified write routes. The first box’s WRITE entry is a specified port, so the one branch visits it in heading S exactly once. Enumeration of all prior input and optional output-read histories verifies that the eastward pass and read route do not enter that cell with the claimed heading. The next row can only share the same whole box as an input; its chronological projection is the already included subsequent READ, not a second WRITE. Therefore it cannot supply a false or duplicate observation. $\square$

40.12.2 Why no other part of the plane can imitate the port

The observation residues in (40.1) are the two translations of that resource under the staggered layers. Apart from the just-accounted next-row shared box, all other body rows have a different vertical phase modulo $2V$. Header, footer, marker turn, and partial-header paths also have different phases. The only normal route rising through old instruction heights is the far-right channel, whose x interval is disjoint from the observer column. The input state changes lie at $y = 54$, the anchor is in $[-144, 76]$, and the actual start has $y = 75$, so initialization cannot satisfy either residue clause.

A partially entered C still executes the full program; its skipped prefix is ignored through $s_1 = 1$. Thus no uncomputed workspace can lead to the observation row. A sound event does not depend on the later completion of the DUP or footer: h was already computed and exactly transported before its WRITE-entry visit.

Theorem 40.12.2 (Acceptance equivalence and simplification). *The event (40.1) with incoming heading S occurs if and only if the source U15 configuration reaches J1. For every valid primary-loaded input, it is equivalent to the single clause*

$$h_t = S, \quad x_t \equiv 258702 \pmod{576000}, \quad y_t \equiv 481225262850 \pmod{481238074400}. \quad (40.9)$$

Proof. The geometric exclusion and Lemma 40.12.1 show that an observed state occurs exactly at a computed $h = 1$. The polarity argument and Theorem 40.6.1 identify this with a computed CA output 21. The layer invariant and (40.3) identify some such output with the valid source machine reaching J1. Conversely, on reaching J1 the corresponding primary CA step computes 21 in the finite active interval, and its complete program reaches the observer’s one branch in finite time.

The first clause is the even physical computation layer’s observer. Such a layer computes an odd CA output from primary inputs, hence only pair codes or zero. None is 21. The second clause detects primary outputs and is the only reachable accepting clause on the loader’s valid runs. Retaining both clauses is a uniform two-phase description with no false event; deleting the unreachable first clause preserves the equivalence. $\square$

This proves the observation portion of Theorem 40.1.1. Neither clause denotes a single absolute lattice site. The ant may accept in any translated occurrence of this fixed residue port, and still continues forever.

40.13 Exact dimensions and conservative physical costs

The global board is specified by finite templates and 958 arithmetic program records. A rectangular period contains

$$576000 \cdot 481238074400 = 277193130854400000$$

lattice positions. No claim is made to enumerate this dense bitmap. The random-access query uses boundedly many local resource candidates, a binary search among the finite program records, and the finite SWAP/COPY arithmetic grammar.

For T computed CA rounds, the successive numbers of C evaluations are $m+1, m+2, \dots, m+T$. Therefore

$$N_C(T, m) = T(m+1) + \frac{T(T-1)}{2}. \quad (40.10)$$

Every finite resource has at most 13197 owned cells and at most two aggregate departures per cell. Set $M = 26394$. Including the explicit long route, a deliberately loose control bound per C is

$$\begin{aligned} B_C &= (R+1)(K+4)M + (K+W+8)M + 2F + 2396 \\ &= 15274393040715416. \end{aligned} \quad (40.11)$$

This intentionally overcounts several shared and partial resource histories. It is a valid upper bound, not an optimized timing claim. Consequently, if U15 reaches J1 after $t \geq 1$ steps, an accepting event is reached within

$$[2t(m+1) + t(2t-1)]B_C \quad (40.12)$$

ant departures. Acceptance can precede completion of its last C.

Through T complete CA rounds, safe coordinate bounds are

$$|x| \leq 576000(m+T+2), \quad -144 \leq y \leq 240619037200(T+1) + 659.$$

Their signed encoding needs $O(\log(m+T+2) + 39)$ bits. Input-coordinate magnitudes grow linearly in m even though support cardinality depends on popcount. These distinctions matter when an external arithmetic representation encodes the initial board as an integer.

40.14 Why one visit cannot realize the same acceptance task

Report 38 [4], included unchanged in the archive, proves the total first-revisit algorithm and exact pattern-query implementation. We give the decisive arguments here and spell out why its input conventions cover this report.

[write] *A re-proof, in summary, of Part I.* Sections 40.14.1–40.14.2 restate and sketch the proofs of Part I's Lemma 3.1 (static shadow), Lemma 3.2 (projected permutation), Proposition 4.1 (the rank-2/1/0 intersection primitive), Proposition 5.2 and Theorem 2.2 (the total first-revisit algorithm) and Theorem 7.1 (exact queries). They are printed here as Report 40 wrote them, as a second, abbreviated route; Part I has the complete proofs and the bit-complexity bounds. Two pieces are new in Report 40: the elementary generalized-CRT decision of heading-and-congruence clauses on a lane (the special case that the acceptance

port needs, without Part I's signed indicator calculus), and the model-match argument of Section 40.14.3 (the reflection between Part I's north-positive and this Part's south-positive y , and the observation that a fixed succinctly specified tile is a constant).

40.14.1 Fresh walks have a finite affine description

Before the first repeated arrival, every departure is from an unmodified cell. The moving-and-flipping ant therefore follows the static walk that always reads the initial periodic board plus finite defects. The two paths agree through the first repeated arrival, inclusive. The next departure is outside this assertion.

On a periodic tile of width u and height v , the static background transition on $(x \bmod u, y \bmod v, h)$ is a permutation of $4uv$ states: given the successor and its heading, subtract the last unit move to recover the predecessor residue; its colour identifies and reverses the turn. Every projected orbit is thus a pure cycle. A defect-free prospective excursion has a finite description

$$p_{r+i+nL} = a_i + n\delta, \quad h_{r+i+nL} = g_i, \quad 0 \leq i < L, \quad n \geq 0, \quad (40.13)$$

with $L \leq 4uv$ and $\delta \in u\mathbb{Z} \times v\mathbb{Z}$. Each phase is an affine lane; a finite prefix is represented by bounding n , without expansion.

To intersect a new lane and an old lane, solve

$$a + nd = b + ke, \quad n, k \geq 0,$$

with their finite bounds when present and strict chronology

$$t_B + k\ell_B < t_A + n\ell_A.$$

The two-by-two spatial system has rank two, one, or zero. In rank two, Cramer's rule and divisibility yield at most one candidate. In rank one, extended Euclid yields all integer solutions

$$n = n_0 + sz, \quad k = k_0 + rz;$$

all domain and chronological inequalities intersect to one integer interval in z , and a suitable endpoint minimizes n . In rank zero, equal spatial bases permit the earliest old index $k = 0$, imposing only a threshold on n ; unequal bases are disjoint. These exhaust all cases, including negative drifts, zero components, and different headings at the repeated site.

At a restart, compute the prospective background excursion, its earliest collision with itself or history, and its earliest encounter with an initial defect. If a collision is no later than the defect, return it. If neither exists, certify a fresh infinite tail. Otherwise the defect arrival is fresh; append the excursion through it, perform its one actual defect departure, and restart. Every processed defect is a new site, so there are at most K such departures and $K + 1$ excursions when K defects are listed. There are at most $(K + 1)4uv$ lanes. Report 38 supplies the polynomial bit bounds on this arithmetic, not merely a bound on the number of large-integer operations. The algorithm requires no one-visit promise and recognizes global freshness itself.

40.14.2 Fixed residue-and-heading events are decidable

On a certified globally fresh run, each time belongs to one of finitely many lanes. A heading condition is constant on a lane. Each coordinate congruence becomes

$$an \equiv b \pmod{q},$$

which is either empty or a single residue class after dividing by $\gcd(a, q)$. Intersect the finitely many such classes by generalized CRT and with the lane's integer interval. Check whether any admissible n remains and take the smallest if desired. A finite union of clauses is handled by taking the minimum over lanes and clauses. This elementary special case already decides (40.9) and both clauses of (40.1).

The companion proves a stronger exact query theorem including finite head-relative colour stencils. Before a first repeated departure, the colour at a site z is its initial colour plus the indicator of an earlier visit, modulo the palette size. Earlier visits are lane relations with strict chronology. Their projection onto the new lane index is an interval congruence atom. Boolean combinations are represented by finite signed sums of atom indicators, using

$$A \wedge B = AB, \quad \neg A = 1 - A, \quad A \vee B = A + B - AB.$$

Generalized CRT gives atom intersections and floor division gives exact counts on an interval. Every resulting Boolean indicator is eventually periodic; a finite exceptional prefix plus one eventual period gives an exact emptiness bracket and binary search gives its minimum. This is a terminating implemented procedure, but distributive growth prevents any general polynomial observation bound from being claimed. The polynomial theorem is specifically for first revisit detection.

40.14.3 The model match is literal

Report 38 permits any explicitly listed finite periodic tile, a finite defect map, the same turn/flip/move rule, and pre-departure queries. Its picture uses positive y north; reflection of coordinate notation identifies its north/south vectors with the present south-positive coordinates without changing the physical RL rule. The input override colours and fixed residue classes transform correspondingly. Fixing the rule, tile, start, and query only restricts its decidable input family.

The upper board is given by a fixed finite colour-query program rather than a supplied dense tile. That does not evade the lower result: its periods and query are constants, and its total colour query computes every one of the finitely many tile cells. The enormous tile can be enumerated once as a mathematical constant. No efficient uniform theorem for arbitrary succinctly specified tiles is inferred. The fixed-background decidability argument needs only effective access to this one fixed finite tile, and is unaffected by its impractical size.

On the one-visit side the certified domain is the whole run. The accepting predicate uses only heading and fixed coordinate congruences, a subset of the companion's query language, with no halt signal, hidden unbounded property, or absolute-site-return interpretation. The upper loader produces a globally two-bounded trajectory on every input, not just on halting ones. These are exactly the hypotheses needed for Corollary 40.1.2.

Proof of Corollary 40.1.2. Suppose an undecidable language reduced computably to the globally one-visit class with this fixed kind of observation. Apply the total first-revisit procedure and then the exact all-time query procedure. The reduction's promised one-visit outputs have no revisit, so the observation answers the language, a contradiction. Conversely Theorem 40.1.1, composed with the published U15 encoding, gives the required halting-equivalent two-visit family. Its predicate may be the single clause (40.9). Every premise concerns the same physical rule, periodic-plus-finite initialization, and pre-departure observation model. $\square$

Report 38 correctly declined to assert a sharp boundary while its literal upper interface was

absent. This report adds that interface and the model comparison; it does not rewrite the earlier document's historical scope statement.

[write] Report 38's historical scope statement is kept, with a dated note, in Part I (Section 1).

40.15 Arithmetic alignment and what remains unpaid

A north-facing coordinate convention used by an existing bounded-board history encoding is obtained by the orientation-preserving rotation and translation

$$X = y - 75, \quad Y = 288650 - x, \quad E \mapsto N.$$

The actual start becomes $(0, 0, N)$ with even checkerboard parity. The periods become $(481238074400, 576000)$ and the accepting residues become

$$(240606225575, 317948, E), \quad (481225262775, 29948, E),$$

each on odd parity. Period-multiple translations preserve these residues and place any finite run prefix inside a sufficiently large odd-width, even-height board.

This alignment is geometric. It does not supply an arithmetic initial word for free. The existing 174-operation complete bounded-board history relation takes an encoded initial board as an external parameter. Composing it with raw U15 words would have to pay for raw-input dilation, periodic phase, finite anchor and background, a sufficient board, and the actual accepting observable. Appendix 40.D supplies a separately paid residue-and-heading endpoint selector, conditional on the existing history ports and theorem. Its exact 42-operation fixed-numeral and 101-operation literal-1/3 ledgers cover that endpoint only. They do not pay the initialization or yield a combined universal count. No 174-operation universal construction, improved universal operation bound, or physical ant halt is claimed.

40.16 Reproducibility and release assurance

The archive contains the literal source packet, the separately paid endpoint packet, all finite numerical maps and complete local histories, the 32-state and 11-bit tables, the 862-gate DAG, the exact compressed physical program, initialization and observer receipts, and the full unchanged Report 38 companion. Third-party complete articles, PDFs, source archives, and article screenshots are excluded. Their official links and source hashes are recorded instead.

[write] *In this repository.* The archive is shipped flattened with the prefix `02-literal-ant-: evidence/X/Y` became `02-literal-ant-X-Y` (the segment `evidence/` dropped), Markdown at the report root, Python in `code/`, everything else in `data/`. The Report 38 companion is Part I and is not shipped twice; the 24 largest regenerable files of `literal-interface/` (156 MB in all, including `ca/radius_half_11_table_le.bin`, `copy/pair_dup.json`, `copy/pair_move_left.json` and `copy/pair_move_right.json`) are not shipped and are rebuilt byte for byte by the shipped generators; the report README lists them and gives the commands. The delivered verifiers check POSIX file modes and the delivered layout, so they do not run on the shipped tree; the README gives a rerun route on a copy. At placement the 28-command literal-interface driver regenerated its 39 outputs byte for byte and the endpoint driver its 10, on a copy.

40.16.1 Authentication before execution

The enclosing release inventories every relative path, byte size, mode, and SHA256, and pins its manifest. It rejects unknown files, missing files, symlinks, special files, duplicate JSON keys, nonfinite numeric values, and noncanonical paths. It also pins the active source inventory and replay plan. A normalized self-pin convention avoids a recursive hash cycle: only the verifier's designated manifest-pin line is replaced by 64 zeroes when that verifier is hashed; all other bytes are exact. The independently supplied final archive digest is the external trust anchor. A self-reported PASS or a newly regenerated manifest is not authentication of a received package.

Scientific scripts execute only after this identity gate. The source driver writes `atlas/fresh_replay.json` under its own root, even though it performs its scientific regeneration in a second temporary tree. Therefore the enclosing replay invokes it only from an authenticated *mutable copy* in an external temporary directory. It never runs that driver in the frozen evidence or original source directory. The source copy's bytes can match again while its receipt's modification time has changed; the release does not claim that mutable worktree metadata stays unchanged. The frozen enclosing release and original source bytes, modes, and modification times are checked separately for preservation.

The driver runs 28 own-code commands and compares 39 regenerated files byte-for-byte with its source data. Its complete receipt is compared with the authenticated expected receipt using exact JSON types and keys; in particular a Boolean cannot masquerade as integer zero or one. No network, third-party source program, dense period bitmap, or complete 601-million-row expansion is needed. The companion has its own pinned active-program allowlist and normal/optimized replay contract.

40.16.2 Optimization is deliberately not a scientific mode

The literal scientific checks use assertions. The 27 explicitly guarded entry points, including the driver, reject optimized Python as required by their contract. Two standalone commands, `copy/translate_anchor.py` and `copy/program_audit/verify_program_index_bounds.py`, lack an explicit optimization guard and are marked unsupported under `-O`; they are not claimed to pass optimized scientific execution. Their supported execution is through the unoptimized guarded driver. The enclosing identity and strict-receipt tools use explicit exceptions and can be tested under optimization, but that is not a claim that optimized source science passes. The different Report 38 code has explicit nonremovable checks and does support its normal/optimized comparison.

40.16.3 Deterministic artifacts and hostile inputs

PDF rebuilding occurs in a new external directory with fixed source date, fixed metadata conventions, and shell escape disabled. Compiler auxiliary files and visual-review images stay outside the sealed release. ZIP construction fixes member order, timestamps, regular-file modes, and compression settings. A second build is compared byte-for-byte before delivery.

Archive checking validates the independently supplied SHA256, exact member inventory and sizes, canonical paths, regular-file modes, permitted compression, and all payload bytes *before* any extraction. It rejects traversal, absolute paths, backslashes, duplicate names, symlinks, special files, encrypted members, and additions. Extraction occurs only in an external temporary directory and is followed by the same identity gate. Regression cases exercise these boundaries as well as modified

sources, metadata, receipts, and output destinations. External QA records the actually executed checks and final hashes; the article does not silently substitute a proposed test plan for passing results.

40.17 Conclusions

The upper construction is now a fixed colour generator and finite perturbation compiler with a fixed actual start and a proved exclusive accepting port. Its small-state logical computation, enormous but finite physical instruction grammar, and infinite growing trajectory are linked by separate exact arguments. The history language, shared ownership, partial headers, marker turns, and long return channel are necessary parts of that link.

The one-visit lower theorem and two-visit upper construction match in their rule, initialization class, global visit quantifier, and finite pre-departure observable. This supplies the stated sharp boundary without changing physical halting into a head-port event by convention. The qualitative universality and two-visit principle retain their published credit, and the arbitrary-program U15 compiler retains its published dependency.

Natural next steps are a proof-assistant formalization of the finite-history composition and arithmetic selector proofs, a materially smaller literal layout, and a complete raw-input arithmetic composition with all initialization costs included. None is required for the present literal U15-to-ant equivalence, and none is claimed as already supplied by it.

40.A Complete primitive rows

Each character denotes one grid location in increasing x order, with successive rows increasing y . Zero and one are owned colours. A dot is unowned. The port coordinates in Section 40.3 use the top-left location as $(0, 0)$. These are complete numerical maps, not schematic circuit symbols.

cable_a	cruceA
000000000	0....0
111111111	10..01
	.1001.
cable_b	..10..
0000..	.0110.
11110.	01..10
....10	1....1
....10	
....10	cruceB
....10	0.....
	10...1
cable_c	.10.10
.....01	..100.
.....01	..110.
.....01	.10.10
.....01	10...1
000001.	
11111..	union
	...01....
box	...01....
..10..01.	...011111
...1001..	000010000
...1001..	111111...
...1001..	
...111...	
111001...	
000.01000	
...011111	
..01.....	
..01.....	

40.B A guide to the exact archived interfaces

Paths in this section are relative to the active literal source packet. The enclosing README identifies that packet and the preserved predecessor lineage.

[write] In this repository, X/Y of this table is shipped as `data/02-literal-ant-literal-interface-X-Y` (`.json`, `.bin`), `code/02-literal-ant-literal-interface-X-Y` (`.py`) or `02-literal-ant-literal-interface-X-Y` (`.md`); for example `ca/physical_program.json` is `data/02-literal-ant-literal-interface-ca-physical_program.json`. Not shipped: `ca/u15_table.json` (a byte copy of a repository file, see the note in Section 40.5) and `ca/radius_half_11_table_le.bin` (regenerable, see the README).

File or directory	Scientific role
<code>common/primitive_maps.json</code>	Complete seven primitive maps
<code>common/primitive_catalog.json</code>	Source asset hashes, ports, legal histories, intermediate boards, halos, and rejected histories
<code>nand/, not/, copy/</code>	Literal composite maps, routes, shared ownership, input/write/read cases, markers, partial headers, anchor
<code>copy/program_audit/</code>	Independent exact grammar, row intervals, column bounds, and arithmetic analysis
<code>ca/u15_table.json</code>	Exact 30-entry fixed machine table
<code>ca/fixed_ca_cell.json</code>	All 862 NAND gate rows, wire references, outputs, and halt wire
<code>ca/radius_one_32_table.bin</code>	32768 primary triple outputs
<code>ca/radius_half_11_table_le.bin</code>	4194304 two-input outputs, little-endian 16-bit entries
<code>ca/physical_program.json</code>	All 958 grammar records and exact prefix array
<code>ca/physical_program.py</code>	Exact row decoder, with no full row expansion
<code>atlas/generator.py</code>	Total fixed periodic colour query
<code>atlas/compile_input.py</code>	Public nearest-head-first finite input loader
<code>atlas/</code>	Complete halt-wire transport and physical polarity check
<code>observer_polarity_receipt.json</code>	Support, dimension, coordinate, and departure bounds
<code>atlas/quantitative_ledger.json</code>	Global construction proof with source scope retained
<code>atlas/PROOF.md</code>	Independent finite and symbolic global geometry review
<code>copy/GLOBAL_GENERATOR_REVIEW.md</code>	Scientific regeneration driver, only to be executed in an isolated mutable copy
<code>replay_all.py</code>	

The delivered Report 38 companion includes its complete proof, active implementation, tests, independent review, and bit-complexity analysis. Its source and article are preserved without edits. Its older statement that a literal upper interface remained open describes that report's then-current scope.

[write] That companion is Part I of this report.

40.C Failure modes explicitly excluded

1. **A true Boolean table without a physical route.** The full-round and global controller proofs supply the missing trajectory and ports
2. **A crossing read in the wrong history.** FIRST-before-SECOND is part of the resource language, and marker branches avoid missing FIRST events
3. **Four visits hidden at an alias.** Complete storage boxes have one merged optional-

WRITE/READ history, never a sum of separate maxima

4. **Uninitialized skipped inputs.** The left-edge mask ignores the skipped state, and W-only plus header-E-only histories handle the untouched physical resources
5. **A malformed CA transition called valid simulation.** The total extension is explicit; unique-head preservation is a separate theorem
6. **A complemented accepting bit.** The exact wire 885 and 8137-row transport are checked at all Boolean inputs before the observer DUP
7. **A false later visit to the accepting cell.** Optional output reads and the next-row alias are included in local exclusivity, while global phase and margin arguments exclude other routes
8. **An anchor run charged to neither past nor future.** The anchor is an explicit 2806-cell net initial patch; the virtual prefix never physically runs, and its setup flips are not duplicated
9. **Zero-only inputs silently shortened.** The right marker's position depends on the actual word lengths even when popcount is zero
10. **An enormous sampled circuit described as exhaustive.** The row decoder has an all-interval arithmetic proof; only the 8137-row observer transfer and bounded local cases are eagerly expanded
11. **A full replay that changes the frozen packet.** The scientific driver is staged in a mutable external copy because its root receipt write changes metadata
12. **Free arithmetic preprocessing.** The raw input, initial board, phase, anchor, bounds, and accepting endpoint are not included in an unproved 174-operation composition

40.D A separately paid arithmetic endpoint

This appendix attaches the single reachable accepting clause to an existing bounded-board history relation. It is a separate endpoint component, not a construction of the arithmetic initial board or a universal combined operation bound. The complete own-code endpoint source, predecessor forms, independent audits, six arithmetic DAGs, and exact ledgers are included in `evidence/endpoint/`.

40.D.1 Inherited interface and counting convention

The parent history relation supplies existing integer ports W , $H = \text{FinalHead}$, and $Z = \text{FinalSignPlus}$. Its theorem guarantees integers w, h, j satisfying

$$w \geq 3 \text{ odd}, \quad h \geq 2 \text{ even}, \quad W = 3^w, \quad Q = W^h, \quad H = 3^j < Q, \quad 0 \leq j < wh.$$

The canonical board coordinates are $x = j \bmod w$ and $y = \lfloor j/w \rfloor$, with east/south-positive coordinates. The final snapshot is after a certified positive number of steps and before the next turn/flip/move. There is no edge wrapping or reflection. The inherited sign is $(Z - 1)/H \in \{0, 1\}$: on a white site $x + y$ odd, zero means east; on a black site it means north. Checkerboard parity is unrelated to the mutable binary ant colour.

The pinned bounded-board component has five positive parameter ports, 61 positive unknowns, and 48 equations at a recorded cost of 174 operations. Those counts and its power/head geometry

theorem are inherited assumptions here; this appendix neither re-proves them nor executes upstream source. W is reused from an existing witness, and H, Z are existing parameter ports. A multiplication costs M , an addition or subtraction costs A , and fixed literals and aliases cost no operation. Asserting equality of already built expressions adds no arithmetic gate. Arithmetic gate outputs are expressions, not separately quantified witnesses, and intermediates may be zero or negative unless positivity is explicitly required.

For the normalized single port set

$$u = 481238074400, \quad x_0 = 481225262775, \quad v = 576000, \quad y_0 = 29948.$$

Both periods are even, x_0 is odd, and y_0 is even. Use prescribed fixed integers

$$K = 3^{x_0}, \quad C = 3^u - 1, \quad D = C - 1,$$

and computed variable powers $A = W^v$, $T = W^{y_0}$. These constant recipes specify exact integers without expanding their enormous digits. In the free-fixed-numeral model K, C, D are fixed coefficients, never existentially chosen values. In the stricter source they are computed from literal 3 and literal 1 with all operations charged.

40.D.2 Three witnesses and three equations

Supply exactly three positive witnesses X_+, Y_+, B . Compute

$$U = CX_+ - D = 1 + C(X_+ - 1), \quad V_* = 1 + (A - 1)(Y_+ - 1), \quad P = KU.$$

Here V_* is an arithmetic factor, distinct from the physical block height V elsewhere in this report. Assert

$$P + B = W, \quad (PT)V_* = H, \quad Z = 1. \quad (40.14)$$

The positive adapters permit zero underlying quotients: $X_+ = 1$ gives $U = 1$, and $Y_+ = 1$ gives $V_* = 1$.

Theorem 40.D.1 (Endpoint equivalence). *Under the inherited history assumptions, positive witnesses for (40.14) exist if and only if the canonical final coordinate satisfies*

$$x \equiv x_0 \pmod{u}, \quad y \equiv y_0 \pmod{v},$$

and its incoming heading is east. No divisibility relation between the board width and either period is assumed.

Proof. Because $X_+, Y_+ \geq 1$, both $U, V_* \geq 1$. The head identity

$$3^j = KTV_*U$$

forces each positive integer factor U, V_* to be a power of 3. Write $U = 3^a$, $V_* = 3^b$ with $a, b \geq 0$. Their defining equations imply

$$3^u - 1 \mid 3^a - 1, \quad 3^{uv} - 1 \mid 3^b - 1.$$

For integers $d > 0, k \geq 0$,

$$3^d - 1 \mid 3^k - 1 \iff d \mid k. \quad (40.15)$$

For the forward direction write $k = qd + r$, $0 \leq r < d$. Reduction modulo $3^d - 1$ leaves $3^r - 1$, which lies between zero and $3^d - 2$, and vanishes only at $r = 0$. The reverse direction is the geometric-sum factorization, including $k = 0$.

Hence $a = um$ and $b = wvn$ for $m, n \geq 0$. Put $x^* = x_0 + um$ and $y^* = y_0 + vn$. Equality of powers in the head equation gives $j = x^* + wy^*$. The strictly positive column slack gives $P < W$, hence $3^{x^*} < 3^w$ and $0 \leq x^* < w$. Thus this is the unique canonical row-major decomposition: $x = x^*$ and $y = y^*$. The inherited $j < wh$ then yields $y < h$, so no extra row slack is needed. Since x^* is odd and y^* even, the site is white; $Z = 1$ selects the inherited zero sign, therefore east.

Conversely suppose the stated residues and east heading hold. Canonical nonnegative coordinates give $x = x_0 + um$, $y = y_0 + vn$ with $m, n \geq 0$. Choose

$$X_+ = 1 + \frac{3^{um} - 1}{3^u - 1}, \quad Y_+ = 1 + \frac{W^{vn} - 1}{W^v - 1}, \quad B = W - 3^x.$$

The geometric-sum identities make the quotients nonnegative integers, including zero. Since $x < w$, $B > 0$. These witnesses make $U = 3^{um}$ and $V_* = W^{vn}$, so all of (40.14) follow from $H = 3^{x+wy}$ and the white/east sign convention. The witnesses are uniquely determined by the inherited history and endpoint. $\square$

Prime factorization in this proof is a mathematical inference, not an uncharged factorization instruction. The column slack is essential: with smaller illustrative parameters $u = v = 2$, $x_0 = 1$, $y_0 = 0$, $w = 3$, and underlying quotients $X_+ - 1 = 10$, $Y_+ - 1 = 0$, the unconstrained exponent is $j = 5$. Its canonical coordinate is $(2, 1)$, outside the intended residues, because the factor representing the column carried into the row. Positive B rules this out.

40.D.3 Fully explicit shared power chains

The joint variable-power chain starts at $P_0 = W$ and squares to P_{19} , costing 19M with $P_i = W^{2^i}$. The binary support of y_0 is

$$\{2, 3, 4, 5, 6, 7, 10, 12, 13, 14\},$$

whose product costs 9M; the support of v is $\{9, 11, 14, 15, 19\}$, costing 4M. Thus both powers T, A cost exactly 32M.

For fixed powers, square literal 3 through level 38, costing 38M. The binary supports of x_0, u are

$$\begin{aligned} \text{bits}(x_0) &= \{0, 1, 2, 4, 5, 7, 9, 10, 14, 15, 17, 22, 24, 25, 27, 36, 37, 38\}, \\ \text{bits}(u) &= \{5, 10, 14, 17, 18, 26, 27, 36, 37, 38\}. \end{aligned}$$

The common eight positions are $\{5, 10, 14, 17, 27, 36, 37, 38\}$. Build their product in 7M and reuse it: multiply by the ten x_0 -only powers in 10M for K , and by the two u -only powers in 2M for 3^u . Both fixed powers cost $38 + 7 + 10 + 2 = 57$ M. Computing $C = 3^u - 1$ and $D = C - 1$ adds 2A. No giant power is materialized during verification; integer exponent propagation proves the exact targets at every multiplication gate.

The folded non-power source has exactly the following ten gates, with $H_y = Y_+ - 1$:

Expression	Cost	Expression	Cost
$H_y = Y_+ - 1$	A	$V_* = 1 + J$	A
$E = CX_+$	M	$P = KU$	M
$U = E - D$	A	$P + B$	A
$A_- = A - 1$	A	PT	M
$J = A_- H_y$	M	$(PT)V_*$	M

Therefore the exact displayed-source ledgers are

$$\begin{aligned} \text{Prescribed } K, C, D : \quad & 32M + (5M + 5A) = 37M + 5A = 42, \\ \text{Only literals } 1, 3 : \quad & (57M + 2A) + 32M + (5M + 5A) = 94M + 7A = 101. \end{aligned} \tag{40.16}$$

These are construction costs, not optimality statements. The historical unfurled fixed-numeral source uses 43 operations. Folding the horizontal expression to $CX_+ - D$ saves one A in the fixed-numeral model; constructing the additional prescribed constant D consumes that saving in the strict model, which remains 101.

40.D.4 Five witnesses with a smaller endpoint degree

Alternatively supply positive U, V_*, U_q, V_q, B and assert

$$\begin{aligned} U + D &= CU_q, \\ (V_* - 1) + (A - 1) &= (A - 1)V_q, \\ KU + B &= W, \\ ((KU)T)V_* &= H, \\ Z &= 1. \end{aligned} \tag{40.17}$$

The first two equations force $U = 1 + C(U_q - 1)$ and $V_* = 1 + (A - 1)(V_q - 1)$. Thus the correspondence $U_q = X_+, V_q = Y_+$, with these forced values of U, V_* and unchanged B , is a bijection with the three-witness solutions. In particular it includes both zero-quotient boundaries and introduces no spurious accepting endpoints.

The non-power gates are $U + D$ (A), CU_q (M), $A - 1$ (A), $V_* - 1$ (A), their sum (A), $(A - 1)V_q$ (M), KU (M), $KU + B$ (A), $(KU)T$ (M), and its product with V_* (M). This again is $5M + 5A$, so both ledgers in (40.16) remain unchanged. Writing the second equality in the displayed form uses only literal 1, reuses $A - 1$, and does not hide a free literal 2.

After expanding the gate expressions, treat the fixed numeral recipes as coefficients and the inherited ports and supplied witnesses as variables. The three endpoint residuals have exact total degrees

$$(1, 605950, 1).$$

The head residual's highest-degree monomial is $W^{v+y_0} X_+ Y_+$ with nonzero coefficient $KC > 0$, so its degree is exactly $v + y_0 + 2$, not merely an upper bound. For the five-witness form the degrees are

$$(1, 576001, 1, 29950, 1).$$

The vertical residual has the nonzero term $-W^v V_q$, degree $v + 1$; the head residual has $KUW^{y_0} V_*$, degree $y_0 + 2$. These are endpoint-only expanded degrees before substituting any parent expression or combining equations into one polynomial.

40.D.5 Verification and remaining obligations

The endpoint verifier checks every source input, dependency, gate opcode, live gate, shared exponent, equation, and fixed-numeral recipe. An independent whole-DAG sparse polynomial expansion uses a formal symbol for base 3 and compares the complete residual polynomials with the stated equations. It verifies nonzero leading coefficients after specializing that symbol to 3, avoiding a false exact-degree claim from an untested symbolic cancellation. Four own commands regenerate six DAGs and four receipts exactly. Supporting regressions cover 17280 small candidate endpoints, horizontal-fold positivity, and 192 witness-correspondence cases. The arbitrary-integer theorem is the proof above, not extrapolation from those finite examples.

This attachment applies to the normalized literal port proved in Section 40.12, and only because the first physical clause is unreachable on valid primary loads. It recognizes an endpoint of an already valid bounded-board history. Constructing the correctly translated periodic board, choosing and encoding its finite rectangle, preparing the initial integer from raw input, and paying any dilation or final single-polynomial combination remain separate tasks. Thus neither $174 + 42$ nor $174 + 101$ is asserted here as a complete universal operation count.

[write] Report 44 attaches this selector in its folded three-witness form (Section 44.4). In Report 48's source the programme later replaced the selector's own power W^{576000} by the initializer's already paid one, deleting nine multiplications (`periodic_ant_endpoint_reuse48.md` [W3]).

References

- [1] Diego Maldonado, Anahí Gajardo, Benjamin Hellouin de Menibus, and Andrés Moreira. *Non-trivial Turmites are Turing-universal*. arXiv:1702.05547v1, 18 February 2017. Theorems 2.1 and 3.1 give the periodic universality construction; Section 5 gives the two-visit property. <https://arxiv.org/abs/1702.05547>. Primary source provenance and hashes are in the frozen packet.
- [2] Turlough Neary and Damien Woods. *Four Small Universal Turing Machines*. Fundamenta Informaticae **91**(1), 123–144, 2009. DOI: 10.3233/FI-2009-0036. Publisher record and author repository: <https://journals.sagepub.com/doi/10.3233/FI-2009-0036>; <https://mural.maynoothuniversity.ie/id/eprint/12416/>. The repository notes that its PDF carries incorrect page numbers; the journal pagination above is authoritative. The arbitrary-program U15 encoding is a published dependency, not an implemented compiler in this release.
- [3] *Literal periodic Langton ant interface proof packet*. Frozen own-code, data, and proof release dated 3 October 2026, together with its separately identified defensive-loader revision. Complete source lineage and SHA256 pins are in the enclosing archive's verification records. The original source manifest is pinned by `a82fdb40349bdb3be08b7454f706232c0a7ac9d855484336abc8bb87497fcd0a`. [write] Shipped beside this report with the prefix `02-literal-ant-literal-interface-` (report README).
- [4] *First Revisit Detection and Exact Pattern Queries for Finite Defect Periodic Turmites*. Research Report 38, 3 October 2026. Full unchanged companion report and authenticated proof/implementation packet included in this archive. The first-revisit polynomial theorem and complete exact query theorem are the lower-side dependency of Section 40.14. [write] Printed as Part I of this report.

- [5] *Checkerboard ant history and toggle-router universality exploration notes*. ProveIt repository at commit `5883b08b7af362077f13bb4afa97a23a90ae4cf8`. Source links and pins are recorded in the literal packet's provenance file. The external-parameter bounded-board relation supplies the conditional inherited assumptions in Appendix 40.D; no combined universal operation count is inferred here. [write] The notes are `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md` and `EXPLORATION_TOGGLE_ROUTER_UNIVERSALITY.md` in the same directory, unchanged since the pin.

Part III

The arithmetic certificate: paid initialization and the complete positive polynomial (Reports 42 and 44)

[write] **This Part** merges two reports. Report 42 builds a fixed polynomial system that turns two raw positive integers into the exact initialized board of Part II’s ant (Theorem 42.1.1); Report 44 joins it with the inherited bounded-history relation and Part II’s endpoint into one polynomial, the Part’s main theorem (Theorem 44.1.1). Report 44’s archive is a *recovered edition*: a filesystem reset before delivery lost the originally verified files, and the edition was rebuilt from retained work (its own edition note, below). Only this edition exists; there is no earlier Report 44 in the repository to compare. At placement its two complete arithmetic streams were regenerated on a copy and reproduced the SHA-256 identities that the earlier audit had recorded (Appendix 44.A). The scope of the recognized language is item (a) of the preface to the four Parts: the fixed U15 sentinel-pair halting language, not an ordinary-input loader for the programme’s input convention. The programme reviewed both reports in commit `cfbe37de7` [W1]: no defect in the read algebra and composition; it independently checked the 1055-row recoder’s 228 residuals and the 174-row history DAG’s closure, count and degree, but did not re-prove the history theorem or regenerate the complete streams.

Report 42: the paid arithmetic initialization

[write] *A Paid Arithmetic Initialization for a Fixed Periodic Langton Ant. Uniform binary word recoding with an explicit Pell macro*, Research Report 42, dated 3 October 2026, author line “Mathematical research report”; batch-88 manuscript 04, archive `Arithmetic_Initialization_for_a_Periodic_Langton_Ant_Package.zip`. Pins: ProveIt `5883b08b7` for the parent history note, Mathlib `ac77769f` for the Pell theorems, and Report 40’s delivered PDF (Appendix 42.A). Numbered 42.*k*; labels `ptr:ai::`; files `03-ant-init-`.

Abstract (Report 42). We give a fixed finite conjunction of polynomial equations that turns two ordinary positive binary-sentinel integers into the exact initialized board and north-facing head required by a bounded-history encoding of one fixed periodic Langton ant. The initialization component has 2306387 arithmetic operations, 234 equations, and 396 new strictly positive internal witnesses, independent of the input word lengths. A fully streamed single-equation conversion uses 2307088 operations and has exact total degree 2304000. Its five inherited expression ports remain explicit. The construction combines a paid periodic-rectangle and finite-patch formula with a fixed all-length binary digit recoder. The latter uses binomial-coefficient parity, coefficient extraction, two residue identities, and fourteen calls to a 70-operation polynomial power macro specialized from pinned constructive Pell theorems. Forward and reversed word values are unique; the existential witness tuple is not. Prescribed enormous background coefficients are specified by a fixed source recipe rather than expanded. A separate literal-1-and-3 ledger charges their astronomical construction. We also give the fixed-length predecessor, an elementary witness-free polynomial obstruction, source-level accounting, and precisely delimited reproducibility checks. This is an initialization theorem conditional on the earlier literal-ant and bounded-history interfaces, not a new global universal arithmetic bound or a re-proof of the inherited Pell theorem.

42.1 What the construction establishes

The gap addressed here is arithmetic initialization. An already encoded board word is not the same input interface as an ordinary binary integer representing a source tape. A periodic colour query is not a free arithmetic instruction either. We therefore pay for the raw word conversion, its orientation and zero padding, the periodic rectangle, the finite anchor and source-field corrections, the board dimensions, and the initial head placement.

The geometric input comes from Research Report 40 [3]: a fixed binary periodic colouring, a fixed 2806-cell startup anchor, and a literal finite input loader for a fixed U15 source machine. The finite-history relation comes from the pinned 174-operation note [1]. Its hypotheses, not new assertions of this report, provide

$$W = 3^w, \quad W_p = 3^{w-1}, \quad Q = W^h, \quad \text{InitialHead}, \quad \text{InitialMemoryPlus}, \quad (42.1)$$

with $w \geq 3$ odd and $h \geq 2$ even. Here InitialMemoryPlus $- 1$ encodes an initially Boolean ternary board, and InitialHead is a north-facing head at an even flattened index. The five displayed quantities are inherited expression ports. They are neither free unconstrained replacements for the parent theorem nor new witnesses counted in this component.

Put

$$u = 481238074400, \quad v = 576000. \quad (42.2)$$

Physical coordinates (a, b) increase east and south. The frozen start is $(288650, 75, E)$. Normalize and translate by

$$X = b - 75, \quad Y = 288650 - a, \quad x = X + u, \quad y = Y + h/2. \quad (42.3)$$

The rotation preserves orientation and makes the initial heading north. The normalized periods are (u, v) .

Theorem 42.1.1 (Fixed all-length initialization). *Under the inherited contracts in (42.1) and the literal interface theorem of Report 40, there is a fixed arithmetic DAG using only $+$, $-$, $\times$, prescribed fixed integer numerals, and strictly positive existential witnesses with the following property. For every two positive raw integers, its equations hold exactly when the inherited initial memory and head are the literal initialized board and start associated with the two sentinel-decoded binary words, placed on a permitted rectangle. If the decoded word lengths are L, R and $m = L + R + 1$, the permitted rectangles are exactly*

$$w = 1 + au \quad (a \geq 2), \quad h = 2kv \quad (k \geq m).$$

The component uses 1153192 multiplications and 1153195 additions or subtractions, hence 2306387 operations, 234 equations, and 396 new positive internal witnesses. It has no gate family indexed by L or R . Every finite prefix of the infinite literal trajectory fits in some permitted rectangle.

The theorem is proved by the word recoder in Sections 42.3–42.5, the geometry and memory formula in Sections 42.6–42.8, and the literal splice in Section 42.9. “Exactly when” describes the initialization relation within the parent’s valid geometry. Without that inherited geometry, the displayed polynomial equations are not claimed to characterize powers of three or a valid bounded history on their own.

The numerals in the headline ledger are fixed independently of both raw inputs. Some are enormous. Section 42.9.2 replaces them by an explicitly counted literal source prefix from 1 and 3 only. We distinguish a finite, exactly specified source from its practical expansion, and a finite regression check from a theorem covering all lengths.

42.2 Arithmetic and raw input conventions

An arithmetic gate evaluates one addition, subtraction, or multiplication of earlier expressions or fixed numerals. Gate outputs are expressions, not automatically quantified unknowns. Equality assertions have zero gate cost. A multiplication by a fixed numeral still costs one multiplication. No simplifier removes gates with zero or repeated operands in the source counted here.

All quantified unknowns are strictly positive integers. For a mathematical quantity $z \geq 0$, introduce $z^+ > 0$ and use the expression $z = z^+ - 1$, charging its subtraction. Intermediate expressions may be zero or negative. In particular, a nonnegative output expression must not silently be reclassified as a positive unknown.

For a nearest-head-first word $d_0, \dots, d_{n-1}$, the ordinary most-significant-bit-first sentinel convention is

$$\text{Raw} = 2^n + \sum_{j=0}^{n-1} d_j 2^{n-1-j}, \quad d_j \in \{0, 1\}. \quad (42.4)$$

Every positive integer has exactly one such word. The empty word has code 1. A length- n all-zero word has code 2^n , so zero padding is retained. The two raw parameters are $\text{Raw}_L, \text{Raw}_R$.

42.2.1 The fixed-length predecessor

If L and R are fixed before constructing a circuit, introduce $L + R$ positive bit adapters. For each bit $b = b^+ - 1$, assert $b(b - 1) = 0$. Reconstruct each raw integer by Horner evaluation in radix 2, starting at 1. This is a correct family indexed by two lengths. It does not, by itself, give a single relation for all raw integers. The fixed-length family is useful both as a direct construction and as a source-polynomial cross-check of the fixed outer formula.

Proposition 42.2.1 (Witness-free polynomial obstruction). *Fix $G = 3^{wv} > 1$ for any valid board width. No finite witness-free circuit over fixed constants using only $+, -, \times$ computes the sentinel-preserving dilation $f(\text{Raw}) = G^n + \sum_{j < n} d_j G^{n-1-j}$ for every positive sentinel input.*

Proof. Such a circuit computes a polynomial f . For the all-zero words, $f(2^n) = G^n$. A nonzero polynomial of degree d has ratio $f(2^{n+1})/f(2^n) \rightarrow 2^d$; for a nonzero constant the ratio is 1. The required ratio is the odd power of three $G > 1$, which is neither possibility. Enormous fixed coefficients do not change this asymptotic argument. $\square$

This obstruction concerns witness-free polynomial evaluation only. The existential construction below has additional witnesses and is fully consistent with it.

42.3 One fixed all-length digit recoder

Fix input ports $C \geq 2$ and $\text{Raw} > 0$. We seek the uniquely determined length n , power $P = C^n$, and either the forward value $\sum_j d_j C^{n-1-j}$ or the reverse value $\sum_j d_j C^j$. The variable length is an existential expression $n = n^+ - 1$; it never controls the number of source gates.

Introduce seven powers

$$\begin{aligned} P &= C^n, & E &= 2^n, & B &= 2^{CP+2}, & F &= B^n, \\ \Lambda &= 2^{U+1}, & Y &= \Lambda^D, & Z &= (\Lambda + 1)^U. \end{aligned} \quad (42.5)$$

Every displayed power will be replaced by the fixed polynomial macro in Section 42.4. In this section $U, D, x, V, q_2, q_C, q, o, r$ are nonnegative; s_x, s_c, s_r, s_V are strictly positive. With $c = 2o + 1$, impose the following nine equations:

$$F - 1 = (B - 1)U, \quad x = \text{Raw} - E, \quad (42.6)$$

$$x + s_x = E, \quad D = q_2(B - 2) + x, \quad (42.7)$$

$$Z = (q\Lambda + c)Y + r, \quad c + s_c = \Lambda, \quad (42.8)$$

$$r + s_r = Y, \quad V + s_V = P, \quad (42.9)$$

$$D = q_C(B - C) + V \quad (\text{forward}), \quad (42.10)$$

with the last equation replaced, for reverse output, by

$$PD = q_C(BC - 1) + CV. \quad (42.11)$$

There are nine equations in either choice, not ten. The separate positive adapter for x matters: the expression $\text{Raw} - E$ alone does not establish its nonnegativity.

Lemma 42.3.1 (Auxiliary radix margins). *The deterministic choice $B = 2^{CP+2}$ satisfies*

$$B > C + P, \quad B > 2^n + 2.$$

Consequently $B - 2 > 2^n - 1$, $B - C > P - 1$, and $CP < BC - 1$.

Proof. Write $t = CP$. Because $C \geq 2$, $P = C^n \geq 1$, we have $t \geq 2$, $t \geq C + P - 1$, and $t \geq 2P \geq 2^{n+1}$. The elementary inequality $2^{t+2} > 2t + 2$ proves both margins. The last conclusion follows from $B > P + 1$ and $C \geq 2$. $\square$

Lemma 42.3.2 (Binary support from coefficient parity). *For B a power of two and $U = \sum_{i=0}^{n-1} B^i$, the coefficient $\binom{U}{D}$ is odd exactly when*

$$D = \sum_{i=0}^{n-1} b_i B^i, \quad b_i \in \{0, 1\}.$$

For $D > U$, the coefficient is interpreted as zero.

Proof. In $\mathbb{F}_2[z]$, repeated squaring gives $(1 + z)^{2^a} = 1 + z^{2^a}$. Since every B^i is a power of two,

$$(1 + z)^U = \prod_{i=0}^{n-1} (1 + z^{B^i}).$$

The subset sums of the distinct place values B^i are distinct for $B \geq 2$. Their coefficients are exactly one, and all other coefficients vanish. This is the needed binary instance of Lucas parity, proved directly rather than used as an unexpanded black box. $\square$

Theorem 42.3.3 (Dilation graph). *After replacing the seven powers by their polynomial macros, equations (42.6)–(42.10), or their reverse variant, have positive witnesses exactly for the power $P = C^n$ and the appropriate value V of the unique sentinel word of Raw . The forward graph uses 524 operations, 196 positive internal witnesses, and 114 equations; the reverse graph uses 527 operations and the same witness and equation counts.*

Soundness. The first equation gives $U = 1 + B + \cdots + B^{n-1}$, including $U = 0$ when $n = 0$. Each coefficient of $(1 + z)^U$ is at most $2^U < \Lambda$. Thus the base- Λ digit at position D in $Z = (1 + \Lambda)^U$ is $\binom{U}{D}$. Equations (42.8)–(42.9) assert $0 < c < \Lambda$, $0 \leq r < Y = \Lambda^D$ and

$$Z = (q\Lambda + c)\Lambda^D + r.$$

They therefore extract that digit exactly. As $c = 2o + 1$, the digit is odd. Lemma 42.3.2 forces $D = \sum_i b_i B^i$ with $b_i \in \{0, 1\}$, and in particular excludes $D > U$ without a separate upper-bound equation.

Modulo $B - 2$, this D equals $\sum_i b_i 2^i$. Both the latter number and x belong to $[0, 2^n)$, which lies below the modulus by Lemma 42.3.1. Hence $x = \sum_i b_i 2^i$. The two equations involving x force $2^n \leq \text{Raw} < 2^{n+1}$, so n is the sentinel length and $b_i = d_{n-1-i}$.

For forward output, reduction modulo $B - C$ gives $D \equiv \sum_i b_i C^i$. Both the candidate and V lie in $[0, P)$, below the modulus, so they agree. For reverse output, $BC \equiv 1 \pmod{BC - 1}$ gives, for $i < n$,

$$C^n B^i \equiv C^{n-i} \pmod{BC - 1}.$$

Thus $PD \equiv C \sum_i b_i C^{n-1-i}$. Both this candidate remainder and CV lie in $[0, CP)$, below $BC - 1$. Equation (42.11) forces the reverse value. No negative exponent is used when $n = 0$. $\square$

Completeness. For a given word, set the seven powers as displayed and set $D = \sum_i b_i B^i$. Its extracted coefficient is odd by Lemma 42.3.2; take $c = \binom{U}{D}$ and $o = (c - 1)/2$. Ordinary Euclidean division gives q, r . The raw and forward quotient differences are nonnegative since $B > C \geq 2$. For the reverse quotient, termwise $C^n B^i \geq C^{n-i}$ for $i < n$ makes the divisible difference nonnegative. Thus all q_2, q_C also have the required signs. The strict remainder and value margins give positive slacks. Positive adapters then represent every quantity that vanishes. For the empty word,

$$\text{Raw} = P = E = F = 1, \quad U = D = V = 0, \quad \Lambda = 2, \quad Y = Z = c = 1.$$

These values satisfy all equations and positive-slack conditions. Completeness of the power macro supplies its witnesses. The operation counts are checked in Section 42.9. $\square$

Functional outputs and nonunique witnesses. The output P, V is unique. The complete existential fiber is not: paired-quotient congruence witnesses in the Pell macro may be simultaneously increased. The relation is not claimed to be single-fold or finite-fold.

42.4 The explicit polynomial power macro

This section supplies the promised fixed source for exponentiation. It specializes the constructive Pell theorems `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in the pinned mathlib source [2]; it does not derive exponentiation from a general undecidability theorem. The source is archived for reading and attribution, not executed or recompiled by this work.

[write] The archive's copy of the cited Mathlib file (`Mathlib/NumberTheory/PellMatiyasevic.lean`, 39,986 bytes, SHA-256 993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a, at Mathlib commit `ac77769fabe23cb237559e7f56578dbead91499f`) and its Apache-2.0 licence file are not shipped in this report; they are third-party files, available at that commit, and the shipped pin record is `data/03-ant-init-science-dependencies-mathlib-primary_source_pin.json`. The repository's own

Lean development uses `Pell.matiyasevic` as well (`Computability/HilbertTenthProblem/Lean/Diophantine/Paper1976/Cor26.lean`); the specialization below is not formalized.

For $a \geq 2$, define Pell coordinates by

$$\begin{aligned} X_0(a) &= 1, & Y_0(a) &= 0, & X_{j+1}(a) &= aX_j(a) + (a^2 - 1)Y_j(a), \\ & & & & Y_{j+1}(a) &= X_j(a) + aY_j(a). \end{aligned}$$

The first cited theorem characterizes an exact indexed pair $(X_k(a), Y_k(a))$ by three Pell equations, divisibility and congruences, with a separate zero-index branch. The second characterizes b^k through that indexed pair and an additional growth Pell equation. We give every polynomial equation of the specialization so that its accounting and domain restrictions are inspectable.

To assert $O = b^e$ for $b \geq 2$, $e \geq 0$, $O > 0$, construct $k = e + 1$ and $m = bO$. Introduce 25 positive unknowns. Eleven are

$$w, A_0, \Theta, g, x, y, p, q, s, t, B_0, \quad a = A_0 + 1, \quad \beta = B_0 + 1.$$

Three more are positive adapters for nonnegative gaps $\delta_{wb}, \delta_{wk}, \delta_{yk}$. A fourth is a positive strict slack S . Two are positive q_b, q_v . The last eight are positive adapters for the nonnegative variables

$$\alpha_1, \alpha_2, \sigma_1, \sigma_2, \tau_1, \tau_2, \rho_1, \rho_2.$$

Impose exactly the following fifteen equations:

$$x^2 = 1 + (a^2 - 1)y^2, \tag{42.12}$$

$$p^2 = 1 + (a^2 - 1)q^2, \tag{42.13}$$

$$s^2 = 1 + (\beta^2 - 1)t^2, \tag{42.14}$$

$$\beta = 1 + 4yq_b, \tag{42.15}$$

$$\beta + p\alpha_1 = a + p\alpha_2, \tag{42.16}$$

$$q = y^2q_v, \tag{42.17}$$

$$s + p\sigma_1 = x + p\sigma_2, \tag{42.18}$$

$$t + 4y\tau_1 = k + 4y\tau_2, \tag{42.19}$$

$$y = k + \delta_{yk}, \tag{42.20}$$

$$w = b + \delta_{wb}, \tag{42.21}$$

$$w = k + \delta_{wk}, \tag{42.22}$$

$$\Theta = m + S, \tag{42.23}$$

$$a^2 = 1 + ((w + 1)^2 - 1)(wg)^2, \tag{42.24}$$

$$2ab = \Theta + (b^2 + 1), \tag{42.25}$$

$$x + \Theta\rho_1 = y(a - b) + m + \Theta\rho_2. \tag{42.26}$$

The paired quotients represent unrestricted signed congruence quotients as differences of two nonnegative integers. No positivity assumption on a signed quotient has been smuggled into these equations.

Lemma 42.4.1 (Polynomial power macro). *The fifteen equations above hold with positive witnesses exactly when $O = b^e$. Their literal source uses 31 multiplications and 39 additions or subtractions, for 70 operations, with 25 positive internal witnesses.*

Soundness from the pinned constructive theorem. Equations (42.12)–(42.20) meet the positive-index, nonzero branch of `Pell.matiyasevic`: $a, \beta \geq 2$, $y \geq k \geq 1$, $q > 0$, and the displayed divisibilities and congruences hold. The omitted branch would require $y = 0$, so $(x, y) = (X_k(a), Y_k(a))$.

The source power theorem uses natural subtraction $a - b$, whereas our arithmetic uses integer subtraction. They coincide here for a proved reason. Equation (42.24), with $w \geq b \geq 2$ and $g \geq 1$, gives the elementary bound

$$a^2 = 1 + (w^2 + 2w)w^2g^2 > w^2.$$

As a, w are positive, $a > w \geq b$, so $a - b$ is nonnegative. The remaining equations now meet `Pell.eq_pow_of_pell`: $w \geq b, k, \Theta = 2ab - b^2 - 1, m < \Theta$, the growth equation, and $x \equiv y(a - b) + m \pmod{\Theta}$. That theorem gives $m = b^k$. Cancelling $b > 0$ from $m = bO$ proves $O = b^e$. $\square$

Completeness and positivity. The cited construction takes $w = \max(b, k)$, $a = X_w(w + 1)$, and $g = Y_w(w + 1)/w$, all satisfying the required positive domains. It takes $x = X_k(a)$ and $y = Y_k(a)$, then uses a larger index for p, q , and a CRT-compatible $\beta \geq 2$, followed by $s = X_k(\beta)$ and $t = Y_k(\beta)$. All these coordinates are positive because $k \geq 1$. Since $\beta \equiv 1 \pmod{4y}$ and $\beta > 1$, its quotient q_b is at least one. Since $q > 0$ and $y^2 \mid q$, $q_v \geq 1$. The source's strict bound $m < \Theta$ provides $S > 0$. All weak gaps and paired congruence quotients admit their stated nonnegative representatives and hence positive adapters. Thus the stricter domains lose no valid power. $\square$

The small Pell examples in the executable checks are regression tests, not a proof that spurious outputs are excluded. The all-input exclusion is supplied by the two cited constructive theorems and their explicit specialization above.

42.5 The pair of word outputs

Take $C = G = W^v$. Apply the reverse recoder to Raw_L and the forward recoder to Raw_R . Denote their outputs by

$$P_L = G^L, \quad V_L = \sum_{j < L} \ell_j G^j, \quad P_R = G^R, \quad V_R = \sum_{j < R} r_j G^{R-1-j}.$$

Build

$$A = P_L P_R, \quad B_{\text{out}} = P_R, \quad T = (G P_R) V_L + V_R. \quad (42.27)$$

This uses three multiplications and one addition. Thus

$$A = G^{L+R}, \quad B_{\text{out}} = G^R, \quad T = G^{R+1} \sum_{j < L} \ell_j G^j + \sum_{j < R} r_j G^{R-1-j}.$$

Equivalently, T is the base- G Horner value of

$$\text{reverse}(\ell), \quad 0, \quad r.$$

The central zero is deliberate: the A0 primary head's nonzero state bit is added separately by the physical initialization formula. B_{out} is not either recoder's auxiliary power-of-two radix B .

There are three distinct interface conventions, whose ledgers must not be mixed:

1. In the inlined interface used here, A, T are expressions and B_{out} aliases P_R . The pair has 1055 operations, 392 positive internal witnesses, and 228 equations
2. Exposing separate ports A, B_{out}, T adds three equality assertions, giving 231 equations with the same operation and internal-witness counts. The last port is nonnegative, not necessarily positive
3. Exposing positive $T^+ = T + 1$ instead adds one operation, giving 1056. If all three ports are separately quantified, count three further positive unknowns outside the 392 internal ones. A consumer needing T must also pay for $T^+ - 1$ unless that expression is already present

The fixed splice uses the first convention; it never turns those three outputs into free raw parameters.

42.6 A paid rectangle and initial head

Set $n = L + R$, $m = n + 1$. The recoder provides the expressions $A = G^n$, $B_{\text{out}} = G^R$, and T . Four additional unknowns are positive:

$$X_{\text{extra}}, \quad H, \quad K, \quad P_{\text{pad}}.$$

The fixed numerals $C_u = 3^u$, $C_x = C_u - 1$ are specified independently of the input. Compute $G = W^v$ once, with its charged binary multiplication chain, and impose

$$W_p - 1 = C_x(X_{\text{extra}} + 1), \tag{42.28}$$

$$H - 1 = (G - 1)K, \tag{42.29}$$

$$Q = H^2, \tag{42.30}$$

$$H = P_{\text{pad}}(AG), \tag{42.31}$$

$$\text{InitialHead} = C_u H. \tag{42.32}$$

The expression $AG = G^{n+1}$ costs one multiplication. Reusing the parent's W_p is an alias, not an uncharged division. A variant exposing only W could introduce a new positive W_p and assert $3W_p = W$, but that would add one witness, equation, and multiplication; it is not the displayed interface.

Lemma 42.6.1 (Exact geometry). *Within the parent geometry, these five equations are equivalent to*

$$w = 1 + au \quad (a \geq 2), \quad h = 2kv \quad (k \geq m), \quad H = W^{kv}, \quad P_{\text{pad}} = G^{k-m}, \quad \text{InitialHead} = 3^u W^{kv}.$$

The remaining X_{extra} and K are uniquely determined positive integers.

Proof. For $q \geq 2$ and $d, e \geq 1$, $q^d - 1$ divides $q^e - 1$ exactly when $d \mid e$: reduce e modulo d and use $0 < q^r - 1 < q^d - 1$ for $0 < r < d$. Apply this first to (42.28). It gives $u \mid w - 1$. The quotient is one for $a = 1$, whereas $X_{\text{extra}} + 1 \geq 2$, so $a \geq 2$; every $a \geq 2$ has a valid positive extra quotient.

Since the inherited h is even and $H > 0$, $H^2 = W^h$ forces $H = W^{h/2}$. Divisibility by $G - 1$ then gives $v \mid h/2$, say $h = 2kv$, with $K = (G^k - 1)/(G - 1) > 0$. Equation (42.31) forces $k \geq m$ and $P_{\text{pad}} = G^{k-m}$. The zero exponent gives the permitted value 1. Equation (42.32) places the initial head at (u, kv) . Its flattened index $u + wkv$ is even, since u, v are even. All converse implications follow by substitution. $\square$

42.6.1 Support containment and arbitrary finite prefixes

The anchor's physical ranges are

$$288617 \leq a \leq 289200, \quad -144 \leq b \leq 76.$$

After (42.3), these become $u - 219 \leq x \leq u + 1$ and $kv - 550 \leq y \leq kv + 33$. All variable source and marker changes have $x = u - 21$, with least row $kv - vm + 23945 \geq 23945$ and greatest row below $kv + 33$. Thus all initialized cells and the head belong to every permitted rectangle: $w \geq 2u + 1$, $k \geq m \geq 1$, and $kv + 33 < 2kv$.

The global literal-interface proof bounds the physical coordinate b below by -144 for every prefix. Hence $X \geq -219$ throughout the infinite trajectory. The fixed shift by u suffices on that side. For any finite prefix the upper X and both extremes of Y are finite. Choosing a and k large enough places it inside a permitted rectangle. Conversely a bounded history that never leaves the rectangle evolves identically to the infinite literal board over its entire duration. Enlarging the rectangle neither creates nor removes an accepting event already inside that prefix.

42.7 The exact periodic background

Let $c(a, b)$ be the frozen total binary colour function before all finite overrides. In normalized coordinates define

$$\mathcal{B}(i, j) = c(288650 - j, i + 75),$$

with periods (u, v) . For $0 \leq j < v$, prescribe

$$\alpha_j = \sum_{i=0}^{u-1} \mathcal{B}(i, j) 3^i, \quad \beta_j = \mathcal{B}(0, j),$$

and the fixed-coefficient polynomials

$$\mathcal{A}(W) = \sum_{j=0}^{v-1} \alpha_j W^j, \quad \mathcal{B}_0(W) = \sum_{j=0}^{v-1} \beta_j W^j. \quad (42.33)$$

These coefficients are fixed by the pinned colour-function recipe. They are neither existential colours nor input-dependent coefficient ports. A complete numerical expansion of an α_j would have roughly 10^{11} decimal digits; none is claimed to have been materialized.

Set $H_x = X_{\text{extra}} + 1$ and $H_y = K(H + 1)$. Construct

$$\text{Background} = H_y(H_x \mathcal{A}(W) + W_p \mathcal{B}_0(W)). \quad (42.34)$$

Lemma 42.7.1 (Rectangle identity). *Equation (42.34) is exactly the row-major ternary encoding of the periodic binary background on the chosen $w \times h$ rectangle.*

Proof. By the width equation,

$$H_x = \frac{3^{au} - 1}{3^u - 1} = \sum_{s=0}^{a-1} 3^{su}.$$

This repeats each complete u -column row tile a times. The $W_p = 3^{au}$ term supplies the first column once more, giving width $au + 1$. The supports are disjoint, so there is no ternary carry. Likewise

$$H_y = K(H + 1) = \frac{Q - 1}{G - 1} = \sum_{t=0}^{2k-1} G^t$$

repeats the v -row tile exactly $2k$ times. With $W = 3^w$, the row supports are disjoint. This explains the extra boundary column: the even horizontal period is not incorrectly required to divide the odd board width. $\square$

42.8 The anchor and input corrections

For each frozen anchor record $(a, b, c_{\text{old}}, c_{\text{new}})$, put $\delta = c_{\text{new}} - c_{\text{old}} \in \{-1, 1\}$. Define the degree-at-most-583 fixed polynomial

$$J(W) = \sum_{\text{anchor}} \delta 3^{b+144} W^{289200-a}. \quad (42.35)$$

The exponents satisfy $0 \leq b + 144 \leq 220$ and $0 \leq 289200 - a \leq 583$. Its 584 signed coefficients are prescribed numerals. The anchor has 2806 distinct cells and already includes the left marker.

Let $C_{\text{base}} = 3^{u-219}$ and $C_{198} = 3^{198}$. Using the recoder expressions, construct

$$E_{\text{tape}} = W^{24000} T + B_{\text{out}}, \quad (42.36)$$

$$Z_{\text{patch}} = W^{23945} + W^{263945} (W^{264000} + 1) E_{\text{tape}}, \quad (42.37)$$

$$D_{\text{patch}} = W^{v-550} A J(W) - C_{198} (W + 1) Z_{\text{patch}}, \quad (42.38)$$

$$\text{InitialMemoryPlus} = \text{Background} + C_{\text{base}} P_{\text{pad}} D_{\text{patch}} + 1. \quad (42.39)$$

All variable-base fixed powers are charged by literal multiplication chains. The final $+1$ is the positive memory adapter.

Theorem 42.8.1 (Exact initialization identity). *The right side of (42.39), minus one, is precisely the fixed literal background after its frozen anchor and the source input changes, translated as in (42.3). Its ternary digits are Boolean.*

Proof. An anchor cell contributes

$$\delta 3^{u+b-75} W^{kv+288650-a} = C_{\text{base}} P_{\text{pad}} \delta 3^{b+144} W^{vm+288650-a}.$$

Since $vm + 288650 - a = (v - 550) + vn + (289200 - a)$, the full anchor contribution is the first term in $C_{\text{base}} P_{\text{pad}} D_{\text{patch}}$.

A source field at slot q of module i changes exactly the two physical cells

$$(vi + 704 + 24000q, 54), \quad (vi + 705 + 24000q, 54)$$

from one to zero. After division by $C_{\text{base}} P_{\text{pad}}$ their negative contribution is

$$-C_{198} (W + 1) W^{vm+287945-vi-24000q}. \quad (42.40)$$

The right marker has $i = m, q = 11$, giving exponent 23945. The left marker is omitted from this correction because its two changed cells already belong to the anchor.

At primary index i , a tape bit one uses fields $(i, 13)$ and $(i + 1, 0)$. In (42.40) their exponents are

$$v(m - 1 - i) + 551945, \quad v(m - 1 - i) + 287945.$$

These are exactly the contribution of that digit to

$$W^{24000}W^{263945}(W^{264000} + 1)T.$$

The primary A0 head has value 2 at index $i = L$, using bit 1 rather than bit 0. Its fields $(L, 14)$ and $(L + 1, 1)$ lower both exponents by 24000; their shared power of G is $G^{m-1-L} = G^R = B_{\text{out}}$. Thus (42.36) includes exactly the head contribution as well.

The frozen interface proves all variable source fields have old colour one. Their only overlap with the anchor is the agreeing left marker, already omitted, and all remaining fields are disjoint. Adding the signed changes to the exact background therefore leaves every ternary digit zero or one. The changed support has cardinality

$$2806 + 4(\text{popcount } \ell + \text{popcount } r + 1) + 2 = 2812 + 4(\text{popcount } \ell + \text{popcount } r).$$

Leading zero symbols still alter m , the endpoint marker and the geometry; they are not erased. $\square$

42.9 Source splice and exact arithmetic ledgers

The source first computes $G = W^v$ once and uses that exact expression in both recoders and the board wrapper. The two recoders have disjoint witness prefixes. Their expressions A, B_{out}, T are substituted into the wrapper before later gates are emitted, with no free dilation ports, no duplicate G chain, and no extra output unknowns. The combined stream validates each opcode, each earlier-wire reference and every equality endpoint.

Component	M	A	Total	Equations	Positive witnesses
Power macro	31	39	70	15	25
Forward recoder	224	300	524	114	196
Reverse recoder	227	300	527	114	196
Pair with inlined outputs	454	601	1055	228	392
Fixed outer wrapper	1152738	1152594	2305332	6	4
Combined initialization	1153192	1153195	2306387	234	396

Table 5: Free fixed numeral convention. Counts concern this initialization component and its stated ports. Equality assertions and expression aliases cost no arithmetic gate.

Each recoder contains seven power calls, contributing $217M + 273A = 490$ operations, 175 witnesses and 105 equations. Its remaining forward part contributes $7M + 27A$, 21 witnesses and nine equations. The reverse formula adds three multiplications. Pair assembly adds $3M + 1A$. There are no coefficients or numbers of gates depending on a raw word length.

42.9.1 Fixed-length family ledger

For $e \geq 0$, let

$$\lambda(e) = \begin{cases} 0, & e = 0, 1, \\ \lfloor \log_2 e \rfloor + \text{popcount}(e) - 1, & e \geq 2. \end{cases}$$

This is the count of the left-to-right binary multiplication chain used in the source. Exponent zero is the constant 1; exponent one is a wire alias. For $n = L + R$, set

$$C_{\text{pow}} = \lambda(v) + \lambda(n) + \lambda(n + 1) + \lambda(R) \\ + \lambda(575450) + \lambda(23945) + \lambda(263945) + \lambda(264000) + \lambda(24000).$$

The original fixed-length source has

$$M = 2(v - 1) + 583 + 3n + C_{\text{pow}} + 18, \quad (42.41)$$

$$A = 2(v - 1) + 583 + 4n + 13, \quad (42.42)$$

with $n + 8$ equations and $n + 4$ positive witnesses. The two $2(v - 1)$ terms are the two background Horner chains. The 583 terms evaluate J . Bit typing, binary sentinel reconstruction and the tape Horner chain contribute respectively $nM + 2nA$, $nM + nA$ and $nM + nA$. No leading-zero coefficient gate is suppressed.

For empty words, the actual streamed source has $1152737M + 1152594A = 2305331$ operations, eight equations and four positive witnesses. The fixed outer wrapper replaces the length-specific G^n, G^R, T by certified expression outputs and computes $G^{n+1} = AG$; its total is one operation larger, 2305332, with six equations. This component comparison is also checked by equality of off-solution residual polynomials after exact substitution.

42.9.2 Literal constants from one and three

The free fixed-numeral ledger does not claim that enormous coefficients appear at zero physical cost. Here is an exact alternative source prefix from literals 1 and 3. First construct $0 = 1 - 1$, $-1 = 0 - 1$, $2 = 1 + 1$. Construct $C_u = 3^u$, $C_{\text{base}} = 3^{u-219}$, and $C_{198} = 3^{198}$ by independent binary chains, and $C_x = C_u - 1$.

For each of the v coefficients α_j , evaluate its prescribed u -digit ternary Horner string, beginning at the first fixed zero or one. This costs $u - 1$ multiplications and $u - 1$ additions even for zero digits. Each β_j is an alias of 0 or 1. For each of the 584 coefficients of J , use its prescribed 221-symbol signed ternary string over $\{-1, 0, 1\}$, costing exactly 220 multiplications and 220 additions. The sign at each position is fixed by the anchor map, never existential.

The bridge-only prefix therefore costs

$$M_{\text{const}} = v(u - 1) + 584 \cdot 220 + \lambda(u) + \lambda(u - 219) + \lambda(198), \\ A_{\text{const}} = v(u - 1) + 584 \cdot 220 + 4,$$

namely

$$M_{\text{const}} = 277193130853952587, \\ A_{\text{const}} = 277193130853952484.$$

The recoder uses only literals 1, 2, 4. The prefix already supplies 2; build $4 = 2 + 2$ once, adding one further addition. Hence the combined literal-1-and-3 prefix costs 554386261707905072 operations, and prefix plus combined initialization costs

$$554386261710211459. \quad (42.43)$$

This enormous finite straight-line source is specified and counted, not expanded or run. The colour recipe is used to generate fixed source digits at compile time, never to answer a variable colour query during arithmetic evaluation. This strict ledger belongs to the initialization alone; it does not retrospectively pay for constants or adapters in any inherited component.

42.9.3 Exact degrees and a paid single equation

For this subsection, all independent ports and positive witnesses have degree one; fixed numerals have degree zero. Let R_i be each equality's polynomial residual.

Proposition 42.9.1 (Standalone recoder degree). *Before substituting $G = W^v$, every one of the six recoder interface variants has maximum residual degree 6, and its direct sum of squared residuals has exact degree 12. For the inlined pair this single polynomial uses $682M + 1056A = 1738$ operations and the same 392 positive witnesses.*

Proof. Every power macro's growth-Pell residual expands as

$$a^2 - 1 - ((w + 1)^2 - 1)(wg)^2 = a^2 - 1 - w^4g^2 - 2w^3g^2.$$

Its unique top-degree term is $-w^4g^2$. All other macro residuals have degree at most four; the possible exponent argument $CP + 2$ is quadratic and occurs linearly where it enters. Coefficient extraction, reverse residue extraction and pair assembly have degree at most three. Thus the maximum is exactly six. On squaring and summing, the degree-twelve part consists of fourteen distinct terms w^8g^4 , each of coefficient one, for a pair. There can be no cancellation.

For E equalities, form every residual with one subtraction, every square with one multiplication and the sum with $E - 1$ additions. This adds $EM + (2E - 1)A = 3E - 1$ operations and no witnesses. The inline pair has $E = 228$, hence 683 additional operations. The separate-output and positive-output variants have $E = 231$, hence 692 extra operations and totals 1747 and 1748 respectively, with their earlier port conventions unchanged. $\square$

This degree-12 statement treats G as an independent degree-one input. It does not survive substitution of the board radix without recalculation.

Proposition 42.9.2 (Initializer degree and single polynomial). *For the 234 initializer residuals, with the inherited initializer ports still independent, the maximum total degree is $2v = 1152000$. Their direct sum of squares*

$$F = \sum_{i=1}^{234} R_i^2$$

has exact degree $4v = 2304000$, and $F = 0$ is equivalent over integers to the original conjunction. Its fully streamed source uses $1153426M + 1153662A = 2307088$ operations, one equation, and the same 396 positive witnesses.

Proof. The exact radix expression $G = W^v$ has degree v . Two residuals, one in each call $\text{EXP}(G, \text{length}, \text{radixPower})$, have the form

$$2aG - (\Theta + G^2 + 1).$$

Each has unique highest homogeneous term $-W^{2v}$. Every other residual has degree strictly less than $2v$. This is checked by propagating an upper degree along each literal gate: add degrees at multiplication and take the maximum at addition or subtraction. More explicitly, the final memory residual has upper degree 1127949, and all remaining residuals have smaller bounds. The memory bound follows from $\deg T = v + 2$, the displayed fixed powers in (42.36)–(42.39), and the degree-583 anchor polynomial. Fixed huge coefficients do not raise any of these degrees.

Consequently the top homogeneous term of F is exactly $2W^{4v}$, proving the stated degree without relying on a cancellation-blind upper bound alone. A sum of integer squares vanishes exactly when each residual vanishes. With 234 residuals, the naive conversion adds $234M + 467A = 701$ operations. The emitted single-polynomial source retains the 234 equalities as expression pairs, asserts only $F = 0$, and introduces no extra witnesses. $\square$

The exact degree here is measured before replacing the inherited ports by any parent-history or endpoint expressions. Thus F alone is not a universal halting polynomial. The literal-1-and-3 constant prefix changes neither its witness count nor its degree, but its enormous operation cost remains separately chargeable. No minimal-degree, sparse-optimal or operation-optimal claim is made.

42.10 Dependencies and the remaining global composition

The initialized board and head can be attached to the bounded-history relation by literal wire identifications. The normalized accepting event retained by the literal interface has residue and heading

$$(481225262775, 29948, E) \pmod{(u, v)}.$$

Both translations preserve those residues because the horizontal shift is u and the vertical shift is kv . The separately frozen endpoint selector thus retains its meaning after this initialization. This report does not re-prove its arithmetic equations.

There are three distinct dependencies:

1. The pinned bounded-history theorem supplies the geometry and finite ant evolution. Its published local ledger is 174 operations, 61 positive unknowns, 48 equations and five positive parameters
2. Report 40 supplies the literal background, anchor, loader geometry and global finite-prefix facts, together with its exact U15 accepting interface
3. The pinned constructive Pell theorems supply the polynomial power macro's soundness and completeness

Report 38 [4] supplies the one-visit boundary context; it is not used to infer a new arithmetic initialization count. The arbitrary-program-to-U15 input compiler remains the separate published dependency identified in Report 40.

Corollary 42.10.1 (Qualitative joined raw input relation). *Assume also the separately proved normalized endpoint selector. Existentially bind the parent history's five non-raw parameter ports*

InitialMemoryPlus, FinalMemoryPlus, InitialHead, FinalHead, FinalSignPlus.

Use the parent's existing W, W_p, Q expressions in the initializer. The resulting one fixed positive Diophantine relation on $(\text{Raw}_L, \text{Raw}_R)$ holds exactly when the pinned U15 machine, started in state A scanning zero with those sentinel-decoded side words, reaches $J1$.

[write] Report 44 makes this relation numerical, with every port bound and the equation combination paid: Theorem 44.1.1.

Proof. For soundness, the parent relation gives a positive number of ordinary no-escape ant steps. The initializer identifies the exact translated restriction of the infinite literal board, so induction identifies its finite trajectory with the literal trajectory. The endpoint predicate has unchanged residues under translations by u and kv . The literal accepting-interface theorem therefore gives U15 halting. Existentially choosing the final memory does not weaken this statement: the parent theorem makes it the actual final Boolean board.

Conversely, a U15 halt yields the accepting physical event after a positive finite number of ant steps. Choose a, k large enough to contain its entire prefix. The initializer supplies its 396 positive witnesses, the parent converse supplies the history witnesses and its genuinely positive parameter ports, and the endpoint converse supplies its own positive witnesses. In particular, both memory ports are one greater than a nonnegative board word, both head ports are positive powers of three, and the white/east accepting endpoint has $\text{FinalSignPlus} = 1$. The literal interface excludes the predicate at initialization, so no zero-time history is required. $\square$

This is a qualitative composition with every non-raw parent port bound. It does not supply an arbitrary-program-to-U15-word compiler or an uncharged numerical sum of all component ledgers.

A new global universal bound would require a new global ledger. In particular, the parent relation's original five external parameters, internal aliases W, W_p, Q , final memory and head/sign ports, endpoint selection, and equation combination must all be accounted for at one consistent boundary. Simply printing $174 + 2306387$ would not do this. Neither the headline theorem nor the archive asserts a minimized universal polynomial or a new record.

The initializer-only single polynomial in Section 42.9.3 pays for its own equation combination. No corresponding combined history-and-endpoint source is counted here. Gate outputs remain expressions under all these ledgers; turning every gate into a separately quantified unknown would be a different witness convention.

42.11 Verification and reproducibility

The accompanying archive contains the full article source, the own arithmetic builders and finite checkers, compact read-only dependency snapshots, reference receipts, a deterministic PDF builder, and an authenticated release inventory. It deliberately does not duplicate the complete large finite-template packet of Report 40 or execute that packet. The colour recipe is included through its compact read closure: unchanged generator and decoder source, unchanged small program data, and exact colour-map projections of the larger template files. The asset manifest records both original and projection hashes. No coordinate or colour is changed; proof histories and unused metadata are omitted. The complete global ant simulation remains a pinned theorem dependency, not something re-certified by replaying an old schedule.

42.11.1 What the arithmetic checks establish

The periodic bridge checker compares exact geometric repetition with dense small periodic boards; compares the compressed patch polynomial against independently assembled literal coordinates; checks sentinel recovery, rejection boundaries, signed off-solution residuals, and structural gate counts; and streams all gates of the full empty-word family and outer wrapper. The raw recoder checker tests 1016 word/radix residue cases, 38 full binomial-extraction cases, 5766 pair-orientation

cases and two tiny explicitly constructed Pell examples. Most residue tests use smaller power-of-two radices meeting the proved margins rather than materializing the deterministic enormous B .

The all-length conjunction and its single-polynomial conversion are streamed as actual source DAGs, not inferred only by adding printed counts. An exact sparse-polynomial expansion independently checks all six standalone recoder degree and sum-of-squares claims; for the much larger initializer, full-stream degree propagation plus its two exact leading monomials establishes the degree claim. Separate independent reviews compare the equations, source pins, domains and theorem specialization. These checks are evidence of implementation agreement with the mathematical construction. They do not replace the universal proofs, instantiate the giant parent history witnesses, simulate the giant physical ant controller, expand the strict constant prefix, or formally verify the Lean source in a proof assistant.

[write] *In this repository*. The archive is shipped flattened with the prefix `03-ant-init-` (Markdown at the report root, Python in `code/`, the rest in `data/`). Not shipped: the three copies of the parent history note (`HISTORY174.md`, byte-identical to `Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md`, unchanged since the pin), the Mathlib file and licence (note in Section 42.4), and the files that duplicate Part II's (shipped once, under `02-literal-ant-`). The colour-map projections of Part II's large templates (`science/recipe_assets/`) are shipped; they are not copies. The delivered verifier refuses NTFS file modes; the report README gives a rerun route on a copy. At placement the whole replay passed on a copy, including the byte-identical regeneration of the dilation evidence.

42.11.2 Replay discipline

An external standard-library inventory check authenticates the release before its own verifier is invoked. The verifier again authenticates the complete inventory before executing any scientific script. Scientific replay takes place in new isolated copies; source bytes, modes and timestamps are checked for preservation. Normal Python replay must reproduce the reference scientific outputs. The recoder checker uses explicit guarded exceptions and also passes optimized Python with all checks active. The six assertion-dependent bridge, splice, degree and folding programs explicitly reject optimized Python before scientific work or outputs; those invocations are checked for that rejection rather than counted as scientific passes. All outputs, logs and TeX caches belong in external QA directories, outside the release tree.

The PDF builder uses three no-shell-escape pdfLaTeX passes with a fixed source epoch. Rebuilding in a fresh external directory must reproduce the packaged PDF byte for byte on the recorded TeX toolchain. The ZIP builder uses a fixed member order, timestamps and modes; its external SHA256 authenticates the archive. The inventory hash itself must be obtained from the enclosing delivery or another trusted channel: a self-consistent replacement of both payload and its checksum is not prevented by an unsigned manifest.

42.11.3 Scope of the completed result

The fixed all-length raw initialization obligation has been closed at the stated component boundary. The resulting bound is intentionally literal and unoptimized. Practical evaluation is not promised, source constants are not materialized, full witness uniqueness is false, and complete universal-certificate bookkeeping remains separate. These distinctions preserve the useful conclusion: ordinary binary sentinel inputs can now be connected to this exact fixed periodic ant initialization by one paid finite polynomial system.

42.12 Concrete further research

The largest present cost is the pair of 576000-row background Horner chains. The literal colour generator has considerable repeated geometry, so a useful next target is a smaller fixed arithmetic description of its coefficient polynomial, with every compression operation and any extra witness charged. A recipe alone does not constitute such a compression.

A second target is a completely merged history, initialization and endpoint source with one consistent port convention, a total positive-witness ledger, fixed-numeral accounting and a paid final equation-combination step. The qualitative corollary already identifies the intended projection, but a numerical bound needs that additional source work.

The raw recoder offers a third concrete optimization problem: reduce its fourteen power calls, witness burden or residue machinery while retaining both orientations, zero padding and all converse bounds. Any such improvement needs its own exact source and independent review; no easy route to single-fold exponentiation is assumed. Finally, formalizing the newly proved recoder and splice in a proof assistant would narrow the gap between source-level arithmetic audits and a machine-checked end-to-end theorem.

42.A Provenance and source identity

The archive’s machine-readable dependency manifest gives full hashes, source locations and read-only status. The principal scientific pins are:

- Parent history note, SHA256 9c26b118aa28f8454204be6fb6f751beeae713a752d18b75943e9862e7de1656
- Literal interface proof, SHA256 a8b4a65ad97a06f3c854a9d133e67dca240d6212749f42ed6c4a92b30318483b
- Frozen 2806-cell anchor JSON, SHA256 cc67f7c924bcb27ca1c4a48c8d3a630d837b639fbff3032d04515ec1d534ed68
- Mathlib Pell source at commit ac77769fabe23cb237559e7f56578dbead91499f, SHA256 993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a

The mathlib file is 39986 bytes; the two central theorem declarations begin at lines 760 and 860. The source header attributes Mario Carneiro and the Apache 2.0 license. Its proof follows the constructive Pell approach associated with Davis. This report claims an explicit specialization and a new paid composition, not discovery of Diophantine exponentiation or a replacement proof of those source theorems.

[write] The parent history note is unchanged at the write (SHA-256 9c26b118... as above); its bibliography entry gives the pinned URL. The literal-interface proof and the anchor JSON are Part II’s 02-literal-ant-literal-interface-atlas-PROOF.md and data/02-literal-ant-literal-interface-copy-fixed_initial_anchor_patch.json.

References

- [1] V. Reshetnikov, *A counted 174-operation bounded ant history*, pinned repository note, commit 5883b08b7af362077f13bb4afa97a23a90ae4cf8. <https://github.com/VladimirReshetnikov>

ikov/ProveIt/blob/5883b08b7af362077f13bb4afa97a23a90ae4cf8/Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md. [write] Unchanged at the write.

- [2] M. Carneiro and the mathlib contributors, *Pell's equation and Matiyasevic's theorem*, Mathlib/NumberTheory/PellMatiyasevic.lean, commit ac77769fabe23cb237559e7f56578dbead91499f; in particular Pell.matiyasevic and Pell.eq_pow_of_pell. <https://github.com/leanprover-community/mathlib4/blob/ac77769fabe23cb237559e7f56578dbead91499f/Mathlib/NumberTheory/PellMatiyasevic.lean>.
- [3] *A Literal Periodic Langton Ant Interface and the One Visit Two Visit Boundary*, Research Report 40, 3 October 2026. Delivered PDF SHA256 32c7de67c5253fd21268ecac5cb6d351b64cdca771663f59f7d40996368e5f39. Prior report supplied with this research series; no public repository location is assumed. [write] Printed as Part II of this report.
- [4] *First Revisit Detection and Exact Pattern Queries for Finite Defect Periodic Turmites*, Research Report 38, 3 October 2026. Delivered PDF SHA256 21879fba45f02483ac7d16abe1130ed5d4e0baa13ec74e04800a1e2a4bc47c1b. Prior report supplied with this research series. [write] Printed as Part I of this report.

Report 44: the complete positive certificate

[write] *A Complete Positive Certificate for the Literal Periodic Ant. A composition addendum to Research Reports 40 and 42*, Research Report 44, recovered edition, dated 4 October 2026, author line “Mathematical research report” (its README title: “Research Report 44: complete positive literal-ant certificate, recovered edition”); batch-88 manuscript 07, archive `Complete_Positive_Certificate_for_the_Literal_Periodic_Ant_Package.zip`. Pins: ProveIt 5883b08b7 (the history note and its base-three and Pell dependencies), Mathlib ac77769f, the Neary–Woods PDF and the endpoint source (Appendix 44.A). Numbered 44.k; labels `ptr:pc::`; files `04-ant-cert-`. Its edition note, after the abstract, is the manuscript’s own.

Abstract (Report 44). We close the arithmetic composition left open in Research Report 42. One fixed integer polynomial recognizes the exact raw-pair halting language of the pinned Neary–Woods U15 machine through the initialized literal periodic ant. Its source quantifies all five formerly external positive history parameters, retains a reconstructed 174-operation bounded-history component, includes the 2306387-operation initializer and 42-operation endpoint, and pays for the final sum of squares. The result has 465 strictly positive witnesses, 2307457 operations, and exact total degree 2304000. A separately paid positive Cantor pairing gives one ordinary positive input, 467 witnesses, and 2307467 operations. The leading homogeneous part is exactly $2W^{2304000}$. We specify the complete literal source, domains, port identifications, coefficients and reproducible checks. The unexpanded literal-1-and-3 variant costs 554386261710212588 operations for two inputs, ten more for one. These conservative counts are not operation records. The positive-history, base-three/Pell, recoder and literal-interface theorems remain explicit dependencies; finite checks do not replace them. No empty-right universality, physical ant cessation or single-fold representation is asserted.

Edition note. The pre-delivery filesystem reset lost the previously verified Report 44 files. This edition reconstructs the manuscript and package from retained work and authenticated earlier reports. It has a new release identity and fresh checks; an earlier PASS receipt is historical evidence, not authentication of reconstructed bytes. The corrected geometric-repunit identity for h_x is retained.

44.1 The exact relation and theorem

For a nearest-head-first finite binary word $d = (d_0, \dots, d_{n-1})$, put

$$\text{sent}(d) = 2^n + \sum_{j=0}^{n-1} d_j 2^{n-1-j}. \quad (44.1)$$

Every positive integer has a unique decoding. The empty word has code 1; explicit zero padding is retained. For $x, y \in \mathbb{Z}_{>0}$, let $H(x, y)$ mean that the pinned U15 machine starts in state A scanning blank 0, with the words decoded from x, y on its left and right respectively, blank elsewhere, and eventually reaches the undefined pair J1. Both side words are nearest-head-first. The table’s J0 instruction is defined; we use that table rather than the contradictory closing sentence of the original simulation discussion [1].

[write] The “contradictory closing sentence” is the last sentence of Section 3.5 of [1], which names c (here 0) in the halting transition, whereas Table 16 leaves (u_{10}, b) , here J1, undefined. Report 40 and the programme’s review make the same observation (note in Section 40.5).

Theorem 44.1.1 (Complete fixed polynomial). *Subject to the inherited theorems identified in Section 44.3, the accompanying literal source defines a fixed polynomial*

$$F_2 \in \mathbb{Z}[x, y, z_1, \dots, z_{465}]$$

such that, for every $x, y > 0$,

$$H(x, y) \iff \exists z_1, \dots, z_{465} \in \mathbb{Z}_{>0} \quad F_2(x, y, z) = 0. \quad (44.2)$$

Its displayed arithmetic source has one asserted equation, exactly 1153586 multiplications and 1153871 additions/subtractions, hence 2307457 operations. The polynomial has exact total degree 2304000. There are no external geometry, board, final-memory, head or sign parameters.

There is also a fixed polynomial $F_1 \in \mathbb{Z}[N, z_1, \dots, z_{467}]$ whose displayed source has 2307467 operations and whose exact degree is the same. For every $N > 0$, it has a positive zero exactly when $H(x, y)$ holds for the unique positive pair satisfying

$$2N = (x + y - 2)(x + y - 1) + 2y. \quad (44.3)$$

An operation is a binary addition, subtraction or multiplication of available expressions. Fixed integer coefficients have degree zero; multiplication by one is still charged. Gate values are expressions, not new quantified coordinates. Storing an equality pair costs no arithmetic, but forming its residual and the final sum of squares is paid. The theorem recognizes an accepting ant observable representing U15 halting. The physical ant continues moving after that event.

[write] One polynomial, successive sources. Theorem 44.1.1 is the recognition theorem of Parts III and IV; it is printed only here. Reports 47 and 48 (Part IV) construct the *same* polynomials F_2 and F_1 , equal at every integer assignment, with the same 465 and 467 positive witnesses, one equation and exact degree 2,304,000; what shrinks is the arithmetic source when every fixed coefficient must be built from the literals 1 and 3. All counts below stay valid for their own grammars; none is an erratum of another.

Source and coefficient convention	F_2 (two inputs)	F_1 (one input)	Where
Report 44, prescribed fixed coefficients	2,307,457	2,307,467	Thm. 44.1.1
Report 44, literals 1 and 3, dense Horner prefix	554,386,261,710,212,588	554,386,261,710,212,598	§44.9
Report 47, literals 1 and 3, structured coefficients	31,388,831	31,388,841	Thm. 47.1.1
Report 48, literals 1 and 3, fused background	14,658,934	14,658,944	Thm. 48.1.1
Programme, 5d7e884a5: Report 48 with the endpoint reusing the paid $W^{576000}(-9)$ and six shared rotation blocks $(-38,146)$	14,620,779	14,620,789	[W3]
Programme, 57a6afbeb: plus a zero-folding and common-subexpression cleanup (-68)	14,620,711	14,620,721	[W4]

The same endpoint reuse lowers the prescribed-coefficient source to 2,307,448 and 2,307,458 operations (`periodic_ant_endpoint_reuse48.md`). The programme’s counts are inherited-plus-delta: the changed components were emitted and checked, but no complete multi-million-row stream was regenerated or hashed. None of these numbers bears on the programme’s 84-operation universal bound (preface to the four Parts).

[write] What the theorem recognizes. The predicate H is the halting language of one fixed machine on sentinel-pair inputs. It is recursively enumerable and complete only through the published Neary–Woods encoding of arbitrary programs as U15 tapes, which no Part implements. In the words of the programme’s review [W1], the certificate is “an ordinary-integer Diophantine recognizer of a fixed recursively enumerable complete language”, “not, by itself, an emitted per-program map from the research tree’s ordinary input variable into the published U15 tape encoding”. It is therefore not an ordinary-input loader for the programme’s input convention, and it is not called a universal polynomial here except with this qualification.

The new contribution is the complete source-level join. It does not repeat Report 42’s long recoder proof [3], replace Report 40’s literal simulation theorem [2], or establish a new qualitative turmite universality theorem. The arbitrary-program-to-U15 reduction remains the published dependency [1].

44.2 Every positive coordinate and shared port

All raw inputs and quantified coordinates range over $\mathbb{Z}_{>0}$. Intermediate expressions may be zero or negative. Subtracting one from a positive adapter produces a nonnegative expression, not another positive input. The complete ordered name list is in each final source receipt; this partition is disjoint and exhaustive.

Positive witness family	Count
Former history parameters, prefix Parent :	5
Bounded-history witnesses, prefix History :	61
Initializer geometry witnesses, prefix Init :	4
Recoder witnesses, prefix Init:Recoder :	392
Endpoint witnesses, prefix Endpoint :	3
Total for F_2	465
Additional internalized x, y in F_1	2

The five parent coordinates are exactly

InitialMemoryPlus, FinalMemoryPlus, InitialHead, FinalHead, FinalSignPlus.

They remain independent positive quantified variables. Their raw memory and sign values are the charged expressions InitialMemoryPlus – 1, FinalMemoryPlus – 1, FinalSignPlus – 1. They are not eliminated using conclusions of the components being joined.

The 61 history witnesses, with the **History**: prefix suppressed, are:

- Thirteen adapters: **APlus**, **BPlus**, **GEPlus**, **GWPlus**, **GNPlus**, **GSPlus**, **DPlus**, **EPlus**, **FPlus**, **GPlus**, **ZPlus**, **SWPlus**, **SNPlus**

- Thirteen geometry coordinates: `q, Q, W, uQ, uW, Gt, Gy, WidthOdd, HeightEven, K, Wp, HeadRoot, HeadQuot`
- Eighteen bounds: `BoundA, BoundB, BoundC, BoundD, BoundE, BoundF, BoundG, BoundGE, BoundGW, BoundGN, BoundGS, BoundTestE, BoundTestW, BoundTestN, BoundTestS, BoundZ, BoundInitial, BoundHead`
- Seventeen Pell coordinates: `a, c, d, f, h, i, j, k, o, r, s, w, tau, eta, zeta, ga, y_aux`

Case distinguishes geometric W from Pell w , and geometric K from Pell k . Computed Pell discriminants are not extra inputs and are distinct from raw history field D .

The initializer has four geometry coordinates:

`HorizontalExtra, HalfRowsPower,
HalfPeriodRepunit, PaddingPower.`

The endpoint has `HxPlus, HyPlus, BoundCol`. The recoder coordinates are the two frozen 196-coordinate modules copied with disjoint left/right names from `data/pair_inline-dag.json`. All 392 names appear in the receipts' `positive_witnesses`; renaming is not inferred from string similarity.

Consumer port	Exact supplier
Initializer W, W_p, Q	<code>History:W, History:Wp, History:Q</code>
Initializer initial memory/head	<code>Parent:InitialMemoryPlus, Parent:InitialHead</code>
Endpoint W	<code>History:W</code>
Endpoint final head/sign	<code>Parent:FinalHead, Parent:FinalSignPlus</code>
Recoder G	Initializer expression W^{576000}
Recoder raw sides	External x, y , or their positive witnesses in F_1

The recoder outputs A, B, T alias already available expressions or coordinates and add no quantified ports. In particular T can be zero and is never required positive. Both initializer and endpoint use the same independent history W . The endpoint conservatively retains its own frozen 32-multiplication joint power chain rather than sharing the initializer's power and claiming a further saving.

44.3 Inherited contracts and proof boundary

44.3.1 Positive bounded histories

The reconstructed history has 174 explicit operations, 48 equations, five positive parameters and 61 positive witnesses. For arbitrary positive parameters, the equations are solvable exactly when there exist odd $w \geq 3$, even $h \geq 2$ and $t \geq 1$ describing an ordinary no-escape ant history on a w -by- h rectangle. On solutions,

$$W = 3^w, \quad W_p = W/3, \quad Q = W^h, \quad q = Q^t. \quad (44.4)$$

The two memory parameters minus one are the exact initial/final ternary Boolean board words. Sites flatten as $x + wy$, east and south positive. The initial head is 3^j at an even index j in the rectangle, with pre-departure heading north. The final head is its actual last head. The value

FinalSignPlus $- 1$ is zero or FinalHead and encodes incoming heading: at an even-index black site zero/one means north/south; at a white site it means east/west. Booleanity, power, range and actual endpoint status are conclusions, not external promises.

The all-integer theorem is inherited from the pinned history and base-three/Pell sources [4]. The new reconstruction compares the actual DAG with separately stated residual polynomials. At zero-based residual 46,

$$R_{46} = S_{46} + S_{45}((2r + 1 + jc)^2 - y_{\text{aux}}^2), \quad R_{45} = S_{45}.$$

This checked unit-triangular replacement preserves the simultaneous zero set over the integers and does not alter witness domains. It is not silently identified with literal equality of the unmodified residuals.

[write] The programme's note states residual 46 literally; this reconstruction replaces it by a unit-triangular combination with the same zero set, and says so. The programme's review checked the 174-row DAG's closure, count and degree, not the history theorem [W1].

44.3.2 No circular power or positivity premise

The inherited proof first uses positive geometry quotients and strict slacks to bound the packed fields, before interpreting digits. The fields are nonnegative; fifteen Horner steps pack sixteen fields with $0 \leq P < q^{16}$. The base-three kernel is applied to kernel input q^4 , whose fourth-power bound is q^{16} . Four squarings construct the latter, and $D_0 = 9q^{16}$, $r = D_0 - 3P - 1$ are charged expressions.

The kernel derives the power-of-three property. Unique factorization and divisibility/repunit equations then give (44.4). Its positive converse requires even packed P : the local parity identity, including the repeated terminal D field, establishes that condition before invoking the Pell construction. Negative intermediate Pell values are permitted. There is no freely imposed upper bound on Pell intermediates.

The precise dependencies are standard integer Pell classification, recurrence and divisibility; the pinned relaxed auxiliary Pell proof; the half-parameter and signed step-down arguments with their required bounds; the base-three kernel's own ratio, congruence and rounding lemmas; and the ternary carry criterion from Kummer's theorem. Their proof sources and exact scope are recorded in `history/CONTRACT.md`. The recoder separately uses the constructive `Pell.matiyasevic` and `Pell.eq_pow_of_pell` specialization [5]. These remain inherited mathematics, not newly mechanized proofs.

44.3.3 Literal ant and initialization

Report 40 supplies the periodic color recipe, 2806-cell startup anchor, literal two-sided loader and accepting-port equivalence. Report 42 supplies the paid all-length recoder, exact board formula and finite-prefix padding lemma. The initializer's normalized local arithmetic/equation hash is the frozen Report 42 hash; its scientific equations are unchanged by this composition.

44.4 Initialized board and endpoint

Set

$$u = 481238074400, \quad v = 576000, \quad x_0 = 481225262775, \quad y_0 = 29948.$$

Physical coordinates (a, b) increase east and south. The frozen start is $(288650, 75, E)$. Normalize and translate by

$$X = b - 75, \quad Y = 288650 - a, \quad x = X + u, \quad y = Y + kv. \quad (44.5)$$

The rotation preserves orientation and sends the initial east heading to north. Permitted rectangles are

$$w = 1 + au \quad (a \geq 2), \quad h = 2kv \quad (k \geq L + R + 1), \quad (44.6)$$

where L, R are the unique decoded word lengths. Since u is even, w is odd. The initial head is $3^u W^{kv}$ with even flattened index. The padding theorem covers both vertical extremes and uses the literal interface's global lower horizontal-support bound.

The recoder takes $G = W^v$ and provides

$$A = G^{L+R}, \quad B = G^R, \quad T = G^{R+1} \sum_{j < L} \ell_j G^j + \sum_{j < R} r_j G^{R-1-j}. \quad (44.7)$$

Only after history establishes $W = 3^w$ is the recoder premise $G \geq 2$ used. Its forward/reverse outputs preserve orientation and zero padding. Six remaining initializer equations enforce (44.6), the head and the periodic background plus anchor and input patch. This is an actual board equality, not a promise that the raw input already represents a suitable board. Tight padding permits `PaddingPower=1`.

In the compact outer formula the positive geometry witnesses are X_g, H, K, P . It computes $h_x = X_g + 1$ and imposes

$$W_p - 1 = (3^u - 1)h_x, \quad H - 1 = (G - 1)K, \quad Q = H^2, \quad H = PAG, \quad \text{InitialHead} = 3^u H.$$

The sixth equation equates `InitialMemoryPlus` with one plus the exact background-and-patch polynomial. All literal Horner and patch expressions are retained; no coefficient oracle is free. On solutions $H = W^{kv}$ and

$$h_x = \frac{3^{au} - 1}{3^u - 1} = 1 + 3^u + \dots + 3^{(a-1)u}.$$

Thus h_x is the geometric repunit for a repeats, not their count. Also $K = (G^k - 1)/(G - 1)$ and $P = G^{k-L-R-1}$. The source's background factor $K(H + 1) = (Q - 1)/(G - 1)$ repeats the v -row tile $2k$ times. The recoder contributes 228 equations and the outer formula six, totaling 234.

The folded endpoint [6] uses prescribed constants

[write] This is the folded form of Report 40's endpoint selector, Theorem 40.D.1 in Appendix 40.D (Part II), with $H_x^+, H_y^+, B_c, K_e, V_e, A_c$ for that appendix's X_+, Y_+, B, K, V_*, P .

$$C = 3^u - 1, \quad D = C - 1, \quad K_e = 3^{x_0}$$

and positive witnesses H_x^+, H_y^+, B_c . Its expressions and equations are

$$\begin{aligned} U &= CH_x^+ - D = C(H_x^+ - 1) + 1 \geq 1, \\ V_e &= 1 + (W^v - 1)(H_y^+ - 1) \geq 1, \end{aligned} \quad (44.8)$$

$$A_c = K_e U, \quad A_c + B_c = W, \quad \text{FinalHead} = (A_c W^{y_0}) V_e, \quad \text{FinalSignPlus} = 1. \quad (44.9)$$

The strict column bound prevents a carry into the row. With the history's actual final-head power, the endpoint theorem selects exactly residues $(x_0, y_0) \bmod (u, v)$ and white/east heading. The zero quotients are retained by $H_x^+ = 1$ or $H_y^+ = 1$. Translation (44.5) preserves these residues.

44.5 Proof of the complete equivalence

Soundness of Theorem 44.1.1. A zero of the final polynomial makes every integer square in the sum of squares zero. All three component systems therefore hold at the same positive coordinates. Apply history first: it gives a real positive-length trajectory, powers (44.4), a north-facing even-index initial head, exact initial/final memories, and no escape.

Now $G = W^v \geq 2$, so the recoder identifies the unique words and outputs (44.7). Initialization identifies the exact translated restriction of the infinite literal ant board, including anchor and finite input patch. Induction identifies the no-escape trajectory with the same infinite literal trajectory prefix: cell color and heading agree, and the turn/write/move rule uniquely fixes each step.

Equations (44.8)–(44.9), with history’s actual final head and sign, give the accepting residue and orientation, unchanged by translation by (u, kv) . The literal-interface theorem yields U15’s J1 event. An existential final memory is not a loophole: history already forces it to be the actual final Boolean board. Thus $\exists z > 0 \ F_2(x, y, z) = 0 \Rightarrow H(x, y)$. $\square$

Completeness. Suppose the decoded U15 input reaches J1. The literal-interface theorem supplies an accepting event after a positive finite number of ant steps and excludes it at initialization. Its finite prefix obeys the global lower horizontal-support bound. Choose $a \geq 2$ and $k \geq L + R + 1$ large enough to contain the entire translated prefix in (44.6). This is Report 42’s two-sided padding statement, not an extra containment assumption.

Initialization supplies its 392 recoder and four positive geometry coordinates. Take InitialMemoryPlus, FinalMemoryPlus to be one plus the actual nonnegative initial/final Boolean board words, the two head parameters to be their positive powers of three, and FinalSignPlus = 1 at the accepting endpoint. These five parameters are positive. The all-positive history converse supplies 61 witnesses, including the Pell tuple and its required packed-word parity. The endpoint converse supplies three positive witnesses, including quotient-zero cases.

Every shared port has the same numerical value in all components. Thus the 285 equations hold simultaneously without a hidden compatibility variable, raw coordinate or uncounted nonnegative adapter. Their paid sum of squares vanishes. Section 44.8 gives the one-input assertion. $\square$

44.6 Literal arithmetic source and complete ledger

The source is a topological stream. External inputs are declared; each positive witness is declared before use. Arithmetic records have an integer output ID, an opcode in $\{+, -, *\}$ and two operands. Integer operands are earlier gate IDs. Strings are positive-coordinate names or prescribed **C**: coefficient labels. Fixed integers in component JSON are translated to coefficient labels before insertion, avoiding confusion with gate IDs.

Residual pairs are metadata, not additional assertions. For every stored pair (a_i, b_i) , the final source computes

$$r_i = a_i - b_i, \quad s_i = r_i r_i, \quad F_2 = \sum_{i=0}^{284} s_i,$$

and asserts only $F_2 = 0$. Every subtraction is paid, even if a simplifier could remove it; each square costs one multiplication and summing the 285 squares costs 284 additions.

Source section	M	A	Total	Pairs	New witnesses
History	72	102	174	48	61
Initialization	1153192	1153195	2306387	234	396
Endpoint	37	5	42	3	3
Conjunction	1153301	1153302	2306603	285	460
Final SOS	285	569	854	—	0
Complete F_2	1153586	1153871	2307457	1 asserted	465

The final witness count includes five parent coordinates in addition to the 460 new component witnesses. The zero-based half-open gate intervals are $[0, 174)$ for history, $[174, 2306561)$ for initialization, $[2306561, 2306603)$ for endpoint, and $[2306603, 2307457)$ for SOS. Gate 2307456 is the final output. These are streamed intervals, not simply sums of printed component totals.

The initializer’s normalized local gate/equation SHA256 is

503215b20d317cfb9e6e40ec82eb26044afbdedf26f37e03029bcd7dc0a04130.

It reproduces the frozen Report 42 component. Complete stream hashes include ordered declarations, gates, stored residual pairs and the sole final equation. They differ from the implementing Python file hash. An optional emitter writes every ordinary free-coefficient gate as JSON lines; default replay generates, validates, counts, hashes and degree-checks the same gates without saving the large stream.

44.7 Exact total degree

Degree means total degree in raw inputs and positive quantified coordinates, with prescribed coefficients degree zero. In particular **History:W** remains an independent degree-one variable; it is not replaced by a power expression.

Literal propagation gives history residual degree at most 104 and endpoint degree at most $v + y_0 + 2 = 605950$. The pairing residual has degree two. Exactly two initializer residuals attain degree bound $2v = 1152000$: local zero-based indices 16 and 130, globally 64 and 178 after the 48 history pairs. Every other residual has bound at most 1127949.

Proposition 44.7.1. *The leading homogeneous part of each final polynomial is exactly*

$$2W^{4v} = 2W^{2304000}.$$

Both therefore have exact total degree 2304000.

Proof. The maximal residuals are the first left/right recoder power-modulus equations. Each has form

$$2aG - (\Theta + G^2 + 1), \quad G = W^v,$$

where a is a positive witness plus one and Θ an independent positive witness. The aG term has degree $v + 1 < 2v$, so the unique top homogeneous term is $-W^{2v}$. Each square contributes W^{4v} ; no other squared residual can reach that degree. Their sum has the nonzero leading coefficient 2. $\square$

This is a surviving-term proof of exactness, not only cancellation-blind propagation. Sharing elsewhere cannot remove those terms. The coefficient-only prefix and degree-two pairing cannot change the leader. No minimum-degree claim follows.

44.8 One ordinary input, with pairing paid

Let $n = x + y - 2 \geq 0$. Equation (44.3) is equivalent to

$$N = \frac{n(n+1)}{2} + y, \quad 1 \leq y \leq n+1.$$

For each n it covers exactly the interval

$$\frac{n(n+1)}{2} + 1 \leq N \leq \frac{(n+1)(n+2)}{2}.$$

These intervals are disjoint, consecutive and exhaust $\mathbb{Z}_{>0}$. The interval containing N determines n , then y , then $x = n + 2 - y \geq 1$. This proves a bijection without a square root, floor, division gate or pairing oracle.

The source internalizes **RawLeft**, **RawRight** as positive witnesses, leaving only **RawInput** external. Its seven gates compute

$$s = x + y, \quad s_1 = s - 1, \quad s_2 = s_1 - 1, \quad p = s_2 s_1, \quad y_2 = 2y, \quad b = p + y_2, \quad a = 2N$$

and store pair (a, b) . This is $3M + 4A$, one residual and two additional witnesses. The enlarged 286-residual SOS costs $286M + 571A$, an extra $1M + 2A$. Hence the entire arity reduction costs $4M + 6A = 10$ operations. The totals are

$$1153590M + 1153877A = 2307467, \quad 467 \text{ positive witnesses.}$$

44.8.1 Why an empty-right shortcut is unsupported

Neary–Woods’ actual initialization puts the dataword to the right of the scanned symbol. Their tape has form

$$\cdots cc \langle B \rangle S^k b [c] D_B(w) cc \cdots,$$

where brackets mark the scanned cell, initial state is u_1 , and the U15 guard is $G = bc$; see Definition 3.1, equations (3)–(4), Table 1 and Section 3.5/Table 16 [1]. Renaming $c \mapsto 0$, $b \mapsto 1$, $u_1 \mapsto A$ yields

$$\ell = \text{reverse}(\beta(\langle B \rangle S^k b)), \quad r = \beta(D_B(w)).$$

The data encodings begin cb , so primary right words begin 01. Their second cell right of the head is 1, whereas an empty-right tape is blank there. The rightmost symbol of G is not the rightmost symbol of the whole finite input.

Therefore **RawRight**=1 does not establish a universal restricted family. We do not assert that family is non-universal; the required argument is absent here. Fixing a universal program permits a fixed-left, varying-right inference, but a numerical certificate would still need its literal fixed data and an explicit paid construction. The pairing avoids both unsupported shortcuts. **one-input/AUDIT.md** records the exact orientation and the J1/table-versus-prose issue.

44.9 Every coefficient from literals 1 and 3

The economical ledger admits prescribed fixed integer coefficients. A separate deliberately unoptimized ledger constructs all coefficient labels by one finite topological prefix whose only initial

literals are 1 and 3. Its enormous expansion is specified, never materialized or executed. Fixed loop endpoints are source-generation parameters, not run-time arithmetic inputs.

For $e \geq 2$, define $\lambda(e) = \lfloor \log_2 e \rfloor + \text{popcount}(e) - 1$, with $\lambda(0) = \lambda(1) = 0$. A power 3^e uses the left-to-right binary multiplication chain: one square per remaining digit, one multiplication by 3 per remaining one bit. The full grammar is:

1. Compute $0 = 1 - 1$, $-1 = 0 - 1$, $2 = 1 + 1$ (three additions/subtractions).
2. Independently construct 3^u , 3^{u-219} and 3^{198} by binary chains, then $C = 3^u - 1$.
3. For $0 \leq j < v$ prescribe $b(i, j) = c(288650 - j, i + 75)$, $0 \leq i < u$, from the pinned total literal color recipe. Construct each row by ternary Horner evaluation of $b(u-1, j), \dots, b(0, j)$, retaining every multiplication and addition, including zero bits. Each row costs $(u-1)M + (u-1)A$. The first-column coefficient aliases $b(0, j)$, hence 0 or 1.
4. For $0 \leq j \leq 583$, prescribe signed anchor corrections $\delta(i, j) \in \{-1, 0, 1\}$ at physical $(289200 - j, i - 144)$ for $0 \leq i \leq 220$. Absent cells have zero correction. Every 221-symbol Horner chain costs $220M + 220A$; signed symbols alias available $-1, 0, 1$.
5. Compute $4 = 2 + 2$, $6 = 3 + 3$, $8 = 4 + 4$, $9 = 3 \cdot 3$. Independently construct $K_e = 3^{x_0}$ and then $D = C - 1$.

The bits are fixed data, not existential coefficients or a variable-dependent color oracle. The compact recipe read closure is hash-pinned under `dependencies/recipe_assets` and never executed by replay. The 2806-cell anchor remains fixed data. Different coefficient labels may denote equal integers; their number is not a count of distinct numerical values.

Steps 1–4 are the Report 42 prefix:

$$\begin{aligned} M_0 &= v(u-1) + 584 \cdot 220 + \lambda(u) + \lambda(u-219) + \lambda(198) \\ &= 277193130853952587, \\ A_0 &= v(u-1) + 584 \cdot 220 + 4 = 277193130853952484. \end{aligned}$$

Step 5 adds 56 multiplications and four additions/subtractions. Thus

$$\begin{aligned} M_{\text{prefix}} &= 277193130853952643, \\ A_{\text{prefix}} &= 277193130853952488, \\ C_{\text{prefix}} &= 554386261707905131. \end{aligned}$$

The history literal 6 and final equation's zero are paid. Initializer and endpoint share C rather than constructing it twice. Main-source multiplications by these constants remain charged. Complete strict totals are

Two-input F_2	554386261710212588
One-input F_1	554386261710212598

The prefix adds no witnesses, equations or variable degree. These totals describe this exact finite grammar, not a practical expansion or optimized literal-source bound.

44.10 Verification, recovery and limits

The reconstructed packet carries the exact arithmetic source, small component specifications, positive-name receipts, dependency contracts, fixed coefficient read closure and fresh verification evidence. Authenticated Reports 40 and 42 remain unchanged. Upstream Python/Lean, physical ant programs, recipe code and saved upstream schedules are read-only provenance; only inspected newly authored reconstruction/audit programs and normal build/archive utilities are executed.

History checks compare all 48 residuals with separately stated integer polynomials, validate each operand and declaration, check the triangular correction and exact residual degree 104. A fresh finite regression examines 768 combinations of Boolean 3-by-2 boards, even starts and one through four requested steps. Exactly 264 prefixes remain on the board. It checks 38 outer-plus-mask equations, positive outer witnesses, Boolean/even packing, strict bounds and the exact ternary central-binomial valuation. Ten mutations are rejected. It does not instantiate an enormous Pell tuple or re-prove the arbitrary-integer theorem.

Complete replay checks every generated gate, declarations, all 285/286 residual pairs, the sole final assertion, hashes, operation counts and degree bounds. Fresh source reviews and the independent checks recorded in the recovered packet are limited to their stated evidence. In particular a recovered old PASS record does not certify a newly generated program merely because its counts agree. Exact stream identity, when established by a matching full-stream hash, is reported separately from implementation-file identity. Implementation, manuscript, PDF and archive hashes are freshly recorded.

[write] *In this repository.* The archive is shipped flattened with the prefix `04-ant-cert-`. Not shipped: the eight research-tree files under `certificate/history/source/` (byte copies of the notes in `Computability/HilbertTenthProblem/Papers/1980/` and of two `Papers/verification/explore_*.py` scripts, cited by path), the Mathlib file and licence (note in Section 42.4), and the files that duplicate Reports 40 and 42 (shipped once, under `02-literal-ant-` and `03-ant-init-`). At placement the complete replay passed on a copy: both complete streams reproduced the hashes of Appendix 44.A, and the history reconstruction and audit reproduced their delivered outputs byte for byte in normal and optimized Python. The delivered verifier refuses NTFS file modes; the report README gives a rerun route on a copy.

44.10.1 Authenticated and isolated reproduction

The delivery supplies the new `MANIFEST.json` SHA256. Before executing bundled code, a standard-library bootstrap must compare that trusted value, authenticate every entry and reject extra files/directories, symlinks, unsafe paths and duplicate keys. The bundled verifier repeats this authentication. A self-consistent replacement of payload and manifest is not prevented by an unsigned manifest without an external trusted hash.

Scientific replay occurs in fresh external copies. The complete generator requires fresh outputs outside its resolved source tree and validates both receipt/stream destinations before opening either. Duplicate destinations, symlinked parents, existing files and dangling symlinks are checked. Original release bytes, modes and timestamps are checked for preservation.

The complete generator and packet verifier reject optimized Python before scientific work because they contain assertions; the boundary checker imports that same guarded generator. These early rejections are guard tests, not optimized scientific passes. History reconstruction and finite-history checking use explicit exceptions and reproduce their reference outputs under both normal Python and `-O`. Main replay uses Python 3.11+ and standard library; exact history checks also need SymPy. No inherited source or recipe directory is executed.

The PDF uses three external no-shell-escape pdfLaTeX passes, fixed UTC source epoch and suppressed dates/trailer ID. Fresh builds must agree byte for byte on the recorded toolchain. Every final page is rendered and visually reviewed. ZIP member order, timestamp, mode and compression settings are fixed; fresh archive builds must agree. All logs and caches stay outside the sealed release. The README gives exact commands and the final binding identifies the freshly checked recovered bytes.

44.10.2 Scope of the completed result

This is one fixed positive-witness Diophantine certificate for the specified raw U15/initialized-ant interface, with aggregation paid. It is not a universal operation record, efficient decision procedure, physical ant cessation, witness uniqueness or a finite-fold theorem. Report 42’s Pell fibers need not be unique. The enormous coefficient prefix is unexpanded. End-to-end formalization and reductions in background Horner cost, witness burden or endpoint sharing would require separate proofs, exact sources and audits.

44.A Source identity and reading guide

Fresh generation of both recovered complete free-coefficient streams reproduces the previously audited SHA256 identities:

- Two inputs: `c16901162e09b07d1e0c27b28e29dcc12a9b6137391fce85e3f150a3fed37eac`
- One input: `85a161972769207e4f5d4212214630535208f69d6736bca2b46a8aff8276a22a`

Both final receipts verify those exact matches. A source-stream identity is not a Python implementation-file or release identity. The recovered history JSON also reproduces SHA256 `2075bb290f3f81d7c04a27b82f50c4f492638d55a3a27f434f66497ae9fb90ea`; reconstructed implementation, manuscript and package files have fresh identities recorded in the new inventory.

In `certificate/`, begin with `PROOF.md`, `history/CONTRACT.md` and `STRICT_GRAMMAR.md`. Both `*-input-receipt.json` files provide all positive names, component intervals, alias maps, residual-degree bounds and data hashes. `merged_source.py` gives the complete arithmetic grammar and `history/history174.json` the fully expanded 174-gate history. Source pins distinguish recovered own code from inherited proof/data. Historical independent QA receipts, where included, are explicitly labeled as such.

The pinned mathlib file is 39986 bytes at commit `ac77769fabe23cb237559e7f56578dbead91499f`, SHA256 `993760c797ad0ff66fa77064a616fce779550bc745cbf6c44e04f392fd0bed0a`; the relevant theorem declarations start at lines 760 and 860. The primary Neary–Woods PDF pin is `6274cb6828579c234bf9f62b8fecc64dea4bb1ae842b4e2e39b9bfc676114c1b`. Its repository warns of incorrect printed pagination; use PDF pages 8 and 17 together with the cited section/table identifiers.

References

- [1] T. Neary and D. Woods, *Four Small Universal Turing Machines*, Fundamenta Informaticae 91(1), 123–144 (2009), DOI 10.3233/FI-2009-0036. Primary author copy: <https://mural.mayn>

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- [2] *A Literal Periodic Langton Ant Interface and the One Visit Two Visit Boundary*, Research Report 40, 3 October 2026. Delivered PDF SHA256 32c7de67c5253fd21268ecac5cb6d351b64cdca771663f59f7d40996368e5f39. Earlier delivered report, recovered from its authenticated archive; no public location assumed. [write] Printed as Part II of this report.
- [3] *A Paid Arithmetic Initialization for a Fixed Periodic Langton Ant*, Research Report 42, 3 October 2026. Authenticated earlier report and initialization packet; normalized local source hash in Section 44.6. Preserved unchanged. [write] Printed as the first half of Part III of this report.
- [4] V. Reshetnikov, *A counted 174-operation bounded ant history*, pinned note and base-three/Pell dependencies, commit 5883b08b7af362077f13bb4afa97a23a90ae4cf8. https://github.com/VladimirReshetnikov/ProveIt/blob/5883b08b7af362077f13bb4afa97a23a90ae4cf8/Computability/HilbertTenthProblem/Papers/1980/EXPLORATION_ANT_CHECKERBOARD_HISTORY.md. Exact dependency scope and individual pins accompany the reconstructed contract.
- [5] M. Carneiro and the mathlib contributors, *Pell's equation and Matiyasevic's theorem*, Mathlib/NumberTheory/PellMatiyasevic.lean, commit ac77769fabe23cb237559e7f56578dbead91499f; especially Pell.matiyasevic and Pell.eq_pow_of_pell. <https://github.com/leanprover-community/mathlib4/blob/ac77769fabe23cb237559e7f56578dbead91499f/Mathlib/NumberTheory/PellMatiyasevic.lean>. Read-only source dependency.
- [6] *Folded three-witness endpoint: independent audit addendum*, endpoint packet, 3 October 2026. Exact recovered 42-gate source SHA256 dd5a29f925e896bfff6759508b4d61c6bee65a18f9e0754d73610f1a0b783d01d; recovered component source and proof are pinned anew in the packet. [write] Report 40's endpoint packet, Appendix 40.D; shipped with the prefix 02-literal-ant-endpoint-.

Part IV

Smaller literal sources for the same polynomial (Reports 47 and 48)

[write] **This Part** merges two reports that keep Part III’s polynomial and shrink the arithmetic source that builds it when every fixed coefficient must come from the literals 1 and 3. Report 47 replaces the dense coefficient prefix of Report 44’s strict grammar by a structured one (Theorem 47.1.1: 31,388,831 operations for two inputs); Report 48 fuses the coefficients with the two background polynomials that consume them (Theorem 48.1.1: 14,658,934). Both prove that the resulting polynomial equals Report 44’s at every integer assignment, so the recognition theorem, Theorem 44.1.1, is not restated as a theorem here: where the sources recall it, a marked pointer stands in its place and keeps the recall’s own remarks. Report 48’s count is the most recent of the two; the programme has since lowered it (the remark after Theorem 44.1.1). The scope of the recognized language is item (a) of the preface to the four Parts. The programme reviewed both reports in commits 5d7e884a5 and cfbe37de7 (no defect in the algebra, the ledger arithmetic or the binding; no complete stream regenerated) [W2, W1].

Report 47: the structured coefficients

[write] *Exact Structured Construction of the Fixed Ant Coefficients. A literal arithmetic refinement of Research Report 44*, Research Report 47, dated 4 October 2026, author line “Mathematical research report”; batch-88 manuscript 09, archive `Exact_Structured_Construction_of_the_Fixed_Ant_Coefficients_Package.zip`. It pins no ProveIt commit of its own and inherits Report 44’s pins; its frozen reference copy of Report 44’s archive is byte-identical to batch-88 manuscript 07 and is not shipped twice. Numbered 47.k; labels `ptr:sc;`; files `05-ant-coeffs-`.

Abstract (Report 47). We replace the dense, deliberately unoptimized fixed-coefficient prefix of Research Report 44 by an exact structured arithmetic construction. All 1,152,598 coefficient labels retain their prescribed integer values. The new prefix has only literal 1 and literal 3 leaves and exactly 29,081,374 paid additions, subtractions and multiplications. Its construction combines row-occurrence polynomials compiled by signed events, finite black-set ownership, division-free geometric sums, and the exact rotated periodic cut. Appending the unchanged variable-dependent source gives 31,388,831 operations for two positive raw inputs or 31,388,841 for one, with the same 465 or 467 positive witnesses, one equation and exact variable degree 2,304,000. We prove the coefficient identities and source splice, state the finite verification boundary, and supply reproducible source and receipts. The old bound remains valid for its original grammar. No global optimality, universal-operation record, new ant-dynamics theorem or stronger universality claim is asserted.

47.1 The result and its scope

Report 44 fixes a polynomial source for the raw-pair halting relation of the pinned Neary–Woods U15 machine and its literal periodic-ant realization [1]. Its variable-dependent source has 2,307,457 arithmetic gates. The separate literal-1/3 prefix in its strict grammar constructs all fixed coefficients

by ternary Horner chains and costs

$$N_{\text{old}} = 554\,386\,261\,707\,905\,131.$$

That count is a valid count of the displayed unoptimized grammar. This report supplies a different prefix for the same coefficients; it does not correct an erroneous old count.

An arithmetic gate is a binary $+$, $-$ or $\times$ of available expressions. We write M for multiplication count and A for addition/subtraction count. The word “addition” in a ledger includes subtraction. Initial constant leaves are exactly 1 and 3. All loop endpoints, template coordinates and fixed JSON bits are compilation data, not free run-time integer leaves. A gate output is an expression, not a quantified witness. Aliasing an available expression costs no gate; computing that expression and every later multiplication by it is charged. This is a unit-cost arithmetic-source model. Operands may have enormous bit length; generating or hashing the source does not evaluate those full integers or imply a comparable numerical running time or memory bound.

Theorem 47.1.1 (Exact coefficient refinement). *There is an explicit finite topological arithmetic prefix P , using only the constant leaves 1, 3, with*

$$M_P = 13\,182\,377, \quad A_P = 15\,898\,997, \quad N_P = 29\,081\,374,$$

and a total alias map to every one of Report 44’s 1,152,598 coefficient labels. Each alias has exactly the prescribed old integer value. The prefix introduces no variable, witness or equation.

Binding the old coefficient operands to these aliases and retaining every old main gate gives the following strict sources:

<i>Raw positive inputs</i>	<i>M</i>	<i>A</i>	<i>Total</i>
<i>Two</i>	<i>14,335,963</i>	<i>17,052,868</i>	<i>31,388,831</i>
<i>One</i>	<i>14,335,967</i>	<i>17,052,874</i>	<i>31,388,841</i>

The resulting polynomials are pointwise identical to the old polynomials on all integer assignments. Their positive witness counts are respectively 465 and 467; each has one asserted equation and exact total variable degree 2,304,000.

The prefix claim is proved in Sections 47.3–47.7. The splice and degree claim is proved in Section 47.9. Counts describe these particular sources, including gates whose values happen to be zero or one. No lower bound or optimality theorem is implicit.

[write] *The recognition statement.* At this point Report 47 recalls, “for completeness”, the inherited recognition theorem: with the sentinel decoding $\text{sent}(d)$ of nearest-head-first binary words and $H(x, y)$ the halting of the pinned U15 machine on the decoded side words, $H(x, y) \iff \exists z \in \mathbb{Z}_{>0}^{465} F_2(x, y, z) = 0$ for every positive x, y , relative to Report 44’s inherited history, Pell, recoder, initialization, endpoint and literal-simulation theorems; the one-input version quantifies x, y as well and pays for the Cantor relation $2N = (x + y - 2)(x + y - 1) + 2y$. This is Theorem 44.1.1 with (44.1)–(44.3), printed only there (Part III). The recall’s own remarks are kept: the one-input version “recognizes the same relation through the unique positive pair associated with $N > 0$ ”; “This is ordinary paid arity reduction, not a claim about empty-right machine inputs. The accepting ant observable represents halting; the ant continues moving.”

47.2 The fixed data and coordinate conventions

Use the following exact constants throughout:

$$\begin{aligned} R &= 601\,547\,591, & S &= 576\,000, & K &= 957, \\ V &= 400(R + 2) = 240\,619\,037\,200, & u &= 2V = 481\,238\,074\,400, & Z &= 3^{400}. \end{aligned}$$

There are 960 width-600 storage slots in a macro, of which slots 1 through 958 are active. A pair index $s \in \{0, \dots, 956\}$ selects physical slots $s + 1, s + 2$. Horizontal coordinates are read modulo S . The next vertical macro is shifted horizontally by $S/2 = 288\,000$. Thus two macros give rectangular period u vertically.

The fixed program JSON contains 958 compressed top-level nodes and specifies exactly R operation rows. Its node forms are **RANGE**, **ROW**, **COPY** and **GATE**. These nodes describe a fixed sequence of adjacent-pair template placements. The new compiler never applies that sequence to a Boolean tape or an ant state. It lowers the finite description algebraically; in particular, it does not read the saved **prefix_rows** field. The old random-access decoder and physical program are retained only as text specifications, not imported or run.

Finite colored maps are functions from finite subsets of $\mathbb{Z}^2$ to $\{0, 1\}$. A colored union is valid when overlapping assignments agree. Its black set contains the coordinates assigned 1; unspecified coordinates have default color 0 in the atlas. It is essential to check agreement of white assignments as well as black union counts. The frozen JSON maps, anchor patch and compressed program are authenticated copies of the old atlas and interface data [3]. Their hashes, exact locations and read closure are recorded in the accompanying provenance and science manifests.

The proof concerns these concrete finite data, not arbitrary template families. Every asserted finite-map equality can be checked by set equality on the supplied cells. The general arguments then extend those exact finite equalities to the full height R , without generating R physical rows or an astronomical bitmap.

47.3 Exact occurrence polynomials from signed events

For $a \in \{\text{DUP}, \text{NAND}, \text{MOVE_LEFT}, \text{MOVE_RIGHT}\}$ and $0 \leq s < K$, define the integer polynomial in an indeterminate z

$$T_{a,s}(z) = \sum_{\substack{0 \leq r < R \\ \text{row } r \text{ has kind } a \text{ and pair } s}} z^r. \quad (47.1)$$

We construct $T_{a,s} = T_{a,s}(Z)$ without expanding the row sequence. These are fixed constants. Their definition does not add a witness, oracle or input-dependent operation.

47.3.1 One monotone run

A run starts at row time n , pair s_0 , direction $d \in \{-1, 1\}$ and length $L > 0$. Its rows have slots $s_0 + dk$ for $0 \leq k < L$. At those slots it contributes z^{n+k} . Create events

$$E(s_0) += z^n, \quad E(s_0 + dL) -= z^{n+L}.$$

Sweep the slots in direction d , with zero just outside the entering boundary, by

$$v(s) = zv(s - d) + E(s). \quad (47.2)$$

Lemma 47.3.1 (Run identity). *The sweep of these two events is z^{n+k} at $s = s_0 + dk$ for $0 \leq k < L$, and zero at every other owned slot. The sum of events from any finite collection of runs gives exactly the sum of their occurrence contributions.*

Proof. Before the start event the recurrence is zero. At s_0 it becomes z^n . Induction multiplies by z once at each next slot, giving z^{n+k} . At the stop position the recurrence produces $z^{n+L} - z^{n+L} = 0$, and stays zero afterward. This argument works for either sweep direction. Equation (47.2) is linear in the event array, so superposition proves the finite-collection assertion over $\mathbb{Z}[z]$, not merely at tested numerical values. A stop event at exterior position -1 or K affects no owned slot. The compiler still pays for creating that event. $\square$

There are separate arrays for each of `MOVE_LEFT`, `MOVE_RIGHT` and each direction. After four sweeps, the two directions for each kind are added. Singleton `DUP` and `NAND` occurrences accumulate directly at their pair indices.

47.3.2 Exact lowering of the finite row specification

The following are word-description identities, not assertions about machine execution. At logical width m and index i , a `DUP` word is a decreasing `MOVE_RIGHT` run

$$(m-1, m-2, \dots, i+1),$$

followed by the singleton (DUP, i) . Its total row length is $m-i$. A `NAND` word is the singleton (NAND, i) followed by an increasing `MOVE_LEFT` run $(i+1, \dots, m-2)$, of total length $m-i-1$. Empty runs contribute no rows or gates.

The interchange word at (m, i) consists of the following 24 pairs (kind, j) , used at index $i+j$, updating the compile-time width by $+1$ after a `DUP` word and -1 after a `NAND` word:

$$\begin{array}{cccccc} (D, 0) & (D, 2) & (D, 1) & (D, 3) & (N, 2) & (D, 2) \\ (N, 1) & (N, 2) & (N, 1) & (D, 1) & (D, 0) & (D, 2) \\ (N, 1) & (D, 1) & (N, 0) & (N, 1) & (N, 0) & (D, 1) \\ (D, 3) & (N, 2) & (D, 2) & (N, 1) & (N, 2) & (N, 1) \end{array}$$

Here D abbreviates `DUP` and N abbreviates `NAND`. The final width is again m . Summing the 24 word lengths with those updates gives exactly $24(m-i) + 14$ rows. This is the same displayed fixed word specification as in the pinned physical-program text.

A `COPY` node (m, i) is one `DUP` word at (m, i) followed by interchanges at

$$(m+1, i+1+t), \quad 0 \leq t < m-i-1.$$

A `GATE` node (m, p, q) is `COPY` (m, p) , then `COPY` $(m+1, q)$, then the singleton (NAND, m) . A `RANGE` node is exactly its stated monotone run; a `ROW` node is its singleton. Induction over these finite descriptions proves that the lowering has exactly the row-kind and pair-index sequence in (47.1). No semantic simulation is required for this syntactic equality.

47.3.3 Paid powers and the occurrence ledger

Maintain a wire $P = Z^n$, initially the literal 1. For each nonempty run, construct

$$P_{\text{next}} = PZ^L$$

with one multiplication. Each distinct needed Z^L is built once by a left-to-right binary chain and cached. For exponent $e \geq 1$ the chain length is

$$\lambda(e) = \lfloor \log_2 e \rfloor + \text{popcount}(e) - 1;$$

$e = 0$ uses the literal 1. An exponent is fixed compilation data, never an exponentiation gate. The invariant $P = Z^n$ follows by induction on runs.

The exact finite lowering gives 5,456,442 nonempty runs: 2,724,814 directional and 2,731,628 singleton. It contains 2,730,765 word lowerings, 113,709 interchanges and 1,749 copies. The accumulated time is exactly R . Its power cache has 845 lengths including 0 and 1; additional cached powers cost 9,705 multiplications. Computing $Z = 3^{400}$ costs 10. Hence

$$\begin{aligned} M_T &= 5\,456\,442 + 9\,705 + 10 + 4K = 5\,469\,985, \\ A_T &= 2(2\,724\,814) + 2\,731\,628 + 4K + 2K + 1 \\ &= 8\,186\,999. \end{aligned}$$

The final 1 in A_T pays for zero as $1 - 1$. Each directional run costs two event updates, including unused exterior updates; each singleton costs one accumulation. The four sweeps and two kind-wise merges are retained in full. These formulas give 13,656,984 gates and agree with independent expansion of the new finite compiler.

47.4 Finite geometric ownership

Put $F_0 = 400(R + 1)$. A macro's vertical fundamental interval is $[50, V + 50)$. Partition it into half-open height-400 slabs:

$$\begin{aligned} \text{header: } & [50, 450), \\ \text{program } r : & [400r + 50, 400r + 450), \quad 1 \leq r \leq R, \\ \text{footer: } & [F_0 + 50, F_0 + 450). \end{aligned}$$

This partition assigns every ordinate exactly once. Profiles $H[x], B[x], F[x]$ encode the black indicators in a header, a baseline program slab and a footer respectively. Their digit 0 is at the lower endpoint of their slab. Thus each is a 400-digit nonnegative ternary integer; the letter F here denotes the footer profile, not the final polynomial.

47.4.1 Bulk and the common seam

The baseline bulk places DELAY in all 958 active slots, current LEFT_TURN and RIGHT_TURN routes, the previous LEFT_TURN tail, and the vertical return wire. All components are clipped to the owned slab. The prior left-turn tail contributes 163 colored cells, 82 black, at local ordinates 50 through 76. The current left turn contributes 401 colored cells, 201 black, at 305 through 449. The current right turn contributes 684 colored cells, 341 black. The vertical wire's black column is $x = S - 600 + 175$, and its white column is one unit to the left.

Every prior DELAY or pair-operation tail at local $y = 450, \dots, 459$ agrees with the next round's incoming interface: 43 colored cells, 23 black, per slot. Those cells are owned by the following canonical slab. They already occur in its initial interface and add no second contribution. The union has no color conflict. In this baseline bulk the retained components are support-disjoint, giving 5,386,966 colored cells and 2,695,878 black cells.

47.4.2 Header ownership and the staggered marker exception

The header contains NOT and COPY in active slots, except that NOT is omitted at slot 481; the header turns; the upper corner and horizontal wire; the part of the vertical wire in the slab; and predecessor-footer tails. Those tails are translated by $S/2$ in x and -400 in local y .

Ordinary predecessor COPY and RIGHT_MARKER tails supply the common 23-black interface. The predecessor LEFT_MARKER is different. Its slot 1 moves under the half-shift to slot 481, exactly where the current NOT is omitted. Its tail spans local $y = 50, \dots, 259$ and contains 1,620 black cells. Relative to NOT, it adds 99 and removes 16 black cells. It overlaps current COPY in 43 colored cells, 23 black, at $y = 250, \dots, 259$, with agreeing colors. The outer slots 0 and 959 receive the common tails even though they have no current header tile. Thus summing raw template black counts would be wrong; taking the compatible set union is necessary.

47.4.3 Footer and wraparound routes

The footer includes its NOT/COPY/marker placements, the common interface from the preceding program round, the preceding left-turn tail, the last 20 black cells of the vertical wire, a 170-black-cell bottom wire and a 10-black-cell east-to-north corner. It has no current turn maps. The LEFT_MARKER tail beyond local $y = 449$ is assigned to the next macro's header. All horizontal coordinates, including the top wire crossing the horizontal boundary, are reduced modulo S before duplicate cells are unioned.

For precision, the finite component lists can be reconstructed from the following rules, where k denotes a storage slot. In the header, use NOT at $1 \leq k \leq 958$, $k \neq 481$, and COPY shifted vertically by 200 in all active slots. In the footer use NOT at $1 \leq k \leq 958$, $k \neq 921$, and shifted COPY at $0 \leq k < 960$ except $k = 1, 921$; place LEFT_MARKER at 1 and RIGHT_MARKER at 921. Insert all prior-slot DELAY tails for the footer. The header receives all predecessor-footer tails, using LEFT_MARKER at 1, RIGHT_MARKER at 921 and shifted COPY elsewhere. Both header and bulk have turns; both bulk and footer have the previous left-turn tail. The supplied finite checker spells out the wire intervals and corner coordinates directly from the pinned atlas text.

Profile	Raw colored	Union colored	Raw black	Union black	Conflicts
Header	5,941,771	5,859,426	2,979,510	2,935,465	0
Bulk	5,386,966	5,386,966	2,695,878	2,695,878	0
Footer	5,944,375	5,862,030	2,980,815	2,936,770	0

These are exhaustive finite-union counts, not random samples. A separate geometric projector checks all 576,000 columns of each profile against independently assembled neighboring macro placements. It also checks the last program band. Synthetic height $R = 2$ suffices for this local ownership check: each finite motif crosses at most a neighboring slab, the long return wire has a translation-invariant interior intersection with each slab, the header/footer seam types are separately included, and the actual R only repeats the interior. This local reduction does not simulate the actual machine or assert any ant behavior.

47.5 Signed operation corrections and the bulk identity

Let D_2 be the black union of DELAY at horizontal offsets 0 and 600. For a pair kind a , let A_a be its black set, using the specified NAND union when $a = \text{NAND}$. In particular NAND is the union of the finite NAND macro with COPY shifted by $(0, 200)$ and $(600, 200)$. Its internal overlaps agree; a naive black total would count 46 shared cells twice. Its exact union has 13,197 colored cells and 6,614 black cells.

For each horizontal offset d , define the signed 400-digit profile

$$Q_{a,d} = \sum_{b=0}^{399} (\mathbf{1}_{A_a}(d, 50 + b) - \mathbf{1}_{D_2}(d, 50 + b)) 3^b. \quad (47.3)$$

It is built from the disjoint plus mask $A_a \setminus D_2$ and minus mask $D_2 \setminus A_a$. Both DELAYs are cell-disjoint; together they have 5,672 black cells.

Kind	Plus cells	Minus cells	Net cells	Nonzero columns
DUP	1,520	1,884	−364	763
NAND	3,447	2,505	942	1,159
MOVE_LEFT	1,424	1,814	−390	678
MOVE_RIGHT	1,495	2,001	−506	772

Every correction point has $0 \leq d < 1200$ and $56 \leq y \leq 449$; NAND is supported even more narrowly in $57 \leq y \leq 297$. The shared incoming/outgoing seam has zero correction. Corrections miss margin routes and the actual outside neighboring DELAY slots at offsets -600 and 1200 . Successive program slabs have disjoint correction supports even when their selected pair changes.

Lemma 47.5.1 (Exact replacement). *In an owned program slab with operation kind a at pair index s , the exact black-indicator profile is the baseline B plus the translate of Q_a by $600(s+1)$. There is no additional ownership subtraction or overlap correction.*

Proof. Within the selected pair's support, subtraction of D_2 removes exactly the baseline black indicators, while addition of A_a inserts exactly the replacement indicators. Their intersection contributes zero to the difference. Outside the pair, every nonzero correction is disjoint from the other retained components. At the following or previous slab boundary, the common compatible seam has zero difference and has already been assigned to its unique slab. Hence the pointwise indicator after replacement is exactly 0 or 1 and is the actual colored-union black indicator. The statement holds also at horizontal period boundaries because translation and set union were taken modulo S . $\square$

Now define

$$G = \sum_{r=0}^{R-1} Z^r, \quad C[x] = \sum_{\substack{a,s,d \\ x \equiv 600(s+1)+d \pmod{S}}} Q_{a,d} T_{a,s}, \quad U[x] = B[x]G + C[x]. \quad (47.4)$$

Proposition 47.5.2 (Exact program column). *$U[x]$ is the exact ternary column of the R consecutive owned program slabs, with the first program slab aligned to digit zero.*

Proof. The baseline repeated R times is $\sum_r B[x]Z^r = B[x]G$. For a fixed row r , exactly its prescribed kind and pair contributes Z^r in (47.1). Distributing the finite sums in (47.4) therefore inserts the unique translated correction from Lemma 47.5.1 in every row. All coordinates and exponents align with that slab's 400-digit origin. This is an identity of integers obtained from a pointwise indicator identity. It does not assume that an arbitrary signed redundant ternary expansion has no carries. The resulting expansion has Boolean digits because it has been identified with the actual compatible black set. $\square$

47.6 Division free sums and the rotated periodic cut

47.6.1 The geometric sum is paid arithmetic

Construct simultaneously

$$(P_n, G_n) = \left(Z^n, \sum_{k=0}^{n-1} Z^k \right), \quad (P_1, G_1) = (Z, 1).$$

A doubling step is

$$P_{2n} = P_n^2, \quad G_{2n} = G_n + P_n G_n;$$

an ensuing odd step is

$$P_{2n+1} = P_{2n}Z, \quad G_{2n+1} = G_{2n} + P_{2n}.$$

Lemma 47.6.1. *The binary grammar above constructs the stated pair for every fixed $n \geq 1$ using only addition and multiplication.*

Proof. The invariant holds at 1. Split the exponent range $0, \dots, 2n - 1$ into $0, \dots, n - 1$ and its translate by n to obtain the doubling sum. Appending the term at exponent $2n$ gives the odd sum. The power identities follow by multiplication. Induction along the bits of n proves the result. No quotient such as $(Z^n - 1)/(Z - 1)$ is evaluated. $\square$

For $n = R$ there are 29 doublings and 16 odd steps, costing 74 multiplications and 45 additions. Five separate binary chains build

$$3^{375}, \quad 3^{V-425}, \quad 3^{V-400}, \quad 3^{V-25}, \quad 3^{u-25},$$

with respective multiplication counts 14, 49, 45, 52, 49. Thus the geometric-sum-and-shifts stage costs exactly $283M + 45A$.

47.6.2 Why the cut is at physical ordinate 75

Reports 42 and 44 use the rotation $X = y - 75$ and $Y = 288650 - x$ [2, 1]. The fixed coefficient `tile:j` is

$$\sum_{i=0}^{u-1} c(288650 - j, i + 75) 3^i, \quad (47.5)$$

where c is the frozen periodic black indicator. Our natural profile origin is 50, so the rotation cuts 25 rows into a header. Split that header directly from its bits:

$$H[x] = H_{\text{low}}[x] + 3^{25} H_{\text{high}}[x],$$

where the low part has 25 digits and the high part 375. Both are separately constructed by Horner evaluation. No division or floor operation is in the arithmetic source.

Define

$$\text{Full}[x] = H[x] + ZU[x] + 3^{V-400}F[x], \quad (47.6)$$

$$\text{Cut}[x] = H_{\text{high}}[x] + 3^{375}U[x] + 3^{V-425}F[x], \quad (47.7)$$

$$\text{Tile}[x] = \text{Cut}[x] + 3^{V-25}\text{Full}[x - S/2] + 3^{u-25}H_{\text{low}}[x]. \quad (47.8)$$

Proposition 47.6.2 (Exact cut and wrap). *For every x modulo S ,*

$$\text{Tile}[x] = \sum_{i=0}^{u-1} c(x, i + 75)3^i.$$

Proof. The header starts at 50, the first program slab at 450 and the footer at $F_0 + 50$. Relative to origin 50 their exponents are 0, 400, $F_0 = V - 400$, proving (47.6). Removing the first 25 header rows and using origin 75 gives exponents 0, 375, $V - 425$, proving (47.7).

The first region in (47.8) has length $V - 25$. At its end starts the complete next macro, with horizontal half-shift; at fixed global x its local coordinate is $x - S/2$. That region has length V and therefore starts at exponent $V - 25$. After it come the 25 wrapped rows of the next header. Two half-shifts equal one full horizontal period, so their local coordinate is again x and their first exponent is $(V - 25) + V = u - 25$. The three regions are disjoint, consecutive, and total $(V - 25) + V + 25 = u$ rows. Their boundary ownership is already fixed by Section 47.4. Concatenating their true Boolean columns gives the claimed equality. $\square$

This proof accounts for the stagger and the cut simultaneously. Treating the period as two unshifted full macros, or replacing the exceptional predecessor marker by an ordinary tail, would give incorrect coefficients.

47.7 Every old coefficient label

For $0 \leq j < S$, put $x_j = (288650 - j) \bmod S$. Bind

$$\text{tile} : j \mapsto \text{Tile}[x_j], \quad \text{first} : j \mapsto \mathbf{1}_{\text{header black}}(x_j, 75).$$

The first alias has the old value by Proposition 47.6.2; the second is precisely digit 25 of the fixed header and aliases the paid zero or literal 1. No bit extraction is an arithmetic operation on a variable or a giant integer; the compile-time finite header bit chooses one of two available wires.

For the 584 anchor labels, define $\delta(i, j)$ as the pinned patch's new-minus-old color at

$$(x, y) = (289200 - j, i - 144), \quad 0 \leq i \leq 220,$$

or zero when no patch entry occurs. The patch has 2,806 entries, of which 1,401 are +1 and 1,405 are -1, with no duplicate coordinate. Construct

$$\text{anchor} : j = \sum_{i=0}^{220} \delta(i, j)3^i$$

by the unchanged 221-symbol signed Horner chain. The symbol -1 is the paid wire $(1 - 1) - 1$. All $220M + 220A$ per anchor row are retained, even when a symbol is zero.

The remaining named constants are exactly

$$\begin{aligned} \mathbf{Cu} &= 3^u, & \mathbf{Cx} &= 3^u - 1, \\ \mathbf{Cbase} &= 3^{u-219}, & \mathbf{C198} &= 3^{198}, \\ \mathbf{EndpointK} &= 3^{481225262775}, & \mathbf{EndpointD} &= 3^u - 2. \end{aligned}$$

The four independent binary chains have lengths 47, 50, 10 and 55. All ordinary main numerals are the eight labels 0, 1, 2, 3, 4, 6, 8, 9. Zero and 2 are already paid; 1 and 3 are the only literal leaves. Compute $4 = 2 + 2$, $6 = 3 + 3$, $8 = 4 + 4$, $9 = 3 \times 3$. The subtractions giving $\mathbf{Cx}$ and $\mathbf{EndpointD}$ are paid. The complete alias inventory is

$$S + S + 584 + 6 + 8 = 1\,152\,598.$$

An internal -1 wire is used for signed profiles and anchors; the main source has no $\mathbf{C}:-1$ port. Equal-valued labels may alias the same expression without identifying their names. These identities establish the value of every port, not merely a sample.

47.8 The exact gate ledger

A nonzero signed profile of width w is formed from its w fixed digits by the complete Horner recursion $a \leftarrow 3a + d$. It costs $(w - 1)M + (w - 1)A$ irrespective of leading zero digits. Identical profiles of identical width share that chain; the all-zero profile aliases the already paid zero. Plus and minus masks are disjoint, so each digit selects one of the existing $-1, 0, 1$ expressions.

The cache has 736 nonzero width-400 profiles, 21 width-25 profiles and 81 width-375 profiles, so

$$736 \cdot 399 + 21 \cdot 24 + 81 \cdot 374 = 324\,462$$

operations of each type build all small profiles. Two further additions/subtractions construct -1 and 2 . The cache counts include operation corrections as well as H, B, F . For clarity, the footer's serialized dictionary has a zero mask available but the actual footer uses 246 distinct masks, all nonzero; dictionary inventory and used-mask inventory are different counts.

For corrections, every nonzero horizontal column of every kind is multiplied by every one of its $K = 957$ occurrence constants and accumulated, even if an occurrence value is zero. Thus this stage costs

$$957(763 + 1159 + 678 + 772) = 3\,227\,004$$

of each type. For each x , forming U costs $1M + 1A$, forming Full costs $2M + 2A$, forming Cut costs $2M + 2A$, and the final Tile costs $2M + 2A$. Hence the period join costs $7S = 4\,032\,000$ of each type. Aliasing the tile/first labels adds no arithmetic.

Stage	M	A	Total
Occurrence constants	5,469,985	8,186,999	13,656,984
Small profiles and basic literals	324,462	324,464	648,926
Geometric sum and five shifts	283	45	328
Operation corrections	3,227,004	3,227,004	6,454,008
Bulk, boundaries and period join	4,032,000	4,032,000	8,064,000
584 anchor rows	128,480	128,480	256,960
Remaining named constants	163	5	168
Prefix total	13,182,377	15,898,997	29,081,374

This verifies the prefix count in Theorem 47.1.1. The saving relative to the old strict prefix is exactly

$$554\,386\,261\,678\,823\,757$$

operations, a reduction by a factor of approximately 19 063 276 092. This comparison is only between the two specified grammars. It is an arithmetic operation-count reduction, not a numerical runtime or bit-complexity speedup: the enormous fixed integers retain exactly the same values and are represented by a shorter source, not materialized or made small. The arithmetic programs differ, while the coefficient identities prove equality of their resulting polynomial. No claim concerns the best possible source, other complete certificates, or operation-record priority.

47.9 Strict binding to the variable source

Let $N_P = 29\,081\,374$, and give prefix gates consecutive identifiers $0, \dots, N_P - 1$. In this source an integer operand denotes an earlier gate identifier, not an integer literal. Only the tokens **one** and **three** are constant leaves. Let $L[k]$ be the prefix expression or literal leaf assigned to old coefficient key k .

For an old main operand t define a fixed substitution σ by four disjoint cases:

1. An old integer gate reference r becomes $N_P + r$.
2. A fixed-coefficient operand $\mathbf{C:k}$ becomes $L[k]$.
3. A declared raw or positive-witness coordinate name is preserved.
4. Every other operand is rejected.

Append to P every old gate $[r, \text{op}, a, b]$ as

$$[N_P + r, \text{op}, \sigma(a), \sigma(b)].$$

Preserve the raw/positive declarations. Transform the operands of residual-pair metadata and of the final equation by the same rule. No $\mathbf{C}$: operand survives. The alias map is a total binding, not an additional input declaration or equation. In particular $L[0]$ is prefix gate 0, which computes $1 - 1$; the final zero side is paid.

Proposition 47.9.1 (Pointwise identity of the splice). *The transformed source is topological and has exactly the old polynomial value for every assignment of all raw and witness coordinates.*

Proof. The prefix is topological. An old main gate r referenced only an earlier gate $q < r$, an old prescribed coefficient or a previously declared coordinate. After substitution, $N_P + q < N_P + r$; every prefix gate reference is less than N_P ; and the declarations remain before their first main use. They may follow the input-independent prefix without changing their meaning. Thus topology is preserved.

Section 47.7 proves equality of each substituted coefficient value. Induct on r : inputs and coefficients agree, and applying the unchanged opcode to equal operand values gives equality of the polynomial, not only equivalence of their zero sets. Intermediate expressions are permitted to be signed; no positivity is imposed on prefix or main gate values. $\square$

The two-input main source has exactly $1\,153\,586M + 1\,153\,871A$; the one-input source has $1\,153\,590M + 1\,153\,877A$. Adding these counts to the prefix, with no binding arithmetic, gives Theorem 47.1.1. The only final equation is respectively gate $31\,388\,830 = \text{gate } 0$ or gate $31\,388\,840 = \text{gate } 0$. The 285 or 286 stored residual pairs are metadata used by the paid final sum of squares, not extra final equations.

The witness domains remain exactly those of Report 44: five positive history parameters, 61 positive history witnesses, four positive initialization-geometry witnesses, 392 positive recoder witnesses and three positive endpoint witnesses. This totals 465. The one-input version adds the two positive side-word codes for 467. No fixed coefficient becomes a variable. Both raw inputs, or the single paired input, range over $\mathbb{Z}_{>0}$.

Every prefix wire has variable degree zero. Pointwise polynomial equality therefore preserves exact degree, rather than just a syntactic upper bound. More explicitly, the inherited main source has exactly two maximal residuals with highest homogeneous term $-W^{1\,152\,000}$; all other residuals have smaller degree. The sum of their squares has highest homogeneous part

$$2W^{2\,304\,000}.$$

It cannot cancel over $\mathbb{Z}$. The pairing residual has degree two and does not affect it. Thus the exact degree remains 2,304,000. No geometric equation is substituted into W when calculating this polynomial degree.

47.10 Evidence and reproducibility

The mathematical proof has three different layers. The general algebraic identities are proved above over integers or polynomial rings. Exact finite-map checks establish the concrete supports, overlaps and coefficient inputs to those identities. Stream checks establish that the supplied arithmetic grammar has the reported operation count, legal leaves, topology and binding. These layers should not be conflated: a stream hash identifies bytes, and a modular comparison tests values modulo its modulus; neither alone proves equality of astronomical integer coefficients.

47.10.1 Prefix identity and independent checks

The complete new prefix has been generated into a checked streaming sink, rather than retained as a large text file. Its canonical gate serialization has SHA-256

6a01d7c1d8ee6b268d49c921b3263b578c152e49c62810d5aa877a4103cb9759.

Its ordered output-alias serialization has SHA-256

```
87c890e3fe7bb138378d09df71125cc5783219b99d171ed426ed9791b0cdcc3e.
```

The independent checker verifies all 29,081,374 gates, the allowed 1/3 leaves, topology, stage counts and the full 1,152,598-label inventory. Its all-label evaluation modulo 1,000,003 is supplementary regression evidence. Exact coefficient equality rests on the event, finite-union and cut proofs. Independent finite geometric projections agree in every profile column with zero color conflicts, and exact symbolic run tests supplement the general proof of Lemma 47.3.1.

47.10.2 Both joined streams independently regenerated

Both complete joined sources were actually generated and hashed in streaming mode. A separately authored independent adapter repeats the binding, checks every gate and domain record, and obtains the same canonical hashes as the construction adapter:

- Two-input source, 31,388,831 gates:

```
b235ba1d2baf0cf089c9658b1995f9c19fbe3ff033dc6ae67c76ec8cd6522d03
```

- One-input source, 31,388,841 gates:

```
737866317089c07465253a0d489813bf1bab50aaba0dda6083acbf78ef45f276
```

In each arity, every one of the 1,152,598 coefficient labels is resolved and actually used. All positive-coordinate names and their ordering agree with Report 44. There are no surviving coefficient ports or free constant declarations. The regenerated original main hashes also agree with the authenticated targets:

- Two-input main:

```
c16901162e09b07d1e0c27b28e29dcc12a9b6137391fce85e3f150a3fed37eac
```

- One-input main:

```
85a161972769207e4f5d4212214630535208f69d6736bca2b46a8aff8276a22a
```

Arithmetic records are tab-separated identifier, opcode and two operands, terminated by a newline; domain, residual and equation records are compact JSON lines in their original order. The prefix precedes all main records. This complete canonical byte convention is reproduced by both adapters. The joined streams were traversed and hashed, but their large flat text files are not retained in the archive. Their exact generators and data are retained. The independent joined addendum supersedes only the earlier prefix audit's pending-join paragraph; it does not alter the prefix verdict or the inherited-theorem limitations.

[write] *In this repository.* The archive is shipped flattened with the prefix `05-ant-coeffs-`. Its `joined_evidence.tex`, printed in the subsection above, is shipped as `data/05-ant-coeffs-joined_evidence.tex`. Not shipped: the frozen Report 44 reference archive (a copy of batch-88 manuscript 07, which is Report 44 in Part III), the files that duplicate Reports 40, 42 and 44 (among them the specification texts `specifications/*.py.txt`, copies of Report 40's generator and physical-program decoder), and the three 1,152,000-byte profile files `science/geometry/profiles/B.u161e`, `F.u161e` and `H.u161e`, which the shipped `code/05-ant-coeffs-science-geometry-build_profiles.py` rebuilds byte for byte in seconds (report README). At placement the complete replay passed on a copy; its one recorded mismatch on Windows is the joined receipt's `pins` keys, written with the operating system's path separator, and is equal after

normalizing the separators (README). The 34 absolute `/workspace/...` paths in the independent audit's `SOURCE_PINS.json` are historical provenance records of the author's workspace and do not resolve here.

47.10.3 What execution does and does not establish

The coefficient compiler executes newly authored arithmetic lowering and finite JSON-set analysis only. It does not execute an upstream physical ant program, color recipe, row decoder, saved row schedule or saved upstream arithmetic schedule. The fixed compressed nodes are compilation data. For the main splice, the authenticated Report 44-owned arithmetic generator may be reused after its import/read closure has been inspected; that narrow reuse is distinct from running the forbidden upstream physical or color programs. The package documents this boundary and retains text-only specifications where appropriate.

The old astronomical Horner prefix is not generated, numerically evaluated or represented as a freshly verified stream. Equality with it is mathematical: its Horner grammar evaluates the same prescribed coefficients proved above. The new prefix's much shorter source can be generated without materializing those coefficient integers. Running a finite checker does not prove the arbitrary-integer Pell or history theorem, nor the literal machine/ant simulation. Those remain the same inherited dependencies as in Report 44.

The accompanying release separates the frozen scientific payload from manuscript/PDF construction and final-release quality checks. Source and data pins authenticate the frozen candidate; archive manifests authenticate delivered files. PDF rebuilding uses local pdfLaTeX with shell escape disabled. Reproduction writes outputs to external new directories and does not mutate the frozen source. Detailed commands, expected hashes and interpretation of each receipt are in the release README.

47.11 Further questions

Several concrete improvements remain open within this fixed problem. Can the repeated copy/interchange description be compiled at a coarser level than 5.46 million runs while still exposing every paid arithmetic operation? Can occurrence polynomials or shifted correction products be shared more aggressively without losing a transparent exact ledger? Can a joint addition chain replace the independent fixed-shift and endpoint powers? These are possible refinements, not savings already claimed here.

A different question is a lower bound for constructing this particular labeled coefficient family in the literal $1/3$ language. No such lower bound is provided. The shortest source, the effect of allowing other literal sets, and comparisons with other complete Diophantine presentations require separately specified cost models. Reducing fixed-coefficient construction also does not reduce the positive witness dimension or prove a stronger machine-input universality statement. Those belong to different mathematical questions.

The present conclusion is exact and limited: the complete fixed coefficient family of Report 44 admits the explicit 29,081,374-operation prefix, and its literal splice preserves the existing polynomial, domains, equation and degree while giving the strict totals in Theorem 47.1.1.

References

- [1] *Research Report 44: A Complete Positive Certificate for the Literal Periodic Ant*, recovered edition, 4 October 2026. Accompanying strict grammar, arithmetic source and inherited contracts. Its delivered recovered PDF and authenticated archive identity are recorded in this release’s provenance. [write] Printed as the second half of Part III of this report.
- [2] *Research Report 42: A Paid Arithmetic Initialization for a Fixed Periodic Langton Ant*, 3 October 2026. Authenticated component source and coefficient conventions retained in Report 44’s dependency package. [write] Printed as the first half of Part III of this report.
- [3] *Research Report 40: A Literal Periodic Langton Ant Interface and the One Visit Two Visit Boundary*, 3 October 2026. Authenticated literal periodic atlas, two-sided loader, startup anchor and accepting-port interface, as retained in Report 44’s dependency package. [write] Printed as Part II of this report.

Report 48: fusion of the background polynomials

[write] *Exact Fusion of the Ant Background Polynomials*. A smaller literal arithmetic source for the same polynomial, Research Report 48, dated 4 October 2026, author line “Mathematical research report”; batch-88 manuscript 10, archive `Exact_Fusion_of_the_Ant_Background_Polynomials_Package.zip`. It pins no ProveIt commit of its own and inherits Report 44’s pins; its frozen reference copy of Report 47’s archive is byte-identical to batch-88 manuscript 09 and is not shipped twice. Numbered 48.k; labels `ptr:fu::`; files `06-ant-fusion-`. Its counts have since been lowered by the programme (the remark after Theorem 44.1.1).

Abstract (Report 48). We fuse the fixed-coefficient construction of Research Report 47 with the two dense spatial Horner expressions in the initializer of Research Report 44. A three-band cyclic-index identity factors the operation corrections into twelve short spatial factors and twenty-four rotated occurrence factors. Exact repeated-block grouping compresses the remaining fixed spatial profiles, with division-free, fully paid geometric weights. The resulting literal-1/3 sources have 14,658,934 arithmetic operations for two positive raw inputs and 14,658,944 for one. They compute exactly the same complete integer polynomial, on every assignment of the same coordinates, as the frozen predecessors. The positive witness counts remain 465 and 467, with one equation and exact variable degree 2,304,000. We prove the new polynomial identities, audit the two precise source replacements and every retained consumer, give the complete arithmetic ledger, and distinguish algebraic proof, finite checks and source reconstruction. No quotient-ring identity, nonzero assumption on the spatial variable, global optimality or operation-record claim is used.

48.1 Result and comparison

Reports 44 and 47 specify a fixed Diophantine polynomial source arising from a periodic-ant realization of the pinned U15 machine [1, 2]. Report 47 reduces the cost of constructing the fixed coefficients. The present refinement avoids first constructing all spatial coefficients separately and then evaluating two dense polynomials from them. Instead, it constructs those two polynomials directly from the same exact coefficient identities.

An arithmetic gate is a binary addition, subtraction or multiplication of already available expressions. We write M for multiplication count and A for the combined addition/subtraction count. The only constant leaves are 1 and 3. Fixed loop indices, mask bits and exponents are compilation data; they do not supply free integer leaves. Aliasing an available expression is free, while every emitted arithmetic operation, including operations on zero or one, is paid. Gates are expressions, not extra quantified witnesses.

Theorem 48.1.1 (Exact fused sources). *There are explicit topological literal-1/3 arithmetic sources with the following counts.*

<i>Raw positive inputs</i>	<i>M</i>	<i>A</i>	<i>Total</i>	<i>Positive witnesses</i>
<i>Two</i>	5,971,120	8,687,814	14,658,934	465
<i>One</i>	5,971,124	8,687,820	14,658,944	467

Each has one final equation and computes the same complete final integer polynomial as its corresponding frozen Report 44 and Report 47 source, for every integer assignment of the declared coordinates. Its exact total variable degree is 2,304,000. All positive domains, names and witness order are unchanged.

The new totals improve Report 47's 31,388,831 and 31,388,841 by exactly 16,729,897 operations each, or approximately 53.2989% for two inputs. Report 47 remains valid for its displayed grammar. The comparison is between these specified sources; it is neither a lower bound nor a priority or global operation-record claim.

[write] Two later programme packets lower these counts further, to 14,620,779 / 14,620,789 and then 14,620,711 / 14,620,721, by sharing inside this source; the remark after Theorem 44.1.1 lists every count of the polynomial.

The equality is stronger than equality of zero sets. It allows substitution into every old consumer without altering any residual polynomial or the final sum of squares. It is also independent of restrictions that particular solutions impose on the spatial coordinate W : the new identities hold as ordinary polynomial identities, including at $W = 0, 1, -1$.

48.2 Inherited inputs and the scope of the new proof

We use distinct symbols for the horizontal period and the physical macro height:

$$\begin{aligned}
 S &= 576\,000 = 600 \cdot 960, & K &= 957, & R &= 601\,547\,591, \\
 L &= 400(R + 2) = 240\,619\,037\,200, & u &= 2L, & Z &= 3^{400}, \\
 c &= 288\,650 = 600 \cdot 481 + 50.
 \end{aligned}$$

The old arithmetic initializer calls S its V ; the Report 47 profile construction calls L its V . These are different dimensions. All spatial Horner polynomials here have S coefficients, not L coefficients.

For each operation kind

$$a \in \{\text{DUP, NAND, MOVE_LEFT, MOVE_RIGHT}\}$$

and each pair index $0 \leq s < K$, Report 47 constructs

$$T_{a,s} = \sum_{\substack{0 \leq r < R \\ \text{row } r \text{ has kind } a \text{ at pair } s}} Z^r.$$

The signed 400-digit ternary profile $Q_{a,d}$, $0 \leq d < 1200$, describes the operation's exact correction to the two baseline DELAY slots. Its physical translation is $600(s + 1) + d$. The unsigned profiles $H[x]$, $B[x]$, $F[x]$, indexed modulo S , encode the owned header, bulk and footer slabs. In this notation $F[x]$ is a footer coefficient, not the final Diophantine polynomial.

The masks define H_{low} as digits 0 through 24 of H , H_{high} as the remaining digits shifted down by 25, and $\text{first}[x]$ as digit 25. Thus

$$H[x] = H_{\text{low}}[x] + 3^{25}H_{\text{high}}[x].$$

Taking bits of the finite compile-time masks merely selects the prescribed digits; the arithmetic source does not divide an enormous coefficient or extract bits of a variable.

The inherited finite ownership theorem gives, with all x indices reduced modulo S ,

$$C[x] = \sum_{\substack{a,s,d \\ x \equiv 600(s+1)+d \pmod{S}}} Q_{a,d} T_{a,s}, \quad (48.1)$$

$$G_Z = \sum_{r=0}^{R-1} Z^r, \quad U[x] = G_Z B[x] + C[x], \quad (48.2)$$

$$\text{Full}[x] = H[x] + ZU[x] + 3^{L-400}F[x], \quad (48.3)$$

$$\text{Cut}[x] = H_{\text{high}}[x] + 3^{375}U[x] + 3^{L-425}F[x], \quad (48.4)$$

$$\text{Tile}[x] = \text{Cut}[x] + 3^{L-25}\text{Full}[x - S/2] + 3^{u-25}H_{\text{low}}[x]. \quad (48.5)$$

The three vertical portions in (48.5) have lengths $L - 25, L, 25$. Their total is u , their cut begins at physical ordinate 75, and the middle macro has the horizontal half-shift. Report 47 proves the exact finite colored unions, signed corrections and boundary ownership underlying these identities. In particular, they are true coefficient identities, not an assumption that arbitrary signed ternary digits never carry.

The old labels are

$$\text{tile:j} = \text{Tile}[(c - j) \bmod S], \quad \text{first:j} = \text{first}[(c - j) \bmod S], \quad 0 \leq j < S. \quad (48.6)$$

This report takes that coefficient theorem and the occurrence-compiler theorem as inherited inputs. It proves the new spatial identities and literal-source substitution in full; it does not repeat the physical template or machine-simulation construction.

Inherited recognition statement. [write] Report 48 recalls here the recognition theorem of Report 44 in its own notation, $\mathcal{H}(x, y)$ for $H(x, y)$ and $\mathcal{P}_2$ for F_2 : $\mathcal{H}(x, y) \iff \exists z \in \mathbb{Z}_{>0}^{465} \mathcal{P}_2(x, y, z) = 0$, relative to the unchanged history, Pell, recoder, initializer, endpoint and literal-simulation theorems of Report 44, with the same sentinel decoding; the one-input source quantifies the two side codes, retains the paid pairing relation $2N = (x + y - 2)(x + y - 1) + 2y$ and has 467 positive witnesses. This is Theorem 44.1.1, printed only there. The recall's own remarks are kept: "This is the same raw-pair recognition statement, not a new assertion about empty-right machine inputs or stronger universality. The accepting ant observable represents halting; the ant itself continues moving."

48.3 The ordinary polynomial cyclic index identity

For any fixed S -array A , any $p \in \{0, S/2\}$ and an unrestricted indeterminate W , define

$$\mathcal{E}_p(A; W) = \sum_{j=0}^{S-1} A[(c - j - p) \bmod S] W^j. \quad (48.7)$$

Permuting the fixed array indices gives the equivalent expression

$$\mathcal{E}_p(A; W) = \sum_{x=0}^{S-1} A[x] W^{(c-p-x) \bmod S}.$$

The modulo operation chooses a fixed coefficient position. It does not reduce a polynomial modulo $W^S - 1$, and it implies no relation $W^S = 1$.

Partition the operation offsets into three bands

$$I_0 = \{0, \dots, 50\}, \quad I_1 = \{51, \dots, 650\}, \quad I_2 = \{651, \dots, 1199\}.$$

Their lengths are 51, 600, 549; all three factors are padded to width 600 in the emitted grammar. For $d \in I_b$ define

$$\ell_b(d) = 50 + 600b - d, \quad k_{b,p}(s) = (480 - b - s - p/600) \bmod 960.$$

Both $p/600 = 0, 480$ are fixed integers.

Lemma 48.3.1 (Exact exponent separation). *For every $0 \leq s < 957$, $d \in I_b$ and $p \in \{0, S/2\}$,*

$$(c - p - 600(s + 1) - d) \bmod S = 600k_{b,p}(s) + \ell_b(d). \quad (48.8)$$

Both sides are ordinary integers in $\{0, \dots, S - 1\}$.

Proof. The three bands give respectively $0 \leq \ell_0 \leq 50$, $0 \leq \ell_1 \leq 599$, and $51 \leq \ell_2 \leq 599$. Hence the right side is nonnegative and at most $600 \cdot 959 + 599 = S - 1$. Before reducing k , its value gives

$$600(480 - b - s - p/600) + 50 + 600b - d = c - p - 600(s + 1) - d.$$

Reducing k changes this by a multiple of $600 \cdot 960 = S$. The unique least nonnegative representative proves (48.8). This includes both band boundaries and all spatial wrap boundaries. $\square$

Define the short and rotated factors by

$$\mathcal{Q}_{a,b}(W) = \sum_{d \in I_b} Q_{a,d} W^{\ell_b(d)}, \quad (48.9)$$

$$\mathcal{L}_{a,b,p}(Y) = \sum_{s=0}^{956} T_{a,s} Y^{k_{b,p}(s)}. \quad (48.10)$$

The first has degree at most 599. The second has degree at most 959, and the map $s \mapsto k_{b,p}(s)$ is injective, since two distinct s values differ by less than 960. Exactly three of its 960 coefficient positions therefore contain the paid zero expression.

Theorem 48.3.2 (Three-band correction factorization). *Treat every coefficient as an independent symbol. In the ordinary polynomial ring*

$$\mathbb{Z}[\{Q_{a,d}\}, \{T_{a,s}\}][W]$$

we have

$$\mathcal{E}_p(C; W) = \sum_a \sum_{b=0}^2 \mathcal{Q}_{a,b}(W) \mathcal{L}_{a,b,p}(W^{600}). \quad (48.11)$$

Consequently the identity holds after substituting the actual inherited integer profiles and occurrence constants.

Proof. Expand (48.1) in the second expression for (48.7). Each term $Q_{a,d}T_{a,s}$ has exponent $(c - p - 600(s + 1) - d) \bmod S$. Apply Lemma 48.3.1, then use $W^{600k+\ell} = W^\ell(W^{600})^k$ and finite distributivity separately on each band. This yields (48.11). If several terms contribute to the same coefficient, their sum is retained on both sides; no disjointness assumption on the correction supports is required. Every exponent already lies in the ordinary spatial range, so no reduction of a product polynomial is involved. $\square$

Paid grammar. Construct $Y = W^{600}$ once by a binary chain of 12 multiplications. Evaluate twelve full length-600 Horner polynomials for (48.9), sharing them between phases. Evaluate twenty-four full length-960 Horner polynomials for (48.10). Multiply each by its short factor, and sum twelve terms per phase. Thus

$$\begin{aligned} M_{\text{short}} &= A_{\text{short}} = 12 \cdot 599 = 7\,188, \\ M_{\text{long}} &= 24 \cdot 959 + 24 = 23\,040, \\ A_{\text{long}} &= 24 \cdot 959 + 2 \cdot 11 = 23\,038. \end{aligned}$$

Leading and internal zero coefficients remain in these dense Horner recurrences. The improvement comes from the proved factorization, not from retroactively deleting paid operations.

48.4 Exact repeated block compression

For each $A \in \{H, B, F, H_{\text{low}}, H_{\text{high}}, \text{first}\}$ and phase p , partition the S spatial coefficient positions into 960 consecutive blocks of width 600. Let

$$a_{k,r} = A[(c - 600k - r - p) \bmod S], \quad 0 \leq k < 960, \quad 0 \leq r < 600.$$

Group k values only when their complete 600-tuples agree. Write J_g for one group of block positions and $b_g[r]$ for its common tuple. The groups must form a disjoint cover of $\{0, \dots, 959\}$.

Proposition 48.4.1 (Exact block identity). *With the grouping above,*

$$\mathcal{E}_p(A; W) = \sum_g \left(\sum_{r=0}^{599} b_g[r] W^r \right) \left(\sum_{k \in J_g} (W^{600})^k \right). \quad (48.12)$$

The half-phase rotates each group's block positions by 480 modulo 960 and leaves its tuple unchanged.

Proof. Write each j uniquely as $600k + r$. Substitute $a_{k,r} = b_g[r]$ for the unique group containing k , and distribute the finite sum. For the second assertion, the phase- $S/2$ tuple at k is the phase-zero tuple at $(k + 480) \bmod 960$. Thus a phase-zero group J_g becomes $\{(k - 480) \bmod 960 : k \in J_g\}$. This is an exact permutation of fixed coefficient positions, not a transformation assuming a value of W . $\square$

The distinct tuple counts, verified in both phases, are

Profile	H	B	F	H_{low}	H_{high}	first
Distinct blocks	5	4	6	3	5	4

Appendix 48.A gives every phase-zero position group. The receipt records every position in both phases and a digest of each complete mask tuple. Verification compares actual entries, not merely digests or a few sample columns.

48.4.1 Division free geometric weights

For a maximal consecutive run $\{k, \dots, k + n - 1\} \subseteq J_g$, its weight is

$$Y^k G_n(Y), \quad G_n(Y) = \sum_{i=0}^{n-1} Y^i.$$

Construct $P_n(Y) = Y^n$ together with G_n by the following finite binary recursion:

$$\begin{aligned} P_1 &= Y, & G_1 &= 1, \\ P_{2n} &= P_n^2, & G_{2n} &= G_n + P_n G_n, \\ P_{2n+1} &= P_{2n} Y, & G_{2n+1} &= G_{2n} + P_{2n}. \end{aligned} \tag{48.13}$$

Lemma 48.4.2 (Paid geometric pair). *Recursion (48.13) computes $(Y^n, \sum_{i=0}^{n-1} Y^i)$ in $\mathbb{Z}[Y]$ for every integer $n \geq 1$.*

Proof. The statement holds at $n = 1$. Multiplication by Y^n shifts the exponents of G_n from $0, \dots, n-1$ to $n, \dots, 2n-1$, proving the even sum and power. The odd step adds the next monomial Y^{2n} and multiplies the power by Y . Induction along the binary digits proves the claim. $\square$

[write] This is Report 47's Lemma 47.6.1, with Y for Z , proved by the same induction along the binary digits. Report 48 states it in $\mathbb{Z}[Y]$ for an indeterminate Y , which its identities at $W = 0, \pm 1$ need, so the statement and its short proof are kept here as a marked second statement.

Each even step costs $2M + 1A$ and each following odd extension $1M + 1A$. The power output remains paid even when only the sum is consumed. Independent powers Y^k use ordinary binary chains. At $k = 0$, the power aliases literal 1; at $n = 1$, the sum aliases literal 1. The source omits a run product only through the explicitly specified branches $k = 0$ or $n = 1$. It retains every group product in (48.12).

No expression divides by $Y - 1$, W or any other expression. Hence no special case or limiting argument is needed at $W = 0, 1, -1$. In particular, when $W = 0$, $Y^0 = 1$ and the finite polynomials evaluate normally; when $W = \pm 1$, $Y = 1$ and $G_n(Y) = n$ by its finite sum.

48.4.2 Sharing and the closed block ledger

One fusion object has a single fixed W and $Y = W^{600}$. It shares identical coefficient-wire tuples as whole spatial Horner blocks; identical position tuples share their weight polynomial. Powers and geometric sums use explicit caches scoped to that same object. The eight actual profile requests, in source order, are

$$(H, S/2), (B, 0), (B, S/2), (F, 0), (F, S/2), (H_{\text{low}}, 0), (H_{\text{high}}, 0), (\text{first}, 0).$$

They use 37 groups in total, 26 distinct spatial coefficient-wire tuples and 18 distinct position tuples. The complete closed ledger is

Fixed-block work	M	A
26 full width-600 spatial Horner blocks	15,574	15,574
37 group products and 29 group-sum additions	37	29
Cached Y powers	119	0
Cached geometric pairs	168	104
13 run products and 7 run-sum additions	13	7
Total	15,911	15,714

The cached exponents are 0, 1, 2, 3, 39, 40, 479, 480, 481, 482, 483, 519, 520, 959; the cached geometric lengths are 1, 2, 36, 440, 477, 478, 479, 480, 919, 957. They are part of the deterministic grammar. Every requested small profile is already in the paid prefix cache, so no hidden profile construction is appended through a stale prefix object during fusion.

48.5 Assembly of the two initializer polynomials

Set

$$\mathcal{U}_p = G_Z \mathcal{E}_p(B; W) + \mathcal{E}_p(C; W), \quad (48.14)$$

$$\mathcal{H}_{1/2} = \mathcal{E}_{S/2}(H; W) + Z\mathcal{U}_{S/2} + 3^{L-400} \mathcal{E}_{S/2}(F; W), \quad (48.15)$$

$$\mathcal{C}_0 = \mathcal{E}_0(H_{\text{high}}; W) + 3^{375} \mathcal{U}_0 + 3^{L-425} \mathcal{E}_0(F; W), \quad (48.16)$$

$$\mathcal{T}(W) = \mathcal{C}_0 + 3^{L-25} \mathcal{H}_{1/2} + 3^{u-25} \mathcal{E}_0(H_{\text{low}}; W), \quad (48.17)$$

$$\mathcal{F}(W) = \mathcal{E}_0(\text{first}; W). \quad (48.18)$$

Proposition 48.5.1 (Exact dense outputs). *The fused expressions satisfy*

$$\mathcal{T}(W) = \sum_{j=0}^{S-1} \text{tile}:j W^j, \quad \mathcal{F}(W) = \sum_{j=0}^{S-1} \text{first}:j W^j$$

as ordinary integer polynomials.

Proof. The operator $\mathcal{E}_p$ is linear in its coefficient array. Apply it to (48.2)–(48.5), using Theorem 48.3.2 and Proposition 48.4.1 for the computed spatial expressions. The middle array $x \mapsto \text{Full}[x - S/2]$ becomes $\mathcal{E}_{S/2}(\text{Full}; W)$ directly from (48.7). This gives (48.14)–(48.17). Equation (48.6) then identifies the old dense coefficients, and the first polynomial follows immediately from its definition. $\square$

After the profile and correction polynomials are available, the two bulk assemblies cost $2M + 2A$ together, the full and cut expressions each cost $2M + 2A$, and the final tile assembly costs $2M + 2A$. The total is $8M + 8A$. The first polynomial needs no further gate beyond its fixed-block construction.

48.6 The retained literal constant prefix

Fusion removes the need for the 1,152,000 individual tile and first coefficient ports, but it does not introduce free fixed constants. The prefix still constructs every required occurrence, profile, shift, anchor and small numeral.

Occurrence constants. The authenticated Report 47 compiler is retained unchanged. It lowers the fixed compressed description into signed run events without executing the ant or Boolean tape. A directional run of length n , beginning at time t and slot s_0 in direction $d \in \{-1, 1\}$, contributes events Z^t at s_0 and $-Z^{t+n}$ at $s_0 + dn$. Sweeping by $v(s) = Zv(s - d) + E(s)$ gives the exact occurrence powers. Linearity combines the runs. The inherited finite lowering and its counts remain $5,469,985M + 8,186,999A = 13,656,984$ gates. This includes the paid zero $1 - 1$, all event updates, cached powers, four sweeps and two kind-wise merges. The arbitrary row count R is encoded by the fixed description and its algebraic lowering, not by an executed physical schedule.

Small profiles. A nonzero signed width- w mask is built by the complete ternary Horner chain from its w fixed digits. It costs $(w - 1)M + (w - 1)A$, including leading zeros. Disjoint plus and minus masks select digits from the existing $-1, 0, 1$ expressions. The zero mask aliases paid zero; identical masks of the same width share a chain. The entire original inventory is retained, even if a profile is not subsequently needed:

$$736 \cdot 399 + 21 \cdot 24 + 81 \cdot 374 = 324\,462$$

of each operation type. The two extra additions/subtractions construct $-1 = (1 - 1) - 1$ and $2 = 1 + 1$. The 838 nonzero cache keys have widths 400, 25 and 375 in counts 736, 21 and 81. The independent cache check proves that all subsequent profile requests are contained in this inventory.

Geometric constant and shifts. The prefix constructs $G_Z = \sum_{r=0}^{R-1} Z^r$ by Lemma 48.4.2, with $Y = Z$, and the five fixed powers

$$3^{375}, \quad 3^{L-425}, \quad 3^{L-400}, \quad 3^{L-25}, \quad 3^{u-25}$$

by paid binary chains. Together they cost $283M + 45A$. An exponent is a finite chain specification, not an exponentiation gate.

All anchors and remaining ports. All 584 anchor rows remain. Each is a signed 221-symbol ternary Horner expression and costs $220M + 220A$, totaling $128,480M + 128,480A$. Their later spatial Horner evaluation also remains in the old main source. The six named constants are

$$\begin{array}{lll} \text{Cu} = 3^u, & \text{Cx} = 3^u - 1, & \text{Cbase} = 3^{u-219}, \\ \text{C198} = 3^{198}, & \text{EndpointK} = 3^{481225262775}, & \text{EndpointD} = 3^u - 2. \end{array}$$

The four independent power chains cost 47, 50, 10, 55 multiplications. Together with $9 = 3 \times 3$, they give 163 multiplications. Constructing $4 = 2 + 2$, $6 = 3 + 3$, $8 = 4 + 4$ and the two named subtractions costs five additions/subtractions. The eight ordinary numeral ports are 0, 1, 2, 3, 4, 6, 8, 9; zero and 2 are already paid, and 1 and 3 are the only constant leaves. There is no main **C**: -1 port. Hence exactly

$$584 + 6 + 8 = 598$$

old constant ports remain, each bound to a paid expression or a permitted literal. The complete input-independent prefix costs 14,563,366 gates.

48.7 Every consumer and the complete source substitution

The change is localized to two exact old-main gate intervals. Old gate identifiers are zero-based, and the intervals below are half-open:

Dense expression	Start	End	Removed gates
Tile Horner	1,264	1,153,262	1,151,998
First Horner	1,153,262	2,305,260	1,151,998

Each expression starts with coefficient $S - 1$ and performs

$$a \leftarrow aW + \mathbf{kind:j}, \quad j = S - 2, S - 3, \dots, 0,$$

using two records per step. Every multiplication and addition in each interval is checked against this exact recurrence. The total removed is

$$4(S - 1) = 2\,303\,996.$$

Only each interval's final output is mapped to its fused replacement. Every other removed output is marked unavailable, and any reference to one from outside its interval causes rejection.

48.7.1 Initializer consumers remain unchanged

To make the scope explicit, let X, h, k, P denote the unchanged positive initializer geometry witnesses, and let A_*, B_*, T_* denote its unchanged recoder outputs. Write W_p for the old coordinate named $\mathbf{Wp}$, and let Q_* be the old geometry coordinate named $\mathbf{Q}$. The following old definitions and residual pairs remain, with their original source order and arithmetic:

$$\begin{aligned} h_x &= X + 1, & W_p - 1 &= \mathbf{Cx} \, h_x, \\ g &= W^S, & h - 1 &= (g - 1)k, \\ Q_* &= h^2, & h_y &= k(h + 1), \\ h &= P(A_*g), & \mathbf{InitialHead} &= \mathbf{Cu} \, h. \end{aligned}$$

The recoder's internal source and residuals are unchanged. The only new operands are the two proved outputs in

$$\mathbf{bg} = h_y(h_x \mathcal{T}(W) + W_p \mathcal{F}(W)). \quad (48.19)$$

In particular W_p is its existing coordinate, not an inferred free power, and neither multiplier h_x nor h_y is dropped.

Let $\mathcal{A}(W) = \sum_{j=0}^{583} \mathbf{anchor:j} \, W^j$ be the retained full anchor Horner polynomial. Every subsequent initializer expression is retained:

$$\begin{aligned} \mathbf{anch} &= (W^{S-550} A_*) \mathcal{A}(W), \\ \mathbf{middle} &= W^{24000} T_* + B_*, \\ \mathbf{tail} &= W^{263945} (W^{264000} + 1) \mathbf{middle}, \\ \mathbf{markerTail} &= W^{23945} + \mathbf{tail}, \\ \mathbf{removed} &= \mathbf{C198} (W + 1) \mathbf{markerTail}, \\ \delta &= \mathbf{Cbase} \, P(\mathbf{anch} - \mathbf{removed}), \\ \mathbf{InitialMemoryPlus} &= \mathbf{bg} + \delta + 1. \end{aligned}$$

These equations summarize the original arithmetic, whose exact parenthesization and individual gates remain unchanged. The five displayed spatial powers are still paid in the old main. The history module, endpoint module, optional one-input pairing module, residual metadata and finalizer are also retained.

48.7.2 Topological binding and polynomial equality

The input-independent prefix is generated first. The old main is then regenerated by the authenticated owned arithmetic generator, whose complete old canonical hash must match the frozen target for each arity. Its declarations are preserved. Fusion is inserted just before old gate 1264, after the referenced W coordinate is available.

Each retained old gate is mapped to a new gate with the same opcode. Its operands bind to an earlier retained gate, an unchanged declared coordinate, one of the 598 paid constant aliases, or one of the two proved fused outputs. There is no residual **C**: operand in the resulting source. The profile-cache closure described in Section 48.6 ensures the fused construction cannot silently append a late constant to the already copied prefix state.

Proof of Theorem 48.1.1, equality and invariants. All prefix aliases have the prescribed old integer values by the inherited occurrence and profile identities and the explicit paid constant chains. Proposition 48.5.1 supplies exactly the old polynomial at each removed interval's final output. Induct on retained old gates: if all mapped operands have the old polynomial values, applying the unchanged opcode gives the same polynomial. The mapping is topological by construction and no removed interior value is referenced. The same induction verifies (48.19), all later initializer consumers, all residual operands and every finalizer gate.

For $m = 285$ in the two-input version, or $m = 286$ in the one-input version, the retained finalizer computes each residual difference $r_i = a_i - b_i$, squares it, and sums all squares. Thus it retains $mM + (2m - 1)A$, namely $285M + 569A$ or $286M + 571A$. Residual pairs are metadata, not extra asserted equations. The single final equation has that sum on the left and paid gate $0 = 1 - 1$ on the right. Coordinate declarations and positive domains are identical, so no new witness is introduced.

The old polynomial's exact degree is $4S = 2,304,000$: its only degree- $2S$ residuals have leading homogeneous term $-W^{2S}$, giving leading final part $2W^{4S}$, as established in Report 44. Equality of the complete polynomial preserves that exact degree. We do not infer it from syntactic intermediate degrees of the factored source, which can ignore cancellations. The operation counts completing the theorem are derived in Section 48.8. $\square$

48.8 Complete arithmetic ledger

Every stage count below is both a count of the specified grammar and a count reproduced by actual source generation. No giant coefficient integer is numerically instantiated.

Stage	M	A	Total
Occurrence constants	5,469,985	8,186,999	13,656,984
Small profiles and basic literals	324,462	324,464	648,926
G_Z and five fixed shifts	283	45	328
All 584 anchor rows	128,480	128,480	256,960
Remaining named constants	163	5	168

Stage	M	A	Total
Retained constant prefix	5,923,373	8,639,993	14,563,366
W^{600}	12	0	12
Twelve short correction factors	7,188	7,188	14,376
Twenty-four long factors and products	23,040	23,038	46,078
Fixed blocks and geometric weights	15,911	15,714	31,625
Fused tile assembly	8	8	16
Fused spatial subtotal	46,159	45,948	92,107
Retained old main for two inputs	1,588	1,873	3,461
Complete two-input source	5,971,120	8,687,814	14,658,934
Complete one-input source	5,971,124	8,687,820	14,658,944

The two-input old main had 2,307,457 gates. Removing precisely 2,303,996 leaves 3,461. The one-input pairing adds $3M + 4A$ and its added residual costs another $1M + 2A$ in the finalizer. Therefore the one-input remainder has $1,592M + 1,879A = 3,471$ gates, exactly ten more overall.

If the actually used outputs of the constant prefix are instead supplied as prescribed fixed coefficients, the remaining body has $92,107 + 3,461 = 95,568$ gates for two inputs and 95,578 for one. This is a separate free-coefficient model: its allowed constants include the profile and occurrence values needed by the fused body, not just the 598 surviving old named ports. The main theorem and headline totals pay for all of those constants from literal 1 and 3.

The old Report 47 prefix's columnwise correction and tile-assembly stages cost respectively 6,454,008 and 8,064,000. These disappear, as do the two old spatial Horner intervals. They are replaced by the 92,107 fused gates, giving the independent subtraction check

$$6\,454\,008 + 8\,064\,000 + 2\,303\,996 - 92\,107 = 16\,729\,897.$$

All other prefix and main work remains paid. This is a unit-cost arithmetic-source comparison. It is not a bit-complexity or numerical runtime estimate: integer values can remain enormous, and a short straight-line program is not an efficient fully expanded numerical representation.

48.9 Independent evidence and reproducibility

The evidence has four separate roles.

1. The generic proofs in Sections 48.3–48.5 establish ordinary polynomial identities for all coefficient substitutions and all W .
2. Exact finite-data checks identify the concrete masks, block partitions and profile-cache closure to which those identities apply.
3. Independent source reconstruction checks actual gates, topology, leaves, metadata, splice bindings, counts and hashes.
4. Modular differential cases provide regression tests. They supplement, rather than replace, the algebraic and exhaustive evidence.

The independent audit [3] reports PASS. It verifies all 9,187,200 formal correction-term identities across four kinds, two phases, 957 pair positions and 1200 offsets, including zero columns. It checks

all 6,912,000 fixed profile entries across the six arrays in both phases. Every block partition is disjoint and exhaustive, and the half-phase is verified as the exact 480-block rotation.

All 18 actual weight polynomials are evaluated as exact sparse polynomials in $\mathbb{Z}[W]$ and compared coefficientwise to $\sum_{k \in J} W^{600k}$. All ten geometric pairs are checked exactly. The independent adapter uses the two fixed intervals of Section 48.7, rather than the candidate’s interval recognizer, validates every removed Horner record, rejects removed-interior uses, and reproduces both complete source streams. It verifies all 598 surviving ports, witness names and order, the 285/286 residual metadata records, paid zero and the sole final equation. All 95 frozen Report 47 files were unchanged in the audit’s before/after hash inventory.

The author’s supplementary checker verifies 2,296,800 exponent identities without repeating identical arithmetic for each kind, and the same 6,912,000 array entries. Seven dense modular comparisons use 3,828 independent synthetic occurrence values. The cases are $(q, W) = (1, 000, 003, 0), (1, 000, 003, 1), (1, 000, 003, -1), (1, 000, 003, 2), (1, 000, 033, 17), (2, 1), (3, 2)$. These are deliberately small finite-field evaluations, not full integer coefficient evaluations or a proof for all occurrence constants.

[write] *In this repository.* The archive is shipped flattened with the prefix `06-ant-fusion-`. Not shipped: the frozen Report 47 reference archive and its loose `Research_Report47.tex`, `.pdf` and `MANIFEST.json` (copies of batch-88 manuscript 09, which is Report 47 in this Part), the three 1,152,000-byte profile files `science/data/profiles/*.u16le` (copies of Report 47’s, which are rebuilt by the shipped `code/05-ant-coeffs-science-geometry-build_profiles.py`; see the report README), and the other files that duplicate Report 47’s. At placement the author replay reproduced `fused-receipt.json` and `checks-receipt.json` byte for byte on a copy, and the relocated independent audit passed after a Windows-only path-separator correction to the adapter’s key spelling (the README describes it). The programme’s two reviews [W2, W3] found no defect in the algebra, the ledger arithmetic or the binding.

Canonical complete-source identities. Arithmetic records use the canonical tab-separated format identifier/opcode/operand/operand with one newline per gate. Metadata is compact JSONL. The source generator streams and hashes the actual complete source, without retaining its roughly 14.7 million records. An integer operand is an earlier gate identifier, never an extra integer literal. The two complete source hashes are

Two inputs. 62cf59e79cd5d3baf61b19ed50ccc8965e8fb4814bbdc33a6d7fad417d379667

One input. 722340689ce84f505af7d010db21d6b48178c5d6008b28966e9836aeca75b25b

Their final equations are respectively gate 14,658,933 = gate 0 and gate 14,658,943 = gate 0. The unchanged old-main target hashes and the input pins are included in the machine-readable receipts. A hash identifies the source bytes and supports reproducibility; it is not itself a mathematical proof.

Execution boundary and limits. The packet executes only inspected, authenticated owned arithmetic/profile code and new finite symbolic checks. It does not run upstream ant, physical-program, color-recipe or saved upstream schedule code. The compressed physical-program JSON is treated as fixed compilation data, not executed as a tape computation, and its saved row schedule is not used. The huge old literal coefficient prefix is not regenerated. Its equality with the new construction rests on the inherited coefficient theorem and the new identities. The checks do not re-prove arbitrary-integer Pell/history results or literal machine/ant dynamics; those inherited assumptions are unchanged.

Reproducible package. The accompanying ZIP contains the LaTeX source and PDF, exact frozen science, independent audit sources and receipts, a frozen Report 47 reference archive, provenance and release manifests, and isolated replay/build instructions. The release verifier authenticates the externally supplied manifest hash before running copied scientific code, uses fresh external output directories and disabled bytecode, and checks that original bytes, modes and modification times are preserved. PDF rebuilding disables shell escape and fixes the source epoch. The README supplies the exact offline commands and current release identities. Rebuilding the PDF requires a compatible local TeX Live installation; science replay requires Python 3. No package download or network execution is part of reproduction.

48.10 Further questions suggested by this construction

The result leaves concrete opportunities, none of which are included in the present count.

1. **Fuse the occurrence compiler more deeply.** The retained occurrence construction uses 13,656,984 gates, about 93% of the new complete source. A direct bivariate or block-valued run construction might avoid materializing every $T_{a,s}$. It would need an exact ordinary-polynomial invariant at the cyclic boundaries and a paid source ledger.
2. **Share rotated factors algebraically.** The twenty-four long Horner expressions are exact and simple but do not exhaust possible sharing. Rotation formulas using prefixes and suffixes might lower their cost; division by W and reduction modulo $W^S - 1$ would not be valid substitutes for an ordinary-polynomial proof.
3. **Refine the paid profile and anchor grammar.** This source deliberately retains the complete old profile cache and all anchor rows. A narrower cache, shared addition chains or a proved anchor factorization may help, provided every remaining consumer and constant port is still paid and independently checked.
4. **Evaluate alternative band widths and block decompositions.** The present 600-column split matches physical slots. Other exact decompositions may trade short-factor cost against occurrence-factor length or cache reuse. Any improvement must include the new powers and geometric weights rather than treating them as free.
5. **Separate arithmetic size from execution cost.** Source-generation memory, validation time and bit complexity of numerical evaluation merit distinct analysis. A lower gate count alone does not settle any of them or establish a lower bound on the best possible source.

The stopping point of this report is the independently reproduced 14,658,934/14,658,944-gate construction and its exact polynomial identity. The frozen earlier reports remain available as independently valid constructions and as the provenance of the inherited mathematical assumptions.

48.A Exact block position groups

Each row below lists the maximal runs of one distinct phase-zero 600-tuple. An interval $a:b$ includes both endpoints; a singleton is written as one integer. Groups are shown in first-occurrence order, matching the deterministic compiler. The contents of each tuple are fixed by the authenticated masks, and their full digests and both phase partitions are recorded in `fused-receipt.json`. For

phase $S/2$, rotate every listed position by -480 modulo 960; split a run at zero when necessary. The exact tuple comparison, rather than the interval notation alone, licenses grouping.

Profile	Group	Block positions
H	1	0
H	2	1:479, 483:959
H	3	480
H	4	481
H	5	482
B	1	0:479, 483:959
B	2	480
B	3	481
B	4	482
F	1	0:478, 483:518, 520:959
F	2	479
F	3	480
F	4	481
F	5	482
F	6	519
H_{low}	1	0
H_{low}	2	1:479, 482:959
H_{low}	3	480:481
H_{high}	1	0
H_{high}	2	1:479, 483:959
H_{high}	3	480
H_{high}	4	481
H_{high}	5	482
first	1	0
first	2	1:479, 482:959
first	3	480
first	4	481

48.B Source provenance and release evidence

The author packet was frozen before manuscript packaging. Its 27 pinned files were copied byte identically; the author confirmed that no scientific or code change followed the independent audit. The original README’s pending-audit language is retained as historical evidence. The later independent audit has a separate PASS verdict and manifest.

Frozen source generator. `science/fusion_source.py` has SHA256
5e433001dcd4a26f7dcb0419a41f124fae983aeb76088d55d4b603802232f60

The complete science manifest has SHA256
c2eb248a007d19b8efac47d385f7b9a820b31dd556b846396eefbaf40716d301

The prefix-only arithmetic-stream SHA256 is

4fc26b236689a78d8e2a1bc6f2eff6ef1e6bc62de337ce70b1f3c8041bc7020a

The science packet’s `INPUT_PINS.json` authenticates 18 local input files: the owned Report 47 occurrence compiler; the owned Report 44 arithmetic generator and its history, recoder and endpoint data; the anchor patch; the fixed compressed program description; and the eleven profile manifest, dictionary, index and correction files. Absolute source paths in that historical pin file record provenance only. Reproduction uses authenticated local copies.

Old canonical main sources. The owned Report 44 generator must regenerate the following targets before the adapter can accept its output:

Two inputs c16901162e09b07d1e0c27b28e29dcc12a9b6137391fce85e3f150a3fed37eac

One input 85a161972769207e4f5d4212214630535208f69d6736bca2b46a8aff8276a22a

The complete fused-stream hashes in Section 48.9 cover the new arithmetic and preserved metadata together. Full ordered witness-name digests are recorded in the receipts.

Independent audit and frozen predecessor. The independent audit manifest has SHA256 6ffaa0336e1fac214a76e9f46d4c652eaabd804c601f42cd6605a7484a0302bb

It pins the generic proof, audit report, independent source checker, cache/weight checker and their exact receipts. Its candidate snapshot used the same source generator and authenticated finite input bytes. The narrow relocation adapter supplied separately for this release changes directory and provenance resolution only; the independent checker bytes remain unchanged.

The included Report 47 release manifest is

317ad533eadbc816f595cb8f5ec262c400ab5a694b7d70559bc50988d4131819

and its complete 1,344,412-byte ZIP has SHA256

fa58914f5334f54a6c648bbaa68d6127bfe3e9de2bf9d711c95370b2310ab57f

This supplies the prior coefficient proof, finite data and authenticated Report 44 reference without relying on the original workspace paths.

The top-level Report 48 manifest inventories the final report, scientific packet, audit and reproduction helpers. Its own hash is delivered separately to avoid a circular self-hash. A manifest identifies the supplied evidence; trust in its origin must come from an external authenticated delivery. Reproduction and visual-review receipts describe the particular checks actually completed, including their preservation boundary.

References

- [1] Research Report 44, recovered authenticated edition, 4 October 2026. Complete raw-pair arithmetic source, inherited recognition theorem and exact degree. Frozen reference archive is included through Report 47. [write] Printed in Part III of this report.
- [2] Research Report 47, *Exact Structured Construction of the Fixed Ant Coefficients*, 4 October 2026. Frozen release and reproducibility archive included in `references/`. [write] Printed as the first half of Part IV of this report; its reference archive is not shipped twice.
- [3] Independent exact-fusion audit, 4 October 2026. `AUDIT.md`, `GENERIC_PROOF.md`, exact finite identities, exact cache/weight checks and complete-source reconstruction, included in `independent_audit/`. [write] Shipped with the prefix `06-ant-fusion-independent_audit-`.

[write] Provenance of Parts II–IV

Sources. Parts II–IV are built from five manuscripts, Research Reports 40, 42, 44, 47 and 48 of the Hilbert’s-tenth programme’s report pipeline, batch-88 manuscripts 03, 04, 07, 09 and 10 of ProveIt’s incoming reports. All five arrived in commit `c5612efa1` (3 October 2026) and were placed by commit `ed7c76266` (batch 88, cluster T1). Each archive’s author line is “Mathematical research report”; none names a person or tool. Part I’s provenance is Appendix C.

Report	Archive, identity, pin, where printed
40 (ms. 03)	<code>Literal_Periodic_Langton_Ant_and_Visit_Boundary_Package.zip</code> , 8,242,235 bytes, SHA-256 <code>0c3acc1e36c0931ea490c7526e053ef1bd8a17ad5ea72ec36b97dc4c3548e47e</code> ; 213 files under <code>Research_Report40/</code> (159,096,079 bytes extracted); main file <code>Research_Report40.tex</code> (830 lines), 32-page PDF, dated 3 October 2026. Pin: ProveIt <code>5883b08b7</code> (Part II, reference [5]). Printed as Part II; files <code>02-literal-ant-</code> .
42 (ms. 04)	<code>Arithmetic_Initialization_for_a_Periodic_Langton_Ant_Package.zip</code> , 761,758 bytes, SHA-256 <code>9ee861d5092370fbfd7157c718833236fab9b6da9dccc3cc41c4798d8d3eea35</code> ; 99 files under <code>ant-initialization-report42/</code> ; main file 557 lines, 18-page PDF, dated 3 October 2026. Pins: ProveIt <code>5883b08b7</code> , Mathlib <code>ac77769f</code> , Report 40’s PDF. Printed in Part III; files <code>03-ant-init-</code> .
44 (ms. 07)	<code>Complete_Positive_Certificate_for_the_Literal_Periodic_Ant_Package.zip</code> , 767,662 bytes, SHA-256 <code>3025a1efd5947e624a23f90a07dea7a826410d25944c3c968ea45edfd71013eb</code> ; 103 files under <code>report44-recovered/</code> ; main file 332 lines, 12-page PDF, dated 4 October 2026; a recovered edition. Pins: ProveIt <code>5883b08b7</code> , Mathlib <code>ac77769f</code> , Neary–Woods PDF, endpoint source. Printed in Part III (main theorem); files <code>04-ant-cert-</code> .
47 (ms. 09)	<code>Exact_Structured_Construction_of_the_Fixed_Ant_Coefficients_Package.zip</code> , 1,344,412 bytes, SHA-256 <code>fa58914f5334f54a6c648bbaa68d6127bfe3e9de2bf9d711c95370b2310ab57f</code> ; 95 files under <code>Research_Report47/</code> ; main file 401 lines, 14-page PDF, dated 4 October 2026. No ProveIt pin of its own (inherits Report 44’s). Printed in Part IV; files <code>05-ant-coeffs-</code> .
48 (ms. 10)	<code>Exact_Fusion_of_the_Ant_Background_Polynomials_Package.zip</code> , 2,162,628 bytes, SHA-256 <code>45f8dcf697012fc67cd9155f76980787fef8ff3cb4f4746ef54c4c3375c7bb65</code> ; 80 files under <code>Research_Report48/</code> ; main file 439 lines, 15-page PDF, dated 4 October 2026. No ProveIt pin of its own. Printed in Part IV; files <code>06-ant-fusion-</code> .

The pin `5883b08b7` exists and is an ancestor of the write; the two research-tree notes it covers are unchanged since. The archives survive in the arrival commit, for example `git show c5612efa1:docs/incoming/Literal_Periodic_Langton_Ant_and_Visit_Boundary_Package.zip`.

What each contributed. Report 40: the literal periodic ant, its two-visit theorem, the accepting port, the sharp boundary and the endpoint selector (Part II). Report 42: the paid all-length initializer, the recoder, the Pell power macro and the qualitative joined relation. Report 44: the complete polynomial F_2 , its one-input form F_1 , the reconstructed 174-operation history, the exact degree and the dense strict ledger (Part III). Report 47: the structured literal-1/3 coefficient prefix; Report 48: the fused background polynomials (Part IV).

What the write changed.

- *Labels.* The 192 delivered labels gained a prefix: 53 of Report 40 `ptr:la:`, 56 of Report 42 `ptr:ai:`, 20 of Report 44 `ptr:pc:`, 26 of Report 47 `ptr:sc:`, 37 of Report 48 `ptr:fu:`; every reference and citation key was rewritten to match. The write added thirteen labels (the four Parts, this section, the preface, its item on scope, the five source introductions and the remark on successive sources). Part I’s 61 labels and all its numbers are unchanged.
- *Numbering.* Section, statement and equation k of Report NN is printed as NN. k , appendix X as NN.X; Part I keeps 1–12, A–C and its equation numbers. Tables and figures are numbered through the whole report.
- *Front matter.* Each source’s title page became a Part heading and an introduction recording its title, subtitle, date and author line; its abstract is printed as a quotation. The tables of contents of Reports 40 and 42 were replaced by one for the whole report. Part I’s abstract, status box and editorial preface follow its Part heading; the report’s title, author line (“reports”), running heads and PDF metadata are new.
- *Included files.* Report 40’s three figure files are input from their shipped copies (`data/02-literal-ant-figures-primitive-*.tex`); Report 47’s `joined_evidence.tex` and Report 48’s `block_groups.tex` and `release_evidence.tex` are printed inline with their labels prefixed (shipped copies in `data/`).
- *Macros.* Renderings unchanged: `\code` of Reports 40, 42 and 44 (`\nolinkurl`) is `\codeu`; `\code` of Report 47 (an `\mbox`) is `\codem`; Report 47’s `\DUP`, `\NAND`, `\ML`, `\MR` are `\sDUP`, `\sNAND`, `\sML`, `\sMR`. No mathematical symbol was renamed.
- *Duplicates.* The recognition theorem is printed once (Theorem 44.1.1). The recalls of it in Report 47’s Section 47.1 and Report 48’s Section 48.2 are replaced by marked pointers that keep the recalls’ own remarks verbatim. Report 48’s Lemma 48.4.2 (Report 47’s Lemma 47.6.1 with Y for Z , stated in $\mathbb{Z}[Y]$) is kept as a marked second statement; Report 40’s Section 40.14 is marked as a summary re-proof of Part I. No other statement, proof, example, remark, non-claim or question of the five manuscripts was removed or changed.
- *Notes.* The notes marked [write] in Parts I–IV, this section and the preface to the four Parts, and the [write] annotations on bibliography entries that name sibling reports.

Where the merge chose. (1) *Host.* The five reports were written into this report, not into a new one, because Report 40 answers Part I’s question 3 nearly verbatim and the other four continue Report 40 alone; a new report *periodic-langton-ant-certificates* was the alternative considered. (2) *Parts.* Report 40 alone is Part II; Reports 42 and 44 form Part III because 44 composes 42; Reports 47 and 48 form Part IV because both refine the same polynomial’s literal source. (3) *The one theorem.* The recognition theorem is printed with Report 44, which proves it; the smaller sources of Reports 47 and 48 and the programme’s later counts are listed beside it. (4) *Order.* The new Parts follow Part I’s appendices and bibliography, so that no number of Part I moves. (5) *Files.* Byte copies of repository files, the five manuscripts, READMEs, PDFs and checksum manifests, a Mathlib file and its licence, in-batch duplicates and 27 large regenerable files are not shipped; the report README lists every one, the commands that rebuild the 27, and how to retrieve the archives.

References

- [W1] [write] Hilbert’s-tenth research programme, *Scoped intake of the literal ant certificates, Reports40/42/44/47/48*, review note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_periodic_ant40_48_intake.md` with its manifest `review_periodic_ant40_48_intake.json`, commit `cfbe37de7`, 3 October 2026.
- [W2] [write] Hilbert’s-tenth research programme, *Scoped intake: ant coefficient construction and fusion, Reports 47/48*, `review_ant47_48_intake.md` in the same directory, commit `5d7e884a5`, 3 October 2026.
- [W3] [write] Hilbert’s-tenth research programme, *Independent review of two disjoint Report48 sharing splices*, `review_periodic_ant_sharing48.md`, with the packets `periodic_ant_end_point_reuse48.md` and `ant48_rotated_occurrence_sharing.md`, same directory, commit `5d7e884a5`, 3 October 2026.
- [W4] [write] Hilbert’s-tenth research programme, *Removing zero arithmetic and repeated expressions from the ant rotations*, `ant48_rotated_occurrence_cleanup.md`, with `review_ant48_rotated_occurrence_cleanup.md`, same directory, commit `57a6afbeb`, 3 October 2026.
- [W5] [write] Hilbert’s-tenth research programme, *Scoped review of the five ant evidence placements*, `review_periodic_ant_placement_ed7c76266.md` with its manifest `review_periodic_ant_placement_ed7c76266.json`, same directory, commit `70bbc9ce2`, 3 October 2026.